Google Generative AI Leader

Prepare for the certification



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Chapter – 1: Artificial Intelligence – Beyond Chatbot

1.1 What is Artificial Intelligence (AI)

Artificial Intelligence (AI) is the ability of computers or machines to perform tasks that typically require human intelligence, like learning, problem-solving, understanding language, recognizing patterns, and making decisions, by processing vast amounts of data to find patterns and adapt their behaviour. It involves algorithms and techniques such as Machine Learning (ML) and Deep Learning (DL) to enable systems to learn from experience, improve autonomously, and act intelligently without rigid, pre-programmed rules.

Key Aspects of AI:

- **Learning:** Acquiring information [Data] and rules for using it.
- **Reasoning & Problem-Solving:** Using logic to solve problems and achieve goals.
- **Perception:** Interpreting visual (Computer Vision) or auditory data.
- **Language Understanding:** Processing and responding to human language (Natural Language Processing - NLP).

How AI Works

Modern AI relies on **Machine Learning (ML)**, a subset of AI where computers use algorithms to identify patterns in data rather than following rigid, pre-written instructions.

- **Deep Learning:** A more advanced subset of ML that uses **neural networks**—layers of algorithms inspired by the human brain—to process massive amounts of unstructured data like images and speech.
- **Generative AI:** A specialized branch of deep learning that creates new content (text, images, music, Video, or code) based on patterns learned from training data.

Everyday AI Applications (2025)

AI is now deeply integrated into daily life and various industries:

- **Personal Life:** Virtual assistants (Siri, Alexa), personalized recommendations (Netflix, Amazon), and real-time navigation (Google Maps).
- **Healthcare:** Early disease detection through medical imaging (e.g., pinpointing cancer on X-rays) and accelerating drug discovery.
- **Finance:** Real-time fraud detection by analyzing transaction patterns and automated stock trading.
- **Professional Work:** AI agents that can automate multi-step workflows, such as summarizing long documents or generating software code.

1.2 What is Generative AI

Generative AI is a type of artificial intelligence that can create **new content and ideas**, including text, images, music, and even code.

Four Primary Ways of using Generative AI are:

- Create
- Summarize
- Discover
- Automate

1.2.1 Create

Using generative AI to create new content, such as: **Writing articles, emails, and social media posts.**

- Generate text responses to customer service inquiries using Conversational AI.
- Draft emails to clients using Gemini in Gmail.

Creating images, videos, and audio

- Create a custom image for a presentation using Gemini in Google Slides.
- Generate multiple design concepts for brainstorming marketing materials such as website pages using Imagen.
- Create video tutorials to guide customers through troubleshooting steps using Google Vids.

Generating code in various programming languages

- Generate unit tests and other test cases for code, reducing the manual effort required for testing while still ensuring the code works as expected.
- Generate code as a starting point for larger software development projects using Gemini Code Assist.

1.2.2 Summarize

Condense large amounts of information into concise summaries, such as:

Summarizing long documents or articles

- Summarize a lengthy financial report without having to read the entire document using NotebookLM Enterprise.
- Summarize customer reviews to understand product feedback and sentiment using Gemini in Looker.

Extracting key takeaways from meetings or presentations

- Project managers can automatically generate meeting notes that highlight key decisions and action items using Gemini in Google Meet.
- Sales teams can use summaries to quickly recap client calls and identify next steps.

1.2.3 Discover

Help your customers and employees find what they need at the right time, such as:

Uncovering hidden patterns and insights in data

- Analyze purchase history and website traffic to predict future demand and optimize inventory using forecasting models on Vertex AI.
- Identify viewing patterns of streaming viewers to recommend personalized content to subscribers, increasing engagement and reducing churn using Vertex AI Search.
- Analyze vast datasets of molecular structures to identify potential drug candidates with higher success rates by fine-tuning models on Vertex AI.

Searching for resources or documents

- Search for a file or ask questions about its contents using Gemini in Google Drive.
- Unlock enterprise expertise for employees with agents that bring together Gemini's advanced reasoning, Google-quality search, and enterprise data, regardless of where it's hosted using Gemini Enterprise.

Monitoring real-time events

- Use anomaly detection to identify fraudulent transactions and prevent financial losses.

1.2.4 Automate

Automate tasks that previously required human intervention, such as:

Automated format conversation

- Convert content to accessible formats in real time by using Google Cloud's APIs, like Text-to-Speech, Speech-to-Text, Translation, and more.

Automated documentation

- Analyze code and automatically generate clear and concise documentation, saving developers time and ensuring consistency using Codey.
- Automate contract review and information extraction with Document AI.

Example: Automated notifications and alarms

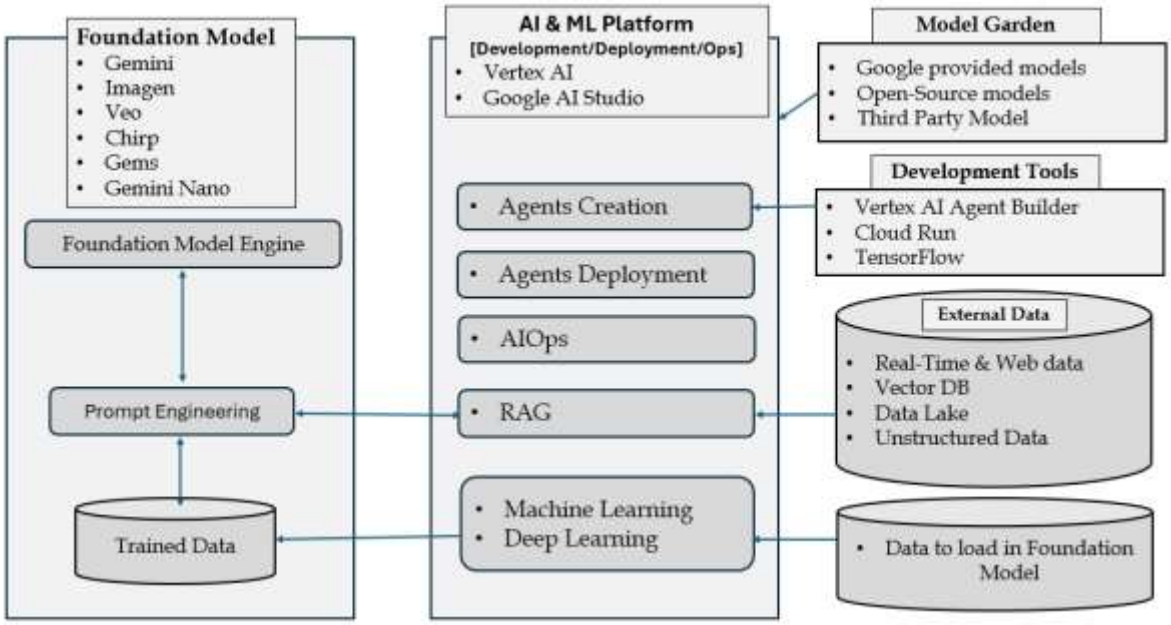
- Automate customer feedback analysis and ticket creation.

Multi-Modal Generative AI

Gen AI can process and integrate information from diverse formats like plain text, images, videos, audio, and PDFs. This capability–known as multimodal gen AI–means that models can process various input types individually or combine multiple kinds of data simultaneously.

1.3 Google AI Platform Architecture

Following diagram shows the Google AI Platform for Development, Deployment and maintenance of AI solutions. Vertex AI is used for development, deployment and AI Ops of AI solutions in google platform.



1.4 Google: An AI-first Company

Since 1998, Google has been at the forefront of technological innovation, pioneering advancements in AI to create groundbreaking products like Search, Gmail, and Android.

And Google continues to lead the way. Google didn't just stop at making new products; they wanted to encourage innovation in a bigger way. Driven by its mission to "organize the world's information and make it universally accessible and useful," Google has continuously shared its research, best practices, and technological advancements with the world.

In 2008, Google launched Google Cloud, making these same cutting-edge technologies available to businesses everywhere. This allows individuals and businesses to benefit from Google's advancements in fields like AI, without having to start from scratch.

2001: ML for spell check

Google begins using machine learning to help with spell check at scale in Google Search.

2006: Google Translate

Google launches Google Translate using machine learning to automatically translate languages.

2012: Deep learning in speech recognition

A new era of AI begins when Google researchers improve speech recognition with deep neural networks, which is a new machine learning architecture loosely modelled after the neural structures of the brain.

2015: Google Search

Google integrates **RankBrain** into Google Search, using machine learning to analyze search queries and deliver more relevant results to users.

2015: Tensorflow

Google introduces a new scalable open source machine learning framework used in speech recognition with the release of TensorFlow.

2016: TPUs

Google announces the Tensor Processing Unit, custom data center silicon built specifically for machine learning.

2017: Transformer

Google releases **"Attention is All You Need,"** a research paper that introduces the transformer, a novel neural network architecture particularly well suited for language understanding, among many other things.

2018: AI Principles

Google publishes a set of guidelines the company follows when developing and using AI. For more information, see Google's AI Principles.

2018: BERT

Google introduces a new technique for natural language processing pre-training called BERT (Bidirectional Encoder Representations from Transformers), helping Search better understand users' queries.

2018: Smart Compose in Gmail

Google announces Smart Compose, a new feature in Gmail that uses AI to help users more quickly reply to their email. Smart Compose builds on Smart Reply, another AI feature.

2023: Bard

Google releases Bard, an early experiment that lets people collaborate with generative AI.

2023: Gemini

Google introduces Gemini, a multimodal AI model capable of understanding and integrating various information types. The following year, Google expands the Gemini ecosystem with Gemini 1.5, bringing it to more products and launching Gemini Advanced, which provides access to Google's most capable AI models.

2024: Gemma

Google announces Gemma, a family of lightweight state-of-the-art open models built from the same research and technology used to create the Gemini models.

1.5 Overview of Google's AI platform

Google AI Platform refers to Google's end-to-end ecosystem for building, training, deploying, and managing machine learning (ML) and generative AI solutions. Today, it is primarily delivered through **Google Cloud's Vertex AI**, combined with **Google DeepMind's foundation models** and supporting AI services.

The platform serves:

- Data scientists & ML engineers
- Application developers building GenAI apps
- Enterprises deploying AI at scale with governance and security

Google Cloud AI Platform Evolution

- **2018**: Launch of **Google Cloud AI Platform** (classic)
    - Focused on training and deploying ML models on GCP.
- **2021**: Introduction of **Vertex AI**
    - Unified AI Platform + AutoML into one managed service.
- **2023–Present**: Shift toward **Generative AI platforms**
    - Vertex AI expanded to support large language models (LLMs), multimodal AI, and foundation models.

Key Strengths of Google AI Platform

- Deep integration with **Google Cloud services**
- Industry-leading **multimodal foundation models**
- Strong **MLOps and governance** capabilities
- Enterprise-grade **security and compliance**
- Open ecosystem (TensorFlow, PyTorch, open models)

1.6 Foundation Model

An **AI foundation model** is a **large, general-purpose machine learning model** that is **trained on vast amounts of data** and can be **adapted to many different tasks** instead of being built for just one specific use.

A foundation model is a base AI model that learns broad patterns from large-scale data and then serves as the foundation for many specialized applications.

Key characteristics

1. Trained on massive data

- Uses huge datasets (text, images, audio, video, or multimodal)
- Learns general knowledge, language, and patterns

2. General-purpose

- Not designed for one task only
- Can perform or be adapted to:
    - Text generation
    - Question answering
    - Image recognition
    - Code generation
    - Speech understanding

3. Adaptable

Foundation models can be specialized for particular domains or use cases through additional, targeted training.

- Can be customized using:
    - Fine-tuning
    - Prompting
    - Few-shot learning
    - Domain-specific data

4. Transfer learning

- Knowledge learned in one context transfers to new tasks
- Saves time, cost, and data compared to training models from scratch

5. Flexible

With foundation models, one AI model can support a wide range of use cases.

Examples of foundation models

Language models

- GPT (OpenAI)
- BERT (Google)
- LLaMA (Meta)

Vision models

- CLIP
- DINO
- Vision Transformers (ViT)

Multimodal models

- GPT-4-type models
- Flamingo
- Gemini-style models

1.7 Google Foundation Models

Google's **foundation models** are large, pre-trained models developed primarily by **Google DeepMind** and Google Research.

List of key Foundation Models are as follows: -

1. Gemini

- Google's flagship multimodal foundation model (text, code, image, audio, video).
- Variants:
    - **Gemini Ultra** – Complex reasoning, enterprise-grade
    - **Gemini Pro** – Scalable general-purpose model
    - **Gemini Nano** – On-device AI (mobile, edge)

2. Imagen

- Text-to-image generation model.
- Used for high-quality image synthesis.

3. Codey

- Code-focused model.
- Supports:
    - Code generation
    - Code completion
    - Code explanation

4. Chirp

- Speech-to-text and audio understanding model.

5. Med-PaLM (Healthcare)

- Specialized LLM for medical and clinical use cases (regulated access).

Use cases: Imagen

PUMA, a global sports brand, partnered with Google Cloud to revolutionize its e-commerce product imagery. By leveraging Imagen, PUMA can now create high-quality images of their products in different settings, colors, and styles – all without the need for expensive photoshoots.

This approach not only saves time and resources, but also allows for greater flexibility and creativity in showcasing their products online. The result is a more engaging and personalized shopping experience for PUMA's customers, driving sales and brand loyalty.

1.8 Google AI Tools and Applications

Over the years google has developed many AI tools and Applications. Few of them are: -

1.8.1 Vertex AI

Vertex AI is Google Cloud's unified machine learning (ML) platform. It empowers you to build, train, and deploy ML models and AI applications. For multiple modalities (text, code, images, speech), Vertex AI gives you access to Google's large generative AI models through Model Garden. You can tune Google's large language models (LLMs) to meet your needs, and then deploy them for use in your AI-powered applications.

Simply put, **Vertex AI** is the key to **creating and deploying custom generative AI solutions** for your business.

1.8.2 Google AI Studio

Google AI Studio is a web-based Integrated Development Environment (IDE). It is designed for quick prototyping and experimentation with Google's generative AI models. It is a primary gateway for developers and researchers to build applications using the Gemini API.

Core Capabilities

- **Model Access:** Provides access to the Gemini 3 Pro and 2.5 families for tasks like coding, reasoning, and creative writing.
- **Multimodal Playground:** Supports inputs beyond text. Users can test and refine prompts using images, video, and audio.
- **Rapid Prototyping:** The "Build" mode can automatically generate code for AI-native apps based on natural language descriptions.

- **Fine-Tuning:** Users can customize models by uploading datasets (CSV or Google Sheets) to train them for specific contexts or industry-specific tasks.
- **API Management:** It is the central hub for generating and managing Gemini API keys at no initial cost for the free tier.

Key Features

- **Prompt Editor:** It is a sandbox to create and test different prompt types, including chat and structured prompts, with adjustable parameters like temperature and token limits.
- **Code Export:** Once a prototype is finalized, it can be exported as production-ready code in languages like Python, Node.js, or cURL.
- **Direct Deployment:** It is integrated with Google Cloud Run. This allows users to deploy prototypes as functional web apps with a single click.
- **System Instructions:** Users can define specific roles (e.g., "You are a coding assistant") and safety filters to control model behaviour.

Accessibility

Google AI Studio is available through aistudio.google.com and is generally free to use within specific usage quotas. It is intended for developers who want a faster, more streamlined path than the enterprise-grade Vertex AI platform.

1.8.3 Google Workspace with Gemini

Google Workspace with Gemini is an AI-powered suite that integrates Google's advanced large language models into everyday productivity tools. It acts as a collaborative assistant to help users brainstorm, write, visualize, and automate tasks in apps such as Gmail, Docs, and Drive.

Key Capabilities

- **Gmail:** Summarizes email threads, drafts responses, and suggests replies based on previous messages.
- **Google Docs:** Generates drafts, proofreads for grammar and style, and summarizes documents.
- **Google Sheets:** Automates data organization and predicts data patterns.
- **Google Slides:** Creates images from text descriptions and removes backgrounds from photos.
- **Google Meet:** Provides audio and video enhancements, generates meeting backgrounds, and automatically captures meeting summaries.
- **Google Vids:** This new AI-powered video creation app helps users draft scripts, generate voiceovers, and edit video clips for work.

1.8.4 Gemini App

Access the power of Gemini directly through a dedicated app for personalized assistance and creative exploration.

1.8.5 Appsheet

This is a No-Code Application Development tool available in Google Cloud environment.

1.8.6 NotebookLM

Google NotebookLM is an AI-powered research and writing tool designed to help you process and understand information by "grounding" a language model in your own specific documents. Originally known as Project Tailwind, it creates a personalized AI assistant that is expert in the sources you provide.

Key Features

- **Source Grounding:** Unlike general chatbots, NotebookLM answers questions using only your uploaded materials—such as PDFs, Google Docs, website URLs, and YouTube transcripts—to reduce inaccuracies and "hallucinations".
- **Audio Overviews:** It can generate a podcast-style audio discussion where two AI hosts summarize, debate, and explain your source materials. Users can even "join" the conversation to ask the hosts questions in real-time.
- **Citations:** Every response includes in-line citations that point directly to the specific passage in your source, allowing you to quickly verify the information.
- **Study Aids:** The tool can automatically generate flashcards, quizzes, study guides, and briefing documents from your notes.
- **Mind Maps:** It offers a visual way to navigate complex topics by creating interactive mind maps that show connections across your different sources.

How to Use It

1. **Upload Sources:** Add documents from Google Drive, local files, or public web links.
2. **Ask Questions:** Use the chat interface to summarize documents, find specific details, or brainstorm new ideas based on the material.
3. **Refine & Organize:** Save key AI responses as notes or use the **Learning Guide** feature for step-by-step tutoring.

Privacy and Pricing

- **Data Privacy:** Google does **not** use the data you upload or your interactions with the AI to train its general AI models.
- **Free & Paid Versions:** NotebookLM is free for individual Google account holders. A premium version, **NotebookLM Plus**, is available for enterprises and included in

the Google One AI Premium plan, offering higher usage limits and advanced administrative controls.

1.8.7 Contact Center as a Service [CCaaS]

Google's **Contact Center as a Service (CCaaS)**, officially known as Contact Center AI (CCAI) Platform, is a cloud-native, AI-powered solution designed to manage all customer interactions across voice and digital channels in a single interface.

Core Components

- **Virtual Agent:** Uses Dialogflow CX to build advanced conversational bots for self-service over voice, chat, and SMS, automating routine inquiries.
- **Agent Assist:** Provides human agents with real-time suggestions, automated summaries, and knowledge base articles during live calls and chats.
- **Insights:** Leverages natural language processing to identify call drivers, customer sentiment, and operational trends from interaction data.
- **Omnichannel Routing:** Automatically directs customers to the most appropriate agent or resource pool based on intent and availability across all communication channels.

Key Features

- **CRM-Centric Design:** Purpose-built to integrate natively with popular CRMs like Salesforce and Zendesk, serving as a "single source of truth" for the customer journey.
- **Mobile-First Experience:** Optimized for smartphones, allowing users to switch between voice and digital channels while sharing photos, videos, or using on-device authentication.
- **Turnkey Deployment:** Offers a "shirt-sized" instance model (Small, Medium, Large) for quick setup through the Google Cloud Console without needing complex on-premises hardware.
- **Advanced Management:** Includes tools for real-time monitoring, workforce management, and detailed performance reporting for supervisors.

Pricing Models

Google provides flexible pricing structures for its CCaaS services:

- **Named Agent Model:** Approximately $50 per agent per month, ideal for full-time staff.
- **Concurrent Agent Model:** Approximately $60 per logged-in agent per month, suitable for shift-based teams.
- **Usage-Based Add-ons:** Costs for AI features like Agent Assist (e.g., $0.06 per chat session) or Insights (e.g., $5 per 100 analyzed conversations).

1.8.8 Cloud Run Function

In August 2024, Google rebranded **Cloud Functions (2nd gen)** to Cloud Run functions, merging its infrastructure with the Cloud Run platform. It is a **Function-as-a-Service (FaaS)** offering that allows you to deploy single-purpose snippets of code that run in response to events or HTTP requests without managing any servers.

Key Capabilities

- **Event-Driven Execution:** Automatically triggers code from over 90 sources via Eventarc, such as file uploads to Cloud Storage, messages in Pub/Sub, or Firebase events.
- **Direct Control:** Because it is built on Cloud Run, developers gain access to advanced configurations like concurrency (up to 1,000 requests per instance), GPU support for AI workloads, and Direct VPC egress for secure networking.
- **Language Support:** Provides managed runtimes for popular languages including Node.js, Python, Go, Java, .NET, PHP, and Ruby.
- **Automatic Scaling:** Scales from zero to "planet-scale" based on demand; you only pay when your code is executing.

Common Use Cases

- **Data Processing:** Generating image thumbnails, transcoding videos, or transforming data before loading it into BigQuery.
- **Webhooks:** Building lightweight APIs or responding to third-party events from services like Stripe, GitHub, or Slack.
- **AI & Automation:** Running sentiment analysis on text files, automating infrastructure labels, or integrating with Dialogflow for virtual assistants.

Pricing & Free Tier

Cloud Run functions follows a pay-as-you-go model based on vCPU-seconds, GiB-seconds, and the number of requests.

- **Free Allowance:** Includes 2 million requests, 400,000 GB-seconds, and 200,000 GHz-seconds of compute time per month.
- **Idle Time:** You pay nothing when the function is idle unless you configure minimum instances to eliminate "cold starts".

1.8.10 Gemini Enterprise

Gemini Enterprise is an AI platform from Google Cloud. It automates business workflows. It was launched in October 2025. It is the successor to "Agentspace". It allows employees to build, deploy, and govern AI agents. These agents can act across enterprise systems.

Core Capabilities

Gemini Enterprise provides a platform for business transformation:

- **AI Agents:** Users can create custom agents. These agents can plan and execute tasks.
- **Enterprise-Wide Search:** It can access and synthesize information from different sources. These sources include Google Workspace, Microsoft 365, Salesforce, SAP, Jira, and ServiceNow.
- **No-Code Workbench:** Employees can build custom agents using plain language via the **Agent Designer**.
- **Pre-built "Made by Google" Agents:** It includes tools like **Deep Research**, **NotebookLM Enterprise**, and **Gemini Code Assist**.

Key Editions (2026)

The platform is offered in three editions:

1. Standard: It is for large organizations. It provides company-wide search and agent controls (approx. $30/month per user).
2. Plus: This tier is for "power users" and developers. It provides access to new models like Gemini 3. It offers agent building and higher storage limits (75 GiB per user).
3. Frontline: This is a lightweight version for field and retail workers.

Comparison with Google Workspace with Gemini

Gemini Enterprise is an independent platform. It connects to external third-party software. It automates cross-application tasks. The Gemini add-on for Google Workspace is an embedded assistant for apps like Gmail and Docs.

Comparison between "Gemini Enterprise" and "Google workspace with Gemini"

Gemini Enterprise is an independent platform. It connects to external third-party software. It automates cross-application tasks. The Gemini add-on for Google Workspace is an embedded assistant for apps like Gmail and Docs.

Feature	Workspace with Gemini	Gemini Enterprise
Primary Goal	Productivity assistant for Google apps	Agentic automation for the whole business
Data Scope	Google Workspace & Public Web	Full enterprise data (Google + 3rd party)
Automation	Minimal (drafts emails, summaries)	High (executes multi-app workflows)
UI	Embedded in Gmail/Docs/Slides	Unified Portal / Hub

Security and Governance

Gemini Enterprise includes security features like Model Armor, VPC Service Controls, and Customer-Managed Encryption Keys (CMEK). User data is not used to train Google's models.

1.9 Prompting Models

Foundation models, like other AI models, take in inputs, which are called prompts, to produce outputs.

While all AI models have inputs or prompts, foundation models are usually much more flexible in what you can prompt. For example, some AI models have strict requirements on the type of inputs they can handle. They may only handle numbers, emails, or images.

Foundation models, and especially multimodal foundation models, can take in a much broader set of inputs.

Some examples of Input [Prompts] and Output for Foundation model is shown below: -

Input [Prompt]	Output
Question	The answer to the question
Request for an image based on a text description	The image
Request a summary of a file	The file summary
Sample code with errors	Corrected code with no errors

The inputs you can prompt your foundation models on are endless. Depending on the foundation model, they can be versatile to many different forms of prompts.

Prompting is one of the most important skills in maximizing the value of gen AI models and tools.

Possible Input types for a Gemini Prompt

- Text description
- Image input
- Code snippet
- Audio file
- Video file

Sample Prompt

- Sample input: "Generate an image of a futuristic city with flying cars and towering skyscrapers."
- Code Snippet: Sample input: "Help me debug my Python code. [Paste the Python code with errors.]"
- Audio File: Sample input: "Summarize the troubleshooting steps from this customer service call. [Upload the audio recording of the customer service call.]"
- Sample input: "Generate a personalized workout plan for a user who wants to [fitness goal] and has [fitness level]."

1.10 Building a successful gen AI strategy

In this section, let us discus on **top-down vision for AI solutions**, **bottom-up employee-driven ideas**, and ensuring initiatives align with your company's business priorities.

1.10.1 Role of Executives and high-level management

Executive sponsorship and a clear vision for gen AI implementation play critical roles in gen AI implementation. Ensure that gen AI initiatives align with overall business goals, receive adequate resources, and gain organization-wide support, while simultaneously empowering employees and teams to experiment.

Prioritize use cases that are feasible to implement, actionable, affordable, and have high anticipated business value and return on investment (ROI).

1.10.2 Role of Mid-level Managers and Individual Contributors

Proximity to daily operations and end-users provides invaluable insight. Champion gen AI adoption by identifying high-impact, feasible solutions that address specific challenges within your workflows. Encourage experimentation and feedback from your teams to ensure gen AI tools are effectively integrated into existing processes, fostering increased productivity and innovation.

- Demonstrate the tangible impact of your gen AI experiments to gain support and drive further adoption. Quantify the achievement, whether it's increased efficiency, improved user satisfaction, or cost savings.
- Articulate how your gen AI initiatives contribute to the organization's overall business goals and objectives to align your efforts with strategic priorities and showcase the value you're delivering.
- Define and track KPIs that measure the effectiveness of your gen AI solutions. This data provides evidence of your impact and helps you refine your approach over time.

1.10.3 Implementing a multi-directional Strategy

Successful gen AI implementation requires a combination of top-down and bottom-up approaches. For both approaches, you want to be strategizing on:

- Strategic focus
- Exploration
- Responsible AI
- Resourcing
- Impact

- Continuous improvement

Strategic Focus – Executives and high levels of management

- Focus on a specific area where gen AI can have a significant impact. Once you've achieved success in that initial domain, expand to other areas of your business. This focused approach maximizes learning and scalability while ensuring tangible business value.
- Prioritize use cases that are feasible to implement, actionable, affordable, and have high anticipated business value and return on investment (ROI).

Strategic Focus – Mid-level managers and ICs

- Identify the most pressing pain points in your daily work or those you hear directly from customers. These real-world challenges are often the best starting point for finding gen AI applications that can have a significant impact.
- Prioritize understanding user needs, their workflows, and their expectations for interacting with AI systems. A human-centered design approach is essential for ensuring that gen AI solutions meet user needs and deliver a positive experience.
- Explore use cases in areas of the business that offer a low-risk environment for initial implementation. This allows for valuable learning and experimentation during the initial stages of adoption, enabling teams to refine processes and maximize benefits as confidence grows.

Exploration – Executives and high levels of management

- Empower employees to experiment with gen AI and identify potential use cases. This can involve hosting hackathons or providing dedicated time and resources for exploration.
- Foster a collaborative environment where employees can share findings, insights, and best practices for using gen AI. This leverages the expertise and creativity of individuals across the organization.

Exploration – Mid-level managers and ICs

- Experiment with different gen AI tools and applications. This practical experience will provide valuable insights into the capabilities and limitations of the technology, helping to identify promising use cases.
- Share your findings and results with your team and colleagues. If initial experiments prove successful, consider expanding the scope of your exploration and bringing your insights to the attention of leadership.

Responsible AI - Executives and high levels of management

- Establish ethical guidelines and implement safety mechanisms to ensure that gen AI is used responsibly and securely across the organization.
- Institute robust data governance practices that support data security and privacy, and follow relevant regulations. Use enterprise tooling like Google Cloud Data Loss Prevention, Cloud Key Management Service, and Chronicle Security Operations to automate data discovery, classification, and de-identification, manage encryption keys, and detect and respond to security threats.
- Implement a content moderation policy that includes employee review of AI-generated content, particularly in sensitive or high-risk applications.
- Regularly monitor the performance and impact of gen AI systems to identify and address any unintended consequences or ethical concerns.

Responsible AI - Mid-level managers and ICs

- Prioritize responsible AI development and deployment by adhering to company AI standards, aligning efforts with organizational values and mitigating potential risks.
- Think proactively about potential pitfalls and unintended consequences that might arise from gen AI implementation.
- Conduct thorough testing and evaluation to identify and address any biases, safety concerns, or ethical issues.

Resourcing - Executives and high levels of management

- Create a robust data strategy to ensure your teams can access high-quality, relevant data that is appropriate for training and fine-tuning gen AI models.
- Avoid reinventing the wheel by leveraging existing tools and platforms where possible.
- Invest in developing the necessary AI talent within the organization by upskilling existing employees and hiring new talent with gen AI expertise.

Resourcing - Mid-level managers and ICs

- Leverage the resources readily available within your team or department. This could include existing datasets, tools, or platforms that can be leveraged for gen AI development.
- If your ideas or experiments require additional resources, don't hesitate to speak to leadership and make a case for the potential value and ROI of your gen AI initiatives.

Impact - Executives and high levels of management

- Set clear goals and define key performance indicators (KPIs) to measure the impact of gen AI, like increased sales or reduced customer complaints.
- Regularly communicate the progress and impact of gen AI initiatives to stakeholders, showcasing the value generated and justifying continued investment.

Impact- Mid-level managers and ICs

- Demonstrate the tangible impact of your gen AI experiments to gain support and drive further adoption. Quantify the achievement, whether it's increased efficiency, improved user satisfaction, or cost savings.
- Articulate how your gen AI initiatives contribute to the organization's overall business goals and objectives to align your efforts with strategic priorities and showcase the value you're delivering.
- Define and track KPIs that measure the effectiveness of your gen AI solutions. This data provides evidence of your impact and helps you refine your approach over time.

Continous Improvement - Executives and high levels of management

- Embrace an iterative development approach, continuously refining gen AI solutions based on feedback, performance data, and evolving business needs.
- Establish a regular evaluation cadence to assess the quality, accuracy, and effectiveness of gen AI outputs. Use this evaluation data to identify areas for improvement and guide future development efforts.
- Implement mechanisms for gathering feedback from users, stakeholders, and other relevant parties to ensure gen AI systems meet user needs and expectations.

Continous Improvement - Mid-level managers and ICs

- Perform continuous testing, measurement, and refinement based on user feedback and real-world performance data. These are essential to ensure gen AI solutions remain effective and valuable.
- Update your gen AI solutions regularly based on new data and feedback to improve performance.
- Continue your own gen AI education by taking courses, reading articles, and experimenting.

1.10.4 Key Takeaway - successful Gen AI strategy

Successfully implementing generative AI requires a combined top-down and bottom-up strategy. Leaders must set a clear vision and provide resources, while empowering individuals and teams to identify practical applications and experiment with the technology. By fostering collaboration and continuous improvement, organizations can effectively leverage generative AI to drive innovation and achieve their business objectives.

1.11 Keeping humans at the forefront of gen AI implementation

When strategizing about gen AI solutions, think about shifting your responsibilities from task execution to strategic thinking.

1.11.1 Augmentation vs Automation

Use gen AI to enhance or **augment** your strategic thinking for:

- **Critical thinking and problem solving:** Gen AI can provide data and insights, but humans are still needed to interpret those insights and make informed decisions.
- **Creativity and innovation:** Gen AI can assist in generating ideas and exploring possibilities, but human ingenuity is still essential for pushing boundaries and developing truly innovative solutions.
- **Relationship building and collaboration:** Gen AI can facilitate communication and information sharing, but strong interpersonal skills are still crucial for building trust, fostering collaboration, and navigating complex human dynamics.
- **Strategic planning and vision:** Gen AI can help with forecasting and trend analysis, but human leadership is essential for setting a long-term vision, defining goals, and charting a course for the future.

Use gen AI to **automate** tasks that are:

- **Repetitive and rule-based:** data entry, information retrieval, content formatting, and basic code generation.
- **Time-consuming and resource-intensive:** research, data analysis, content summarization, and initial draft creation.

Even for task automation, humans-in-the-loop are a necessary component for the gen AI implementation and continuous improvement. Use people for:

- **Data selection and preparation:** ensuring that gen AI models are trained on high-quality, relevant data that is representative of the intended use cases.
- **Prompt design and refinement:** crafting prompts that elicit accurate and useful responses from gen AI models.
- **Output evaluation and refinement:** reviewing and editing gen AI-generated content to ensure accuracy, relevance, and alignment with brand guidelines.
- **Continuous monitoring and feedback:** providing feedback on gen AI performance and identifying areas for improvement.

1.11.2 Use Case: Warner Brothers AI-Assisted Captioning on Google Cloud

Let's take a look at how Warner Brothers is augmenting human captioning work with AI-assisted captioning, along with humans-in-the-loop, to reduce overall costs by 50% and reduce the time it takes to manually caption a file by 80% without the use of machine learning.

It's crucial to remember that generative AI solutions always involve human interaction in some capacity. Whether it's your employees using gen AI tools to streamline their workflow or your customers engaging with AI-powered chatbots for support, the human element is always present. This is why you always need humans in your process so you can design and implement solutions in a way that enhances user experience and fosters positive interactions with the people involved.

1.11.3 Example of Automation and Augmentation

Example 1: Your sales team spends a lot of time answering basic questions about product pricing and availability. You decide to implement a gen AI chatbot to handle these initial inquiries. Is this **augmentation** or **automation**?

Answer: Automation

Answering frequently asked questions is a repetitive task that can be easily handled by a gen AI chatbot. This frees up the sales team to focus on more complex customer interactions and closing deals.

Example 2: You're a financial analyst who uses gen AI to analyze large datasets and generate reports summarizing market trends. You then use this information to advise clients on investment strategies. Is this **augmentation** or **automation**?

Answer: Augmentation

While gen AI is automating the data analysis and report generation, your role as a financial analyst is enhanced. You're using the AI-generated insights to inform your strategic thinking and provide valuable advice to clients. This aligns with the text's emphasis on using gen AI to enhance critical thinking and problem-solving.

Example 3: An online retailer uses gen AI to generate product descriptions for its vast catalogue of items. They provide the AI with key product attributes (size, material, features, etc.) and the AI creates unique and informative descriptions for each item. Is this **augmentation** or **automation**?

Answer: Automation

Gen AI automates the labor-intensive task of writing product descriptions for a large catalogue, increasing efficiency and reducing the need for human writers to create basic product information. This allows the retailer to allocate resources to other important areas like marketing and customer service.

Example 4: A software development team uses gen AI to generate code for standard functions and components within their applications. They then review and integrate this code into their larger projects. Is this **augmentation** or **automation**?

Answer: **Augmentation**

The gen AI automates code generation, but it's augmenting the developers' capabilities by increasing their efficiency and allowing them to focus on more complex tasks. Human oversight remains essential for reviewing and integrating the AI-generated code, ensuring quality and alignment with project goals.

1.12 Question Paper – 1 [40 Questions – 80 Minutes]

1. A CEO is hesitant to invest in generative AI because they believe it's just a technology for building chatbots. Which of the following examples demonstrate that generative AI can offer their business much more?
    A) Creating photorealistic images of new product prototypes based on text descriptions.
    B) Counting the number of website visitors and their locations.
    C) Providing generic responses to all customer inquiries, regardless of the specific issue.
    D) Scheduling meetings and managing calendars to help employees improve their time management.

2. Which of the following statements accurately describe Gemini? Select two.
    A) Gemini is an AI assistant that can help you be more creative and productive.
    B) Gemini is a type of artificial intelligence that can create new content, including text, images, music, and even code.
    C) Gemini is a generative AI model (or family of models) developed by Google.
    D) Gemini is a type of generative AI specifically designed for generating images.
    E) Gemini is a chatbot that can answer questions and generate creative content.

3. Which of the following defines generative AI?
    A) A type of artificial intelligence that focuses on automating repetitive tasks and improving efficiency.
    B) A specific application, like a chatbot, that utilizes AI technology.
    C) A type of artificial intelligence that analyzes existing data to identify patterns and make predictions.
    D) A type of artificial intelligence that can create new content, including images, text and music.

4. Which of the following describes a multimodal gen AI application?
    A) Using NotebookLM Business to summarize a financial report.
    B) Using Imagen to create an image for a website.

C) Using Gemini in Gmail to write an email.
D) Using gen AI to analyze customer sentiment in video testimonials and survey data.

5. How do foundation models and prompt engineering work together to create value in generative AI?
   A) Foundation models provide the computing power for generative AI, while prompt engineering directs that power to complete specific tasks.
   B) Foundation models offer a vast knowledge base, and prompt engineering guides the model to use this knowledge in responses.
   C) Prompt engineering trains foundation models on specific tasks, allowing them to generate highly specialized content and insights.
   D) Foundation models ensure the ethical use of generative AI, while prompt engineering focuses on improving the quality and creativity of outputs.

6. Which of the following is a primary feature of foundation models?
   A) Adaptability across a wide range of tasks.
   B) Specialization for a single task
   C) Limited to their initial pre-trained function.
   D) Training on small, specific datasets.

7. Which of the following defines a foundation model?
   A) Small, very specialized AI models trained on narrow datasets in order to perform specific tasks.
   B) Traditional machine learning algorithms that rely on explicitly defined rules.
   C) Large AI models trained on a vast quantity of data, capable of adapting to a variety of tasks.
   D) Hardware infrastructure used to train and deploy AI models.

8. What is the purpose of a prompt in the context of foundation models?
   A) To train the model on new data.
   B) To fine-tune the model for a specific task.
   C) To provide input to the model and trigger an output.
   D) To evaluate the model's performance.

9. What is the primary difference between foundation models and traditional AI models?
   A) Foundation models are trained on massive amounts of diverse data for various tasks, while traditional models are trained on specific data for a single task.
   B) Foundation models are trained on specific data for a single task, while traditional models are trained on diverse data for various tasks.
   C) Foundation models are only trained on text data, while traditional models use images and code.
   D) Foundation models cannot be adapted to new tasks, while traditional models can.

10. Which of the following is a benefit of using Google Cloud for gen AI development?
    A) It offers a limited set of pre-trained AI models for specific industry use cases.
    B) It requires extensive upfront hardware investments for running complex AI workloads.
    C) It provides comprehensive services, including scalable infrastructure, enterprise-grade governance and security.
    D) It focuses primarily on individual developer tools without robust governance or scalability features.

11. Why is it important for mid-level managers and individual contributors to be involved in gen AI adoption?
    A) Their proximity to workflows allows them to identify impactful gen AI solutions.
    B) They have limited understanding of daily operations and user needs.
    C) They should avoid any involvement in gen AI experimentation.
    D) Their focus should be solely on technical implementation details.

12. What is a key advantage of using Google Cloud's gen AI ecosystem for businesses?
    A) It allows businesses to leverage Google's AI advancements without starting from scratch.
    B) It assures businesses that they will achieve immediate and widespread success in all of their generative AI endeavours.
    C) It eliminates the need for companies to cultivate internal gen AI knowledge and experimentation across their organizations.
    D) It forces each and every businesses to become an "AI-first" company.

13. What is the recommended approach for businesses to effectively implement gen AI?
    A) Combine a top-down strategic vision with bottom-up input from teams.
    B) Adopt the latest gen AI trends.
    C) Focus on a bottom-up approach with no overarching plan.
    D) Focus on a top-down strategy driven by executives.

14. Your Sales team spends a lot of time answering basic questions about product pricing and availability. You decide to implement a gen AI chatbot to handle these initial enquiries. Is this?
    A) Augmentation
    B) Automation
    C) Manual process
    D) Neither of Augmentation nor Automation
15. What is Vertex AI
    A) A Foundation model
    B) Unified machine learning (ML) platform from Google
    C) An end-user AI product from google
    D) A component of NotebookLM

16. **What best defines Generative AI?**
    A) AI that only analyzes structured numerical data

B) AI that creates new content such as text, images, music, or code
C) AI designed solely for automation of workflows
D) AI that replaces traditional machine learning models

17. **Which of the following is NOT one of the four primary ways of using Generative AI?**
    A) Create
    B) Summarize
    C) Predict
    D) Automate

18. **Which activity is an example of using Generative AI to *create* content?**
    A) Detecting fraudulent transactions
    B) Generating unit test cases for code
    C) Monitoring real-time system logs
    D) Encrypting enterprise data

19. **Summarizing a long financial report without reading the entire document is an example of which Gen AI capability?**
    A) Create
    B) Discover
    C) Summarize
    D) Automate

20. **Identifying hidden patterns in customer purchase history to predict future demand falls under which Gen AI category?**
    A) Create
    B) Discover
    C) Summarize
    D) Automate

21. **Converting content into accessible formats using Text-to-Speech or Translation APIs is an example of:**
    A) Discover
    B) Create
    C) Summarize
    D) Automate

22. **What does "multimodal generative AI" mean?**
    A) AI that uses only text-based inputs
    B) AI that can process only images and videos
    C) AI that integrates text, images, audio, video, and PDFs
    D) AI that operates across multiple cloud platforms

23. **What is Vertex AI primarily used for?**
    A) Email automation
    B) Building and deploying custom generative AI solutions

C) Data encryption and compliance management
D) Social media content generation

24. **Which is a key feature of foundation models?**
A) They are trained on small, task-specific datasets
B) They support only one specific use case
C) They can be adapted to multiple downstream tasks
D) They require no prompts

25. **Which foundation model is primarily trained on images and text descriptions?**
A) Gemini
B) Chirp
C) Imagen
D) BERT

26. **In generative AI, what is a "prompt"?**
A) The output generated by the model
B) The training dataset
C) The input provided to the model
D) The model evaluation metric

27. **Which of the following is NOT a possible input type for a Gemini prompt?**
A) Text description
B) Image
C) Audio file
D) Database index

28. **Why is prompting considered an important skill in generative AI?**
A) It replaces the need for training models
B) It helps guide models to produce desired outputs
C) It eliminates the need for human involvement
D) It ensures models never make errors

29. **Which approach is recommended for successful gen AI implementation?**
A) Only top-down leadership strategy
B) Only bottom-up experimentation
C) A combination of top-down and bottom-up approaches
D) Outsourcing all AI decisions

30. **What is the purpose of humans-in-the-loop (HITL) in gen AI systems?**
A) To eliminate automation
B) To continuously monitor, refine, and evaluate AI outputs
C) To slow down AI deployment
D) To replace AI models with manual processes

31. **Which task is most suitable for automation using generative AI?**

A) Strategic vision setting
B) Relationship building
C) Repetitive, rule-based data entry
D) Long-term leadership planning

32. **Which Google initiative provides guidelines for ethical and responsible AI development?**
    A) TensorFlow
    B) Secure AI Framework (SAIF)
    C) RankBrain
    D) Vertex AI Model Garden

33. Which of them is not a Foundation Model
    A) ChatGPT
    B) Gemini
    C) Google Workspace
    D) Imagen

34. Which Tool can be used to train the Gemini Foundation Model
    A) Vertex AI
    B) NotebookLM
    C) Gemini Enterprise
    D) RAG
35. You are planning to develop a AI system that will translate Audio from one language to another. Which foundation model you would prefer to minimize the cost.
    A) Gemini
    B) ChatGPT
    C) Chirp
    D) Imagen

36. What input file you can attach for analysis in Gemini Prompt. Choose all that apply.
    A) Binary File
    B) Audio File
    C) Video File
    D) Code Snippet

37. What is Gemma
    A) An open-source light weight foundation model from Google
    B) A Paid Foundation model from Google
    C) A platform to develop AI solution
    D) An agent to access external data like weather report

38. What is TensorFlow and PyTorch**?**
    A) Foundation Model
    B) Open-source machine learning frameworks
    C) Infrastructure component to optimize the hardware in which Google AI platform runs

D) AI Agents to translate languages

39. Google Sheet and Google Meet tools are part of
    A) Google Workspace
    B) Gemini
    C) Vertex AI
    D) Google AI Studio

40. What is Veo
    A) Veo is an AI Agent to analyze a Video
    B) Google's Text-to-Video AI Model
    C) Veo is an AI search engine
    D) Veo is used to train AI model

Chapter – 2: Unlock Foundational Concepts

2.1 Overview

Gen AI: Unlock Foundational Concepts is the second course of the Gen AI Leader learning path. In this course, you unlock the foundational concepts of generative AI by exploring the differences between AI, ML, and gen AI, and understanding how various data types enable generative AI to address business challenges. You also gain insights into Google Cloud strategies to address the limitations of foundation models and the key challenges for responsible and secure AI development and deployment.

2.2 Understanding of AI, ML and Generative AI

- **Artificial intelligence (AI):** The broad field of building machines that can perform tasks requiring human intelligence.
- **Machine learning (ML):** A subset of AI where machines learn from data.
- **Generative AI:** An application of AI that creates new content.

2.3 Understanding the importance of data in AI

We talked about what ML, AI, and gen AI is all about. But the question is, how do AI systems, powered by ML, actually accomplish these tasks? It all boils down to the data they're fed. Essentially, machine learning models predict the future based on existing data, much like how humans use experience to make educated guesses. However, while humans might rely on intuition or gut feelings, these models use probability.

Choosing the right data is crucial for AI. Let us discuss some of the parameters for quality of data:-

2.3.1 Data quality

ML models analyze the data they've been given, identify patterns, and then calculate the likelihood of different outcomes when presented with new information.

This is why the quality and quantity of the data ML models learn from is absolutely crucial to how well they perform. There are five factors to focus on when thinking about data quality.

- **Accuracy**

  If the data is inaccurate, the model will learn incorrect patterns and make faulty predictions. Imagine teaching a child about animals using a book with mislabelled pictures—they'd learn the wrong things. The same applies to AI.

- **Completeness**

  Completeness refers to the size of a dataset as well as representation within the dataset. The size of the dataset is important because the model needs enough to make an accurate

prediction. If a meteorologist tries to predict weather only based on the data of the past day, it will be a much worse prediction than if it used a much larger sample size.

- **Representative**

Data needs to be representative and inclusive, otherwise it can lead to skewed samples and biased outcomes. If a dataset about customer preferences is missing information about a certain demographic, the model might make inaccurate or biased generalizations about that group.

- **Consistency**

Inconsistent data formats or labeling can confuse the model and hinder its ability to learn effectively. Imagine trying to assemble a puzzle where some pieces are labeled with numbers and others with letters—it would be a mess.

- **Relevance**

The data must be relevant to the task the AI is designed to perform. For example, data about traffic patterns in London is unlikely to be helpful for predicting crop yields in Kansas.

2.3.2 Data accessibility

The ability of AI systems to effectively utilize this data is directly tied to data accessibility. Data accessibility ensures that the necessary data is readily available, usable, and of high quality, allowing for comprehensive model training and reducing potential biases. Without accessible data, even the most sophisticated algorithms are limited in their ability to learn and provide accurate predictions.

- **Availability**
  If the necessary data simply isn't available, the AI model can't be trained. For some problems, the data might exist, but it might be locked behind paywalls or restricted due to privacy concerns.
- **Cost**
  Gathering and cleaning data can be expensive. The cost of acquiring high-quality data can be a significant barrier to AI development, especially for smaller organizations.
- **Format**
  Data needs to be in a format that the AI model can understand and process. Converting data into the appropriate format can be time-consuming and technically challenging.

2.3.3 Data types

Data is central to many business processes and various data types can be used in machine learning, but data isn’t just about numbers and files. Business datasets help organizations understand customers, optimize operations, and drive strategic decisions.

Larger datasets frequently improve model performance, particularly for complex generative AI models that produce diverse content. However, a large dataset isn't the only thing that matters because model performance is heavily dependent on the type of data used for training. This includes the data's volume and its organization and structure.

Understanding your company's data, its quality, availability, and form is essential for understanding the realm of what will be possible in terms of using that data for AI.

Data forms

Data comes in various forms, just like information itself. We can broadly categorize this data into two main types: structured and unstructured. Let's explore this through a fictitious company that sells eco-friendly cleaning supplies called Cymbal Cleaning.

The data they store for customer orders can include information like:

- Customer ID
- Customer name
- Purchase date
- Order cost
- Delivery address
- Product image
- Feedback (on a 1-5 star scale)

Examples

A product review left on an ecommerce website (e.g., "This phone is great! The battery life is amazing."). Is it structured or unstructured data?

- Unstructured Data

Sales data in a CSV file with columns like "Date," "Product ID," "Quantity Sold," and "Price." Is it structured or unstructured data?

- Structured Data

An audio recording of a customer service call. Is it structured or unstructured data?

- Unstructured Data

A digital image of a cat. Is it structured or unstructured data?

- Unstructured Data

A medical record with a patient's date of birth, address, and vaccination history. Is it structured or unstructured data?

- Structured Data

2.3.4 Key Takeaway

Data is the foundation of any AI system, with data quality and accessibility being essential for effective AI development. Data can be structured or unstructured, each requiring different analysis techniques. Understanding the types and quality of your data, including accuracy, completeness, consistency, relevance, availability, cost, and format is crucial for successful AI initiatives.

2.4 Machine learning approaches and data requirements

The quality, accessibility, and form of data are fundamental, but how that data is used for ML depends on the specific learning method. Machine learning has three primary learning approaches: supervised, unsupervised, and reinforcement learning, each with its own data requirements.

Supervised models rely on labelled data, unsupervised models work with unlabelled data, and reinforcement learning learns through interaction and feedback.

2.4.1 Labelled versus unlabelled data

Labelled Data

Labelled data has tags, such as a name, type, or number. These tags, whether applied manually or by automated systems, assign meaning to the data.

For instance, an image dataset for training a cat-detection model would label each picture as either a cat or dog. Similarly, a set of customer reviews might be labeled as positive, negative, or neutral. These labels enable algorithms to learn relationships and make accurate predictions.

Unlabelled data

Unlabelled data is simply data that is not tagged or labelled in any way. It's raw, unprocessed information without inherent meaning.

Examples of unlabelled data include a collection of unorganized photos, a stream of audio recordings, or website traffic logs without user categorization. In these cases, algorithms must independently discover patterns and structures within the data, as there are no pre-existing labels to guide the learning process.

2.4.2 Supervised versus unsupervised learning

Supervised machine learning

Supervised machine learning trains models on labeled data, where each input is paired with its correct output, allowing the model to learn the relationship between them. The

model's goal is to identify patterns and relationships within this labeled data, enabling it to accurately predict outputs for new, unseen inputs.

Predicting housing prices is a common example of supervised learning. A model is trained on a dataset where each house has labeled data, such as its size, number of bedrooms, location, and the corresponding sale price. This labeled data allows the algorithm to learn the relationship between the features of a house and its price. Once trained, the model can then predict the price of a new house based on its features.

Unsupervised ML models

Unsupervised ML models deal with raw, unlabeled data to find natural groupings. Instead of learning from labeled data, it dives headfirst into a sea of unlabeled data.

For example, an unsupervised learning algorithm could analyze customer purchase history from your company's database. It might uncover hidden segments of customers with similar buying habits, even though you never explicitly labeled those segments beforehand. This can be incredibly valuable for targeted marketing or product recommendations.

Think of it as exploratory analysis. Unsupervised learning helps you understand the underlying structure of your data and uncover insights that you might not have even known to look for.

2.4.3 Reinforcement learning

Reinforcement learning is all about learning through interaction and feedback. Imagine a robot learning to navigate a maze. It starts with no knowledge of the maze's layout. As it explores and interacts with the maze, it collects data—bumping into walls (negative feedback) or finding shortcuts (positive feedback). Through this process of trial and error, the algorithm learns which actions lead to the best outcomes.

It's like training a pet. You reward good behavior and discourage bad behavior. And over time, the pet learns to perform the desired actions. Similarly, in reinforcement learning, the algorithm learns to maximize rewards and minimize penalties by interacting with its environment.

This type of learning is particularly useful in situations where you can't provide explicit instructions or labeled data. For example, you could use reinforcement learning to train a self-driving car to navigate complex traffic situations or to optimize the performance of a robot in a manufacturing plant.

2.4.4. Examples of machine learning approaches on Google Cloud

Now, let's explore some use cases for different machine learning approaches on Google Cloud.

- **Predictive maintenance with Vertex AI (supervised learning)**
  How it works
  By training a model on sensor data from machines like temperature, pressure, and vibration, Vertex AI can predict when a machine is likely to fail, enabling proactive maintenance and reducing downtime.

- **Anomaly Detection with BigQuery ML (unsupervised learning)**
  How it works
  BigQuery ML can analyze historical transaction data (amount, location, time, etc.) to identify patterns and flag unusual transactions that deviate significantly from the norm. This helps prevent fraud and minimize financial losses.

- Product recommendations with Vertex AI (reinforcement learning)
  How it works
  Vertex AI can train a reinforcement learning model to recommend products to users based on their browsing history, purchase behavior, and other factors. The model learns to maximize user engagement and sales by continuously refining its recommendations.

2.5 Deep Learning

2.5.1 About Deep Learning

In previous section, we explored the connection between artificial intelligence and machine learning, as well as the core types of machine learning: supervised, unsupervised, and reinforcement learning.

Now, let's see how deep learning fits into this larger picture.

Machine Learning is a broad field that encompasses many different techniques, one of which is deep learning (DL).

Deep Learning is a powerful subset of machine learning, distinguished by its use of artificial neural networks. These networks enable the processing of highly complex patterns and the generation of sophisticated predictions.

Neural networks can leverage both labeled and unlabeled data, a strategy known as semi-supervised learning. They train on a blend of a small amount of labeled data and a large amount of unlabeled data. That way, they learn foundational concepts and generalize effectively to novel examples.

Generative AI uses the power of deep learning to create new content spanning text, images, audio, and beyond. Deep learning techniques, particularly those centered on neural networks, are the engine behind these generative models.

2.5.2 Deep Learning in Foundation Model

Remember foundation models from course one? Foundation models use deep learning. They are trained on massive datasets that allow them to learn complex patterns and perform a

variety of tasks across different domains. They are incredibly powerful machine learning models trained on a massive scale, often using vast amounts of unlabeled data. This training allows them to develop a broad understanding of the world, capturing intricate patterns and relationships within the data they consume.

Think of a foundation model as a highly advanced learner. It's like a student who has read everything in an entire library, absorbing knowledge from countless books, articles, and websites. This deep and extensive learning allows foundation models to be adapted to a wide range of tasks.

2.5.3 Large language models (LLMs)

One particularly exciting type of foundation model is the LLM. These models are specifically designed to understand and generate human language.

They can translate languages, write different kinds of creative content, and answer your questions in an informative way, even if they are open ended, challenging, or strange. This is likely the most common foundation model you've encountered, such as in popular generative AI chatbots like Gemini. They also help power many search engines you use today.

2.5.4 Diffusion Models

Diffusion models are another type of foundational model. They excel in generating high-quality images, audio, and even video by iteratively refining noise (or unstructured/random data and patterns) into structured data.

2.5.5 Key takeaway

To summarize, deep learning provides the core technology, foundation models are the powerful architectures built on deep learning, and generative AI is the application of these models to create new and original content.

2.6 Choosing a Model

2.6.1 Factors when choosing a model

When picking a model for your gen AI use case, it's important to start to think about some criteria for different models. Here are some important factors to keep in mind.

Modality

When selecting a generative AI model, it's crucial to consider the modality of your input and output. Modality refers to the type of data the model can process and generate, such as text, images, video, or audio. If your application focuses on a single data type, like generating text-

based articles or creating audio files, you'll want to choose a model optimized for that specific modality. For applications that require handling multiple data types, such as generating image captions (processing images and producing text) or creating video with accompanying audio, you'll need a multimodal model. These models can understand and synthesize information across different modalities.

Context Window

The context window refers to the amount of information a model can consider at one time when generating a response. A larger context window allows the model to "remember" more of the conversation or document, leading to more coherent and relevant outputs, especially for longer texts or complex tasks. However, larger context windows often come with increased computational costs. You need to balance the need for context with the practical limitations of your resources.

Security

Security is paramount, especially when dealing with sensitive data. Consider the model's security features, including data encryption, access controls, and vulnerability management. Ensure the model complies with relevant security standards and regulations for your industry.

Availability and Reliability

The availability and reliability of the model are crucial for production applications. Choose a model that is consistently available and performs reliably under load. Consider factors like uptime guarantees, redundancy, and disaster recovery mechanisms.

Cost

Generative AI models can vary significantly in cost. Consider the pricing model, which might be based on usage, compute time, or other factors. Evaluate the cost-effectiveness of the model in relation to your budget and the expected value of your application. This is where selecting the right model for the right task is important. Be sure to match the model to the task; bigger isn't always better, and multi-modal capabilities aren't always necessary.

Performance

The performance of the model, including its accuracy, speed, and efficiency, is a critical factor. Evaluate the model's performance on relevant benchmarks and datasets. Consider the trade-offs between performance and cost.

Fine-tuning and Customization

Some models can be fine-tuned or customized for specific tasks or domains. If you have a specialized use case, consider models that offer fine-tuning capabilities. This often involves training the model further on a specific dataset related to your use case.

Ease of Integration

The ease of integrating the model into your existing systems and workflows is an important consideration. Look for models that offer well-documented APIs and SDKs.

2.6.2 Google Cloud's ML Models

Vertex AI streamlines the integration of advanced artificial intelligence capabilities into business applications, allowing for seamless discovery, deployment, and customization. These models empower businesses to leverage cutting-edge AI, providing the flexibility to work with many different models without the need for extensive in-house model development.

With Vertex AI you can access models developed by Google including **Gemini, Gemma, Imagen**, and **Veo**. You can also access proprietary third-party models, and openly available models.

Gemini

Gemini, a multimodal model, can understand and operate across diverse data formats, such as text, images, audio, and video. Gemini's multimodal design supports applications that require complex multimodal understanding, advanced conversational AI, content creation, and nuanced question answering.

Gemma

A family of lightweight, open models is built upon the research and technology behind Gemini. They offer developers a user-friendly and customizable solution for local deployments and specialized AI applications.

Imagen

A powerful text-to-image diffusion model, it excels at generating high-quality images from textual descriptions. This makes it invaluable for creative design, ecommerce visualization, and content creation.

Veo

A model capable of generating video content. It can produce videos based on textual descriptions or still images. Its functionality allows for the creation of moving images for applications such as film production, advertising, and online content.

Collectively, these Google Cloud foundation models empower businesses to enhance customer experiences through intelligent chatbots and personalized content. They increase productivity by automating tasks and improving information retrieval. They foster innovation by generating new ideas and designs. They also derive data-driven insights for improved decision-making.

2.6.3 Key takeaway

When choosing a generative AI model, consider factors such as modality, context window, security, availability, cost, performance, fine-tuning, and ease of integration. Google Cloud offers a suite of foundation models, including Gemini, Gemma, Imagen, and Veo, each with unique strengths and capabilities.

These models can be accessed and customized through Vertex AI, empowering businesses to enhance customer experiences, increase productivity, foster innovation, and improve decision-making.

2.7 Google strategies for foundation model limitations

2.7.1 Foundation model limitations

Foundation models, while groundbreaking, aren't without limitations. Recognizing these limitations is essential for the responsible and effective utilization of these powerful tools.

Data Dependency

The performance of foundation models is heavily data-dependent. They require large datasets, and any biases or incompleteness in that data will inevitably seep into their outputs. It's like asking a student to write an essay on a book they haven't read. If the data or questions are inaccurate or biased, the AI's performance will suffer.

Knowledge cutoff

Knowledge cutoff is the last date that an AI model was trained on new information. Models with older knowledge cutoffs may not know about recent events or discoveries. This can lead to incorrect or outdated answers, since AI models don't automatically update with the latest happenings around the world.

For example, if an AI tool's last training date was in 2022, it wouldn't be able to provide information about events or information that happened after 2022.

Bias

An LLM learns from large amounts of data, which may contain biases. You can think of bias as an unbalanced dataset in LLMs. Due to their statistical learning nature, they can sometimes amplify existing biases present in the data. Even subtle biases in the training data can be magnified in the model's outputs.

Fairness

Even with perfectly balanced data, defining what constitutes fairness in an LLM's output is a complex task. Fairness can be interpreted in various ways. Fairness assessments for generative AI models, while valuable, have inherent limitations. These evaluations typically focus on specific categories of bias, potentially overlooking other forms of prejudice. Consequently, these benchmarks do not provide a complete picture of all potential risks associated with the models' outputs, highlighting the ongoing challenge of achieving truly equitable AI.

Hallucination

Foundation models can sometimes experience hallucinations, which means they produce outputs that aren't accurate or based on real information. Because foundation models can't verify information against external sources, they may generate factually incorrect or nonsensical responses. These cause significant concern in accuracy-critical applications. The

responses might sound convincing, but they are completely wrong. We will cover this in more detail below.

Edge Cases

Rare and atypical scenarios can expose a model's weaknesses, leading to errors, misinterpretations, and unexpected results.

2.7.2 Techniques to overcome Foundation model limitations

Several techniques can significantly overcome foundation model limitations. Let's explore some of the key approaches.

- **Grounding**

Generative AI models are amazing at creating content, but sometimes they hallucinate. Grounding is the process of connecting the AI's output to verifiable sources of information—like giving AI a reality check. By providing the model with access to specific data sources, we tether its output to real-world information, reducing the risk of invented content.

Grounding is essential for building trustworthy and reliable AI applications. By connecting your models to verifiable data, you ensure accuracy and build confidence. It offers several key benefits, including reducing hallucinations, which prevents the AI from generating false or fictional information. Grounding also anchors responses, ensuring the AI's answers are rooted in your provided data sources. Furthermore, it builds trust by enhancing the trustworthiness of the AI's output by providing citations and confidence scores, allowing you to verify the information.

There are many different options on how you can ground in data. For example, you can ground in enterprise data or you can ground using Google Search.

- **RAG [Retrieval-augmented Generation]**

One common grounding method to do this is with **retrieval-augmented generation**, or **RAG**.

RAG is a grounding method that uses search to find relevant information from a knowledge base and provides that information to the LLM, giving it necessary context.

The first step is retrieval. When you ask an AI a question, RAG uses a search engine to find relevant information. This search engine uses an index that understands the semantic meaning of the text, not just keywords. This means it finds information based on meaning, ensuring higher relevance.

The retrieved information is then added to the prompt given to the AI. This is the augmentation phase.

The AI then uses this augmented prompt, along with its existing knowledge, to generate a response. This is referred to as the generation phase.

- **Prompt Engineering**

  Prompting offers the most rapid and straightforward approach to supplying supplementary background information to models. This involves crafting precise prompts to guide the model towards desired outputs. It refines results by understanding the factors that influence a model's responses. However, prompting is limited by the model's existing knowledge; it can't conjure information it hasn't learned.

- **Fine-tuning**

When prompt engineering doesn't deliver the desired outcomes, fine-tuning can enhance your model's performance. Pre-trained models are powerful, but they're designed for general purposes. Tuning helps them excel in specific areas. This process is particularly useful for specific tasks or when you need to enforce specific output formats, especially if you have examples of the desired output.

Tuning involves further training a pre-trained or foundation model on a new dataset specific to your task. This process adjusts the model's parameters, making it more specialized for your needs. Google Cloud Vertex AI provides tooling to facilitate tuning.

Here are some examples of how tuning can be used:

- Fine-tuning a language model to generate creative content in a specific style.
- Fine-tuning a code generation model to generate code in a particular programming language.
- Fine-tuning a translation model to translate between specific languages or domains.

- **<u>Humans in the loop (HITL)</u>**

Beyond these techniques, we must remember the invaluable role of humans in the loop (HITL). Machine learning models are powerful, but they sometimes need a human touch. For tasks requiring judgment, context, or handling incomplete data, human expertise is essential. HITL systems integrate human input and feedback directly into the ML process. This collaboration makes models more adaptable, especially in areas like the following.

  - **Content Moderation** - HITL ensures accurate and contextually appropriate moderation of user-generated content, filtering out harmful or inappropriate material that algorithms alone might miss.
  - **Sensitive Applications** - In fields like healthcare or finance, HITL provides oversight for critical decisions, ensuring accuracy and mitigating risks associated with automated systems.
  - **High-risk decision-making** - When ML models inform decisions with significant consequences, such as medical diagnoses or criminal justice

assessments, HITL acts as a safeguard, providing a layer of human review and accountability.

- **Pre-generation Review** - Before deploying ML-generated content or decisions, human experts can review and validate the outputs, catching potential errors or biases before they impact users.
- **Postg-generation review** - After ML outputs are deployed, continuous human review and feedback help identify areas for improvement, enabling models to adapt to evolving contexts and user needs.

2.7.3 Comparison of overcoming Tools

Feature	RAG	Fine-tuning	Grounding
Definition	Augments LLMs by retrieving relevant information from external knowledge bases and adding it to the prompt.	Further trains a pre-trained model on a new dataset to adapt it to a specific task or domain.	Connects an AI model's output to verifiable sources of information.
Process	Retrieve relevant information. → Add it to the prompt. → Generate a response.	Select a pre-trained model. → Gather data. → Train the model. → Evaluate and refine.	Provide access to data sources. → Use RAG or fine-tuning to connect the output.
Data sources	External knowledge bases (databases, documents, internet).	Task- or domain-specific datasets.	External knowledge bases or specific datasets.
Relationship to grounding	A specific technique for achieving grounding.	Improves a model's ability to be grounded in specific domains.	The overarching goal, achieved through techniques like RAG and fine-tuning.

2.7.4 Key Takeaway

Foundation models have limitations, including data dependency, knowledge cut-offs, biases, potential for hallucinations, and issues with fairness and edge cases. Techniques like grounding, prompt engineering, fine-tuning, and human-in-the-loop systems can address these challenges and improve the accuracy, reliability, and fairness of foundation models.

2.8 Secure AI

2.8.1. Introduction

AI systems are enabling computers to see, understand, and interpret the world in ways that were unimaginable a few years ago. And these systems are developing at an extraordinary pace. Yet, despite these remarkable advancements, AI, like anything else, is not infallible. Let's hear a quick introduction from the course creators on building responsible AI.

2.8.2 What does Secure AI mean?

Secure AI is about preventing intentional harm being done to your applications. This is about protecting AI systems from malicious attacks and misuse. For all applications, including AI, you need to ensure security throughout the full lifecycle from development through deployment. This includes considering the data, infrastructure, and how and where applications are deployed.

To better understand how these security principles apply, let's reexamine the machine learning lifecycle from an earlier lesson.

- **Gather your data**
  The foundation of any robust AI system is secure data. Therefore, data must be protected and secured at all times. Access controls must be implemented to restrict who can access the data. They are also necessary to restrict who can add or input to the data. This is crucial to prevent data poisoning, a malicious attack where bad actors corrupt your AI model's training data by introducing manipulated data. This can cause your model to learn incorrect patterns and make flawed (biased, inaccurate, or even harmful) predictions. It's akin to someone secretly swapping out vital ingredients in a recipe, leading to a disastrous final dish.

- **Prepare your Data**
  During data preparation, special attention should be paid to confidential or sensitive data that might be present in the training data. Where feasible, anonymization techniques should be applied to this data, mitigating potential privacy risks. Data also should be rigorously validated, using integrity checks and anomaly detection to prevent poisoning. Data should be securely processed with encryption both at rest and in use. Logging and real-time monitoring should also be performed.

- **Train your Model**
  Securing the model and the training process is paramount. This means safeguarding both the training data and model parameters from unauthorized access or modification. Model theft, where attackers steal proprietary or sensitive AI models, is a significant threat. Beyond gaining a competitive advantage, the stolen models could be used for malicious purposes, exploiting vulnerabilities or replicating sensitive functionality.

- **Deploy and Predict**
  Once your model is trained and ready for action, it's crucial to safeguard it within its operating environment. This means controlling access to the model, determining who

can interact with it and how. If you're using a pre-built model, it's essential to verify its source and check for any potential vulnerabilities.

- **Manage your Model**
  Continuous monitoring and maintenance of the model's security are essential. Stay up to date on the latest security best practices in the field, specifically for your platform and model, and regularly update to patch vulnerabilities. Monitor model performance and outputs for anomalies or signs of tampering, and regularly review access permissions.

  Think of it like installing a high-tech security system in your office. You wouldn't give everyone the master key, right? Similarly, you need to restrict access to your model and filter the information it receives and sends out.

  One major risk here is adversarial attacks, where attackers try to trick the model with misleading information. Imagine someone trying to bypass your security system with a fake ID—that's essentially what an adversarial attack does to your AI. Therefore, it's critical to have safeguards in place to filter, sanitize, and check the data coming into your model, ensuring it's not being fed misleading information. Likewise, you need to monitor the information your model sends out. Think of it as screening outgoing messages to prevent any accidental leaks of sensitive data or the spread of harmful content.

2.8.3 Stay Informed

Keeping AI applications secure demands constant vigilance and staying informed about the latest security threats and research.

2.8.4 Applying the Secure AI Framework (SAIF)

Google has developed the Secure AI Framework, or SAIF, to establish security standards for building and deploying AI systems responsibly. This comprehensive approach to AI/ML model risk management addresses the key concerns of security professionals in the rapidly evolving landscape of AI. Following the security practices outlined in SAIF can help your organization find and stop threats, automatically strengthen its defenses, and manage the unique risks of each AI system. The framework is designed to integrate with your company's existing security, ensuring that AI models are secure by default. For more information, explore Google's Secure AI Framework (SAIF).

2.8.5 Google Cloud security tools

Platforms such as Google Cloud can facilitate secure development. In addition to the Secure AI Framework, Google Cloud offers a range of tools to ensure applications remain secure throughout their lifecycle. Google Cloud has built security into its core through a secure-by-design infrastructure, encompassing its global network and hardware, as well as robust encryption in transit and at rest. It provides customers with detailed control over access and usage of cloud resources through Identity and Access Management (IAM). Google Cloud Security Command Center provides a centralized view of your security posture across your

entire Google Cloud environment, and Google Cloud also offers tools for monitoring various workloads.

2.8.6 Key Takeaway

AI offers significant benefits, but it also introduces security risks like data poisoning, model theft, and prompt injection. To address these challenges, a secure foundation for AI applications is essential. Google Cloud's SAIF framework, combined with security tools, can help as you build and maintain secure AI systems.

2.9 Responsible AI

Responsible AI means ensuring your AI applications avoid intentional and unintentional harm. It's important to consider ethical development and positive outcomes for the software you develop. The same holds true, and perhaps even more so, for AI applications.

2.9.1 Building on Security

The foundation of responsible AI is security. Secure applications protect both your company and your users. Think of it like building a house: if the foundation is weak, the entire structure is compromised, no matter how beautiful the design. Just as a strong foundation ensures a stable and safe house, robust security forms the essential foundation for building truly responsible AI.

2.9.2 Transparency is key

Transparency is paramount for responsible applications. Users need to understand how their information is being used and how the AI system works. This transparency should extend to all aspects of the AI's operation, including data handling, decision-making processes, and potential biases.

2.9.3 Privacy in the age of AI

Protecting privacy often involves anonymizing or pseudonymizing data, ensuring individuals can't be easily identified. AI models can sometimes inadvertently leak sensitive information from their training data, so it's crucial to implement safeguards to prevent this.

2.9.4 Data quality, bias, and fairness

- **Ethical AI requires high quality data**

  Machine learning and generative AI applications are fundamentally based on data. Therefore, responsible AI requires high-quality data. Inaccurate or incomplete data can lead to biased outcomes. Remember, technology often reflects existing societal biases. Without careful consideration, AI can amplify these biases, leading to unfair and discriminatory results. Beyond data quality, it's crucial to consider the responsible use of the data itself. Was it collected consensually? Could it perpetuate harmful biases?

- **Understanding and mitigating biases**

  AI systems are not independent of the world they are built in. They can inherit and amplify existing societal biases. This can lead to unfair outcomes, such as a resume-screening tool

that favors a certain demographic of candidates due to historical biases in hiring data. It's like training a dog with biased commands; the dog will learn and replicate those biases. To counter this, fairness must be a core principle in AI development.

2.9.5 Accountability and explainability

Fairness requires accountability.

We need to know who is responsible for the AI's actions and understand how it makes decisions. This is where explainability comes in. Explainable AI makes the decision-making processes of AI models transparent and understandable. This is crucial for building trust, debugging errors, and uncovering hidden biases. Think of it like a judge explaining their verdict; without a clear explanation, it's hard to trust the decision. Tools like Google Cloud's Vertex Explainable AI can help understand model outputs and identify potential biases. Understanding how your application uses and interprets the AI's output is crucial for ensuring responsible use.

2.9.6 Legal implications

Beyond considerations like fairness and bias, AI development is increasingly governed by legal frameworks. Key areas include data privacy, non-discrimination, intellectual property, and product liability.

- Laws mandate responsible data handling, bias mitigation, and transparency in algorithmic decision-making.
- Organizations must also adhere to the specific rules and limitations of the AI models they employ, ensuring compliance with licensing agreements and legal standards.
- The legal landscape is rapidly evolving, requiring organizations to stay informed and seek legal counsel to ensure compliance.
- Legal compliance is not just a regulatory hurdle. Navigating these legal implications is crucial for building trustworthy AI.

2.10 Question Paper – 2 [50 Questions: 100 Minutes]

1. Which of the following statements accurately describes the difference between Gemini and Gemma?
    A) Gemini is a family of lightweight, open models suitable for local deployments, while Gemma is a multimodal model that can process and generate various data types, including text, images, audio, and video.
    B) Gemini is a model capable of generating video content from text or images, while Gemma is a model that specializes in code generation
    C) Both Gemini and Gemma are text-to-image diffusion models that excel at generating high-quality images from textual descriptions.
    D) Gemini is a multimodal model that can process and generate various data types, including text, images, audio, and video, while Gemma is a family of

lightweight, open models suitable for local deployments and specialized AI applications.

2. Which of the following statements explains the key difference between Imagen and Veo?
    A) Imagen is a family of lightweight, open models, while Veo is a text-to-image diffusion model.
    B) Imagen is a text-to-video model, while Veo is a text-to-image model.
    C) Imagen is a text-to-image diffusion model that generates images from text, while Veo is a model capable of generating video content from text or images.
    D) Both Imagen and Veo are multimodal models that can process various data types, including text, images, audio, and video.

3. What type of technique is this - adapting a model with task specific data
    A) Prompt Engineering
    B) Fune-tuning
    C) RAG
    D) All of the above

4. What type of technique is this – Crafting precise instructions for desired output
    A) Prompt Engineering
    B) Fune-tuning
    C) RAG
    D) All of the above

5. What type of technique is this – Augmenting the model with external knowledge
    A) Prompt Engineering
    B) Fune-tuning
    C) RAG
    D) All of the above

6. Which of the following *best* describes the purpose of grounding in generative AI?
    A) To encourage the AI to generate more creative and imaginative outputs.
    B) To reduce the complexity of AI models and make them easier to train.
    C) To improve the speed and efficiency of AI inference.
    D) To enhance the accuracy and reliability of AI-generated content by connecting it to verifiable sources.

7. What is the primary role of humans in the loop (HITL) in machine learning?
    A) To automate all decision-making processes.
    B) To integrate human expertise into the ML process, especially for tasks requiring judgment or context.
    C) To eliminate the need for data collection and model training.
    D) To replace AI algorithms entirely.

8. Why is fine-tuning and customization an important factor when choosing a model?
    A) It reduces the cost of using the model.

B) It allows the model to be adapted for specific tasks or domains.
C) It increases the model's context window.
D) It ensures the model is always available.

9. Which type of foundation model is specifically designed to understand and generate human language?
    A) Predictive model
    B) Large language model (LLM)
    C) Multimodal model
    D) Diffusion model

10. Which type of foundation model would be most suitable for generating photorealistic images from textual descriptions?
    A) Multimodal model
    B) Diffusion model
    C) Classification model
    D) Large language model (LLM)

11. Which techniques can be used to overcome the limitations of foundation model performance?
    A) Grounding, prompt engineering, fine-tuning, and humans in the loop (HITL).
    B) Increased processing power and faster hardware.
    C) Advanced algorithms and data structures.
    D) Preventing hallucinations by restricting the AI model's access to external knowledge sources.

12. What is the primary purpose of grounding in generative AI?
    A) To connect the AI's output to verifiable sources of information.
    B) To increase the speed and efficiency of AI model training.
    C) To improve the AI's ability to generate creative content.
    D) To reduce the cost of deploying AI applications.

13. What are some key legal considerations in AI development?
    A) Data Privacy
    B) Non-discrimination
    C) Intellectual property
    D) All of the above

14. Why is data quality crucial for responsible AI?
    A) To avoid biased or discriminatory outcomes.
    B) To improve performance on data retrieval
    C) To implement better security system
    D) All of the above

15. What is the purpose of explainable AI?
    A) To explain the purpose of the AI solution
    B) To make AI decision-making transparent and understandable.

C) To create summary of a document
D) None of the above

16. What is the primary goal of the Secure AI Framework (SAIF)?
    A) To promote innovation and accelerate AI development without considering security risks.
    B) To focus solely on preventing external attacks on AI systems.
    C) To establish security standards for building and deploying AI responsibly, addressing the unique challenges and threats in the AI landscape.
    D) To restrict AI development and deployment due to concerns about potential security risks.

17. What is a potential consequence of using inaccurate or incomplete data in AI training?
    A) It eliminates the need for fine-tuning and further model training.
    B) It results in an inefficient user interface.
    C) It introduces biased outcomes and unfair results.
    D) It results in data storage inefficiency.

18. You're developing an AI-powered loan application assessment system. Which steps would you take to ensure your system is developed responsibly? Select two options.
    A) Prioritize speed and efficiency over fairness and transparency, as these factors are more important for business success.
    B) Deploy the model without any human oversight, trusting its ability to make fair and unbiased decisions
    C) Use a complex "black box" model that makes highly accurate predictions, even if its decision-making process is difficult to understand.
    D) Regularly audit the model's performance to identify and mitigate any biases that may emerge over time.
    E) Train the model on a dataset that includes a diverse range of applicants, ensuring representation across different demographics and socioeconomic backgrounds.

19. What is the primary goal of ethical AI development?
    A) To comply with all legal and regulatory requirements.
    B) To ensure AI systems are used responsibly and do not cause harm
    C) To promote transparency and accountability in AI systems
    D) To maximize AI capabilities regardless of societal impact.

20. Which of the following is a key aspect of securing the "model training" phase?
    A) Optimizing the model for faster processing.
    B) Safeguarding training data and model parameters from unauthorized access.
    C) Minimizing the amount of training data used.
    D) Ensuring the model's output is visually appealing.

21. Which statement best defines Artificial Intelligence (AI)?

A) A technique that generates new content
B) A subset of machine learning
C) The broad field of building machines that perform tasks requiring human intelligence
D) A system that only works with structured data

22. How is Machine Learning (ML) best described?

A) A replacement for artificial intelligence
B) A subset of AI where machines learn from data
C) A rules-based expert system
D) A data storage mechanism

23. What distinguishes Generative AI from traditional AI systems?

A) It relies only on unlabeled data
B) It creates new content such as text, images, or audio
C) It cannot learn from examples
D) It only supports supervised learning

24. Why is data quality critical to machine learning performance?

A) It reduces infrastructure costs
B) It determines how quickly a model trains
C) It ensures models learn correct patterns and make accurate prediction
D) It eliminates the need for human validation

25. Which of the following is NOT a data quality dimension ?

A) Accuracy
B) Completeness
C) Visualization
D) Consistency

26. What risk arises when training data is not representative?

A) Increased computational cost
B) Slower model deployment
C) Biased or skewed predictions
D) Higher storage requirements

27. What does data accessibility ensure for AI systems?

A) Faster model inference
B) Availability, usability, and proper format of data
C) Elimination of bias
D) Reduced need for training data

28. Which example represents unstructured data?

A) Sales data in a CSV file
B) A medical record with fixed fields
C) An audio recording of a customer service call
D) A database table

29. Which of the following is an example of structured data?

A) Product review text
B) Digital image
C) Audio recording
D) Sales data with defined columns

30. What type of data includes tags or annotations?

A) Raw data
B) Unstructured data
C) Unlabeled data
D) Labeled data

31. Which learning approach relies on labeled data?

A) Reinforcement learning
B) Unsupervised learning
C) Supervised learning
D) Exploratory learning

32. Discovering hidden customer segments from purchase history is an example of:

A) Supervised learning
B) Reinforcement learning
C) Unsupervised learning
D) Transfer learning

33. What is the core idea behind reinforcement learning?

A) Learning from labeled examples
B) Learning from historical trends
C) Learning through interaction, rewards, and penalties
D) Learning through static datasets

34. Which Google Cloud service supports predictive maintenance using supervised learning?

A) BigQuery ML
B) Cloud Storage
C) Vertex AI
D) Pub/Sub

35. Deep learning is best described as:

A) A replacement for machine learning

B) A subset of machine learning using neural networks
C) A data preprocessing technique
D) A form of unsupervised learning only

36. What enables foundation models to perform a wide variety of tasks?

A) Rule-based logic
B) Small labeled datasets
C) Training on massive datasets using deep learning
D) Manual configuration

37. Which foundation model is designed specifically for language understanding and generation?

A) Imagen
B) Veo
C) Gemini
D) Diffusion Model

38. What is the primary strength of diffusion models?

A) Language translation
B) Numerical forecasting
C) Generating high-quality images, audio, and video
D) Data labelling

39. Which factor refers to the amount of information a model can process at once?

A) Modality
B) Context window
C) Fine-tuning
D) Grounding

40. What is model hallucination?

A) Increased inference speed
B) Generating outputs not based on real or verifiable information
C) Bias in training data
D) Knowledge cutoff

41. What is the purpose of grounding in generative AI?

A) Increasing creativity
B) Reducing training time
C) Connecting outputs to verifiable data sources
D) Improving hardware efficiency

42. What does RAG (Retrieval-Augmented Generation) primarily do?

A) Replace foundation models
B) Add retrieved external information to model prompts
C) Eliminate hallucinations completely

D) Encrypt AI outputs

43. When is fine-tuning most useful?

A) When no data is available
B) When prompt engineering fails to meet specific needs
C) When models must be smaller
D) When using multimodal data

44. Why are Humans in the Loop (HITL) important?

A) To fully replace AI systems
B) To automate all decisions
C) To provide oversight in sensitive or high-risk decisions
D) To increase data volume

45. What is the main goal of Secure AI?

A) Improve model accuracy
B) Prevent intentional harm and misuse of AI systems
C) Increase AI creativity
D) Reduce operational costs

46. What is the main goal of Secure AI?

A) Improve model accuracy
B) Prevent intentional harm and misuse of AI systems
C) Increase AI creativity
D) Reduce operational costs

47. Which security risk involves corrupting training data intentionally?

A) Prompt injection
B) Model theft
C) Data poisoning
D) Hallucination

48. . What is the purpose of Google's Secure AI Framework (SAIF)?

A) Improve model performance
B) Define UI standards
C) Establish security standards for AI systems
D) Replace IAM

49. What principle is foundational to Responsible AI?

A) Automation
B) Speed
C) Security
D) Model size

50. Why is explainable AI important?

A) It improves inference speed
B) It reduces training data
C) It makes AI decisions transparent and accountable
D) It removes bias automatically

Chapter – 3: Navigating Google Gen AI Landscape

3.1 Overview

Gen AI is changing how we work and interact with the world around us. But as a leader, how can you harness its power to drive real business outcomes?

In this course, you get answers to:

- What are the different layers of building gen AI solutions?
- How does Google Cloud offer you different options for building and using gen AI?
- When should you choose to use one solution over another?

Gen AI agents can process information, reason over complex concepts, and take actions. They can also be used in a variety of applications, including customer service, employee productivity, and creative tasks.

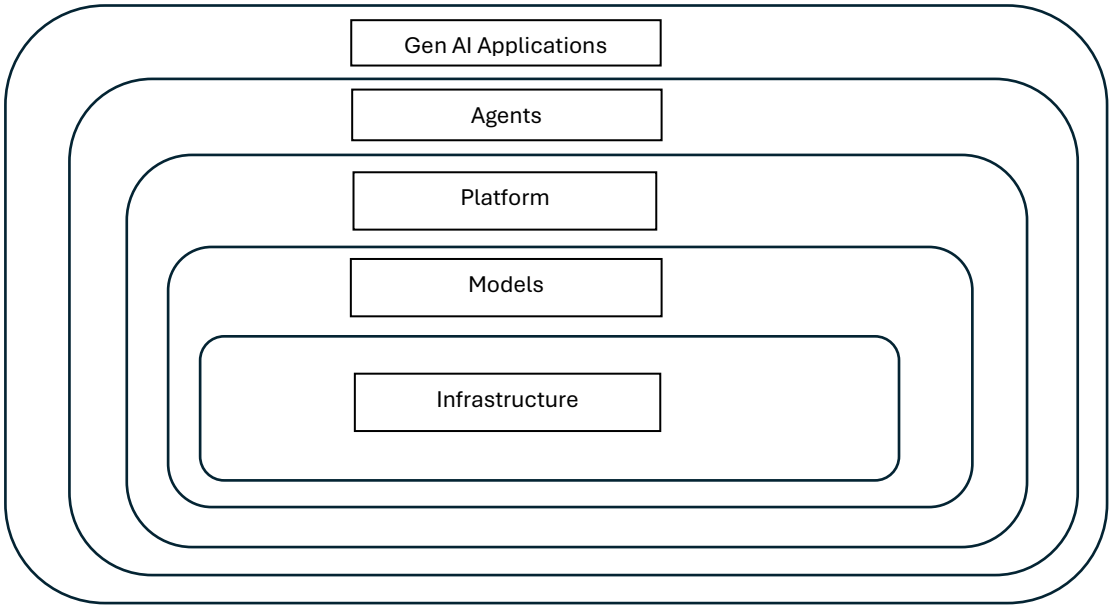
A gen **AI agent** is an application that tries to **achieve a goal by observing the world and acting upon it** using the tools it has at its disposal.

With generative AI, you have the flexibility to leverage either pre-built agents and applications, or create custom ones tailored to your specific needs. No matter which path you choose, you can apply generative AI's capabilities in a way that best aligns with your objectives.

Google Cloud tools like AI Applications help teams create AI agents using natural language rather than having to write complex code to do the same thing.

3.2 The building blocks of generative AI

Generative AI is a powerful technology composed of interconnected layers, each playing a vital role in its capabilities. Let's explore these layers—from the foundational infrastructure to the user-facing applications—to gain a deeper understanding of how generative AI works.



Infrastructure

The foundation of any AI system, comprising the hardware (physical servers, cloud computing resources, specialized chips like GPUs and TPUs) and software (operating systems, networking) that provide the necessary computing power, storage, and connectivity to train, deploy, and scale AI models.

Models

The model is where the brains of the AI system reside. It comprises various algorithms that learn patterns from data and can make predictions or generate new content. Examples are large language models (LLMs) like Gemini, image recognition models, and recommendation systems.

Platform

This is the layer that consists of tools and services that help with building and deploying AI models. This includes model training platforms like Vertex AI and data management tools.

Agents

This layer utilizes the capabilities of the model layer to perform more complex actions. This layer interacts with the environment, gathers information and makes decisions and executes actions based on the information received.

Gen AI powered application

This is the layer that delivers the AI capability to users through interfaces.

Key takeaway

Each layer plays a crucial role in enabling the capabilities of generative AI, from the underlying computing power to the user-facing tools. By grasping the interplay between these layers, you'll be better equipped to make informed decisions, foster innovation, and unlock the transformative potential of generative AI for your business.

3.3 Agents

3.3.1 About AI Agents

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Google Cloud tools like AI Applications help teams create AI agents using natural language rather than having to write complex code to do the same thing.

3.3.2 How do Agents relate to gen AI powered applications?

Think of agents as the intelligent pieces within a larger gen AI powered application. They bring specific capabilities to the table.

- **Understanding and responding to natural language**

  This allows applications to have more intuitive interfaces, like chatbots that can actually understand complex requests.

- **Automating Complex Task**
  Agents can handle multi-step processes within an application, such as gathering information, making decisions based on that information, and taking action.

- Personalization
  Agents can learn user preferences and tailor the application experience accordingly.

3.3.3 Role of Agents in gen AI powered Application

Think of gen AI powered applications as the user-facing layer. They provide the structure and context for agents to operate within. They define the user interface, the overall goals, and the specific tasks that agents will help with. Often, multiple agents with different specializations work together within a single application to create a richer, more dynamic user experience. This is known as a multi-agent system. You can review some examples below.

A Travel Booking App

An agent could handle the complex task of finding the best flights and hotels based on user preferences, while another agent might specialize in suggesting relevant activities and attractions at the destination. Then, the application provides the user with the interface for browsing options and making reservations.

A customer Support App

An agent could answer common questions, troubleshoot problems, and escalate complex issues to human representatives. The application would provide the chat interface and integrate with other support systems.

Personalized Learning App

An agent could assess a student's knowledge, recommend relevant learning materials, and even generate personalized exercises. The application would provide the structure for lessons and track progress.

Key takeaway

Agents enhance applications by adding intelligence and automation and the applications provide the framework and purpose for agents.

As you start to think about building custom agents for your organization you will begin to understand how they can really enhance your organization.

3.3.4 How agents work

While there are many ways to use gen AI agents, most will fall under two categories: conversational agents or workflow agents.

3.3.4.1 Conversational agents

Conversational agents are designed to understand what you mean, not just what you say, and respond in a way that makes sense.

- You provide input: You can type a message or speak to the agent.
- The agent understands: Using powerful AI, it figures out the meaning and intention behind your words.
- The agent calls a tool: Based on your request, the agent might need to gather additional information or perform an action. This could involve searching the web, accessing a database, or interacting with another software application.
- The agent generates a response: It formulates an answer that's relevant to your request and sounds natural.
- The agent delivers the response: You'll see or hear the answer depending on how you interacted with it.

Example

- Answering questions: Imagine you're curious about the weather in Paris. You ask a conversational agent, "What's the weather like in Paris?" And it instantly provides you with the current temperature, forecast, and even suggests what to pack if you're traveling there.
- Casual conversation: Feeling bored? A conversational agent can engage in light-hearted banter, tell you jokes, or even discuss your favorite books and movies, just like a friend would.
- Accessing information: Need to know the capital of Australia? Or maybe you want a quick summary of the French Revolution? A conversational agent can access a vast amount of world knowledge and provide you with concise and accurate information.

3.3.4.2 Workflow agents

How it works

Workflow agents are designed to streamline your work and make sure things get done efficiently and correctly by automating tasks or going through complex processes.

- You provide input: You define a task or trigger a process like submitting a form, uploading a file, initiating a scheduled event, or even ordering a product online.
- The agent understands: The agent is the software that automates those steps. It interprets the task's requirements and defines the series of steps needed to complete the task.
- The agent calls a tool: Based on the workflow's definition, the agent executes a series of actions. This could involve data transformation, file transfer, sending notifications, integrating with external systems, or initiating other automated processes using APIs.
- The agent generates a result/output: It compiles the outcome of the executed actions, which might be a report, a data file, a confirmation message, or an updated status within a system.
- The agent delivers the result/output: The agent delivers the output to the designated recipient(s) or system(s), such as via email, a dashboard, a database update, or a file storage location.

Examples

- Ecommerce order fulfilment: An agent automatically processes orders, updates inventory, sends shipping notifications, and handles returns.
- Customer onboarding: An agent guides new customers through account setup, provides tutorials, and answers frequently asked questions.
- Automated research: An agent can conduct in-depth research on a given topic by autonomously browsing the web, summarizing relevant content, and generating comprehensive reports. (Try this out with Gemini Deep Research.)
- Security Log Parsing: An agent that inspects incoming security logs for abnormalities and can flag them, open a ticket, begin triage, and assign to humans for review when necessary.

3.3.5 Agent use cases

Both conversational and workflow agents can cover a broad range of use cases:

Customer Service Agents

Customer service agents can answer questions, resolve issues, and provide personalized recommendations.

Employee productivity agents

Employee productivity agents can help workers find information, manage tasks, and automate workflows.

Creative Agents

Creative agents can generate new ideas, create content, and translate languages.

Code Agents

Code agents can assist developers in writing, reviewing, and debugging code, and even generating code from natural language descriptions.

Data Agents

Data agents can analyze large datasets, identify trends, and extract insights from data.

Security Agents

Security agents can automate tasks like summarizing case information, researching threats, generating security-related queries, and even taking actions like creating detection rules.

3.3.6 Advanced prompt engineering frameworks

The reasoning loop often utilizes advanced prompt engineering frameworks to guide its decision-making process. These frameworks can include:

- Simple rule-based calculations
- Complex thought chains
- Machine learning algorithms
- Probabilistic reasoning techniques

Examples of such frameworks include ReAct or chain-of-thought (CoT) prompting, which we discuss in more detail in Course 5 of this learning path.

Q & A: Answer to confirm between Model and Agent

- Organizing and formatting the content according to specific guidelines
  A) Agent ➔ Correct
  B) Model
- Evaluating a student's short-answer response to an open-ended question, considering factors like factual accuracy, completeness, and clarity of expression
  A) Agent
  B) Model ➔ Correct
- Providing customized learning materials and exercise
  A) Agent ➔ Correct
  B) Model
- Tracking the students' progress and adapting the learning plan
  A) Agent ➔ Correct
  B) Model

Key takeaway

By combining reasoning abilities with access to tools, gen AI agents can tackle complex problems and manage intricate workflows, far surpassing the capabilities of standalone models. Whether it's a conversational agent providing seamless customer service or a workflow agent streamlining operations, understanding their potential is crucial for leaders looking to drive innovation and efficiency within their organizations.

3.4 The platform layer

Platform layer plays a crucial role in building and scaling successful AI initiatives. The platform layer provides the foundation for building and scaling your AI initiatives. Vertex AI platform is the main platform from google for building, deploying and managing AI solution.

3.4.1 Vertex AI

Vertex AI is Google Cloud's unified machine learning (ML) platform designed to streamline the entire ML workflow. It provides the infrastructure, tools, and pre-trained models you need to build, deploy, and manage your ML and generative AI solutions.

Features of Vertex AI

- **Open and Flexible** - Vertex AI supports open frameworks, allowing you to use your preferred tools and models without vendor lock-in.
- **Comprehensive tooling** - Develop, deploy, and manage your models with ease using Vertex AI's integrated tools.
- **Powerful Infrastructure** - Leverage Google Cloud's scalable and reliable infrastructure to handle your ML workloads.
- **Customization** - Fine-tune and adapt models to your specific needs using the built-in IDE.
- **Pre-trained Models** - Choose from a wide variety of pre-trained models or build your own.
- **Easy-Integration** - Integrate your models into applications seamlessly using available APIs.

MLOps for efficient workflows

Operationalizing model training can be challenging and involves thinking about infrastructure and security. By using Vertex AI, you get a fully managed compute infrastructure, high-performance ML optimized training jobs, distributed training, hyperparameter optimization, and enterprise security.

Vertex AI's MLOps tools

Vertex AI also includes ML operations (MLOps) tools built in to help you orchestrate end-to-end ML workflows, perform feature engineering, run experiments, manage and iterate your models, track ML metadata, and monitor and evaluate model quality. MLOps helps automate, standardize, and manage the ML project lifecycle, enabling better collaboration and continuous improvement. Some of the Vertex AI tools are as follows:

- **Feature store -** Share, serve, and reuse ML features to maintain consistency and efficiency.
- **Model Registry –** Manage model versions, track changes, and organize your models throughout their lifecycle.
- **Model Evaluation** - Quickly evaluate and compare model performance to identify the best model for your use case.
- **Workflow Orchestration** - Automate ML workflows, from data preprocessing to deployment, using Vertex AI Pipelines.
- **Model Monitoring** - Monitor models for performance degradation, detect input skew and drift, and trigger updates or retraining.

Q1. Choose the Vertex AI tool that suite your need – You need to track experiments, manage different versions of your model, and keep a record of changes.

E) Model Monitoring
F) Workflow Orchestration
G) Model Registry

Q2. Choose the Vertex AI tool that suite your need – Your deployed models' performance is declining and you need to identify the cause and retrain it.

A) Model Monitoring
B) Workflow Orchestration
C) Model Registry

Q3. Choose the Vertex AI tool that suite your need –You want to automate your entire ML pipeline from data processing to model deployment.

A) Model Monitoring
B) Workflow Orchestration
C) Model Registry

3.5 The model layer

3.5.1 What is an AI model?

At the heart of every AI and machine learning system lies the **model.** Think of it as the brain of the operation. These aren't just any algorithms. They're sophisticated mathematical structures trained on massive amounts of data. This training process allows them to learn patterns and relationships, ultimately enabling them to perform a wide range of tasks, such as generating content, analyzing data, and classifying information.

Vertex AI: A model hub

Within the Vertex AI platform, you have multiple options for how to handle AI models for your project. You can choose to build a model from scratch or use an existing model.

Use models with Model Garden

For many use cases, your organization won't need to develop a new model. There are many AI models that already exist today that you can fine-tune and leverage for so many different use cases. Model Garden on Vertex AI is a service that lets you discover, customize, and deploy existing models from Google and Google partners.

Model Garden gives you options to pick from over 160 models and offers options of Google models, third-party models, and open-source models. With Vertex AI, you can customize these models with your own data and deploy them to your custom applications. Additionally, some pre-trained models can be used out-of-the-box without tuning.

✓ **First-Party Foundation Model**

Leverage state-of-the-art models from Google including Gemini models and task-specific models for image generation, speech-to-text, and more.

- ✓ Gemini Pro, Flash, and more. View Gemini specs ↗
- ✓ Imagen for text-to-image. View Imagen specs ↗
- ✓ Veo for text-to-video and image-to-video. View Veo specs ↗
- ✓ Chirp 2.0 for speech-to-text. View Chirp specs ↗

- **Open Models**

Access a variety of enterprise-ready open models.

- **View Gemma 2 specs ↗**
- **View CodeGemma specs ↗**
- **View PaliGemma specs ↗**
- **View Meta's Llama 3.1 specs ↗**
- **View Meta's Llama 3.2 specs ↗**

- View Mistral AI specs ↗
- View Mistral AI21specs ↗
- **TII's Falcon BERT, FLAN-T5, ViT, EfficientNet.** View TII's Falcon specs ↗

- **First-Party Pre-Trained APs**

Build and deploy AI applications faster with our pre-trained APIs powered by the best Google AI research and technology.

- ✓ View Speech-to-Text specs ↗
- ✓ View Natural Language Processing (NLP) specs ↗
- ✓ View Translation specs ↗
- ✓ View Vision specs ↗

- **Third-Party Models**
  - ✓ Model Garden supports third-party models from partners with foundation models.
  - ✓ View Anthropic's Claude Model Family specs ↗

3.5.2 Build Models with Vertex AI

With Vertex AI, you have two options for training and using your own models. You can go fully custom and create and train models at scale using any ML framework (PyTorch, TensorFlow, scikit-learn, or XGBoost), or there are also options for using AutoML to create and train models with minimal technical knowledge and effort.

AutoML allows you to build these types of models:

Data type	Supported objectives
Image Data	Classification, object detection.
Video Data	Action recognition, classification, object tracking.
Tabular data	Classification/regression, forecasting.

Follow this standard workflow when creating your models in Vertex AI.

- **Gather your data**
  Determine the data you need for training and testing your model based on the outcome you want to achieve.

- **Prepare your data**

Make sure your data is properly formatted and labeled.

- **Train**
  Set parameters and build your model.

- **Manage your model**
  Review model metrics to evaluate the model's performance and monitor changes over time.

- **Deploy and predict**
  Make your model available to use.

Question 1 → which is the AI model that best suits the needs.

- You need a model to translate languages for your global e-commerce platform. You want a readily available, high-performing solution and have a moderate budget.
  A) Vertex AI - build your own custom model
  B) Vertex AI Model Garden

Question 2 → which is the AI model that best suits the needs.

You're an experienced AI researcher developing a cutting-edge model for protein folding prediction. You need complete control over the model architecture and training process.

A) Vertex AI - build your own custom model
B) Vertex AI Model Garden

Key takeaway on Models
The model lies at the core of any AI system, acting as its "brain" and enabling a wide range of functions from content generation to data analysis. Vertex AI provides a comprehensive platform for leveraging the power of models, offering access to a vast repository of pre-built models in Model Garden and the ability to create custom models. By understanding the different types of models and the model development workflow, leaders can effectively harness AI to drive innovation and solve complex business challenges.

3.6 The infrastructure layer

3.6.1 The foundation of gen AI solutions

You've seen the incredible potential of generative AI. But have you considered the engine that drives this transformative technology? Let's dive into the crucial role of infrastructure in powering generative AI solutions.
The "infrastructure layer" is the foundation upon which any AI system is built. It's the combination of hardware and software that provides the necessary computing power, storage, and networking capabilities to train, deploy, and scale AI models. This layer is crucial for handling the massive datasets and complex computations involved in modern AI, especially with the rise of large language models and generative AI.

Google has invested heavily in building a robust and cutting-edge AI infrastructure, designed to handle the most demanding AI workloads. Here are some key components.

3.6.2 High performance computing

Imagine your company wants to undertake a new, bold gen AI project. However, the engineering team hits a major roadblock. Training the complex model requires significant processing power, far beyond the capabilities of your company's in-house servers. It takes hours, if not days, to train your specialized models and the long training times have started to hinder the team's progress to the point where it's delaying the product launch. Your existing infrastructure simply can't keep up, and the cost of scaling their own hardware is prohibitive. This is where high-performance computing, the heart of Google's AI infrastructure, comes in.

- **GPUs and TPUs**

  These specialized processors are the workhorses of AI. They excel at parallel processing, or doing multiple things at once, which is crucial for training large neural networks quickly and efficiently. GPUs were originally designed for graphics rendering, but since they offer general-purpose parallel processing power they've increasingly been used in the AI domain. TPUs are Google's custom-designed chips specifically optimized for AI tasks.

- **Hypercomputers**

  A hypercomputer is essentially a supercomputer built by connecting many individual computers (nodes) together. These nodes contain GPUs and TPUs. By linking them with high-speed networks, they work together as one massive computing unit. They provide the massive scale necessary for training and running the most demanding generative AI models.

3.6.3 High performance storage

Having high-performance storage is essential when building gen AI apps. Gen AI models are data hungry because they learn by analyzing massive datasets, often petabytes in size, containing text, images, code, and other forms of data. This storage provides the capacity and speed to store and access these enormous datasets efficiently. Without it, training would be incredibly slow or even impossible.

- **Large-scale storage system**

  Google Cloud's storage infrastructure is optimized for AI workloads, offering high throughput (the amount of work done within a specific period of time), scalability, and the ability to create dense compute clusters for faster training and inference. As AI models and datasets grow, the system can scale seamlessly to accommodate the increasing demands. Losing data due to storage failures can be catastrophic, so the storage is highly reliable to ensure data durability.

- **Fast storage access**

The storage systems also need to provide fast read and write speeds to keep up with the demands of the training process. Otherwise, if storage access is slow, it can significantly hinder the training process.

3.6.4 Networking

Fast and efficient communication is essential for coordinating the work all of the processors in a high-performance computing cluster are undertaking. Google's global fiber network provides high-bandwidth, low-latency connectivity, ensuring smooth data flow and efficient communication between different parts of the AI infrastructure.

Using Google Cloud infrastructure can benefit teams in several ways, particularly with respect to scalability, performance, and cost-effectiveness.

Key takeaway
Google Cloud's robust infrastructure, with its high-performance computing, storage, and networking capabilities, provides the foundation for handling the massive datasets and complex computations required by today's advanced AI models. As leaders, understanding the critical role of infrastructure in AI initiatives is crucial for making informed decisions about resource allocation, scalability, and ultimately, the successful implementation of generative AI solutions within your organizations.

3.7 Edge computing

While hosting infrastructure on the cloud is powerful, it's not always the ideal solution. Imagine a self-driving car needing to make split-second decisions. It can't wait for data to travel to the cloud and back. That's where edge computing comes in.

Google provides the tools to deploy your AI models in these different locations, giving you more control and flexibility.

Edge computing runs AI on devices or servers **closer to the data source** or point of need.

3.7.1 Why go local or edge?

Imagine a drone navigating a complex environment. It needs to react instantly to obstacles, making cloud processing too slow. Running AI locally on the drone ensures real-time responsiveness. Other benefits include increased data privacy and reduced reliance on internet connectivity.

To run powerful AI models on edge devices and mobile phones, Google provides tools like **Lite Runtime (LiteRT)**. Think of LiteRT as a platform that helps machine learning models work efficiently on your device. To learn more about Google's high-performance runtime for on-device AI, check out the article, Lite Runtime (LiteRT) Overview.

One example of an AI model designed for edge is **Gemini Nano**.

3.7.2 Gemini Nano

Gemini Nano is Google's most efficient and compact AI model, specifically designed to run on the edge on devices like smartphones and embedded systems. It's part of the larger

Gemini family of models. Think of Gemini Nano as a miniature version of the powerful AI that usually lives in Google's data centers. This "on-device" or edge approach offers several benefits.

- Privacy - Your data stays on your device, enhancing your privacy.
- Speed - You get fast responses since there's no need to send data to the cloud.
- Offline Access - Gemini Nano can work even without an internet connection.

Gemini Nano brings the power of AI directly to your devices, making them smarter, more helpful, and more privacy-focused. It is currently being integrated into various Google products and services, including:

- **Pixel phones -** Gemini Nano powers features like Call Notes, which summarizes phone conversations and Pixel Recorder, which summarizes voice recordings.
- **Android** - Gemini Nano is available to Android developers through the **AI Edge SDK**, enabling them to build innovative AI experiences into their apps.

Even when your goal is to deploy on the edge, Vertex AI provides a powerful platform for building, training, and refining your models. You can leverage Vertex AI's capabilities for tasks like data preparation, model training, tuning, model evaluation, collaboration, and MLOps.

Once your model is ready for the edge, Vertex AI offers tools to streamline the deployment process.

You can:

- **Convert models:** Convert your models to Lite Runtime (LiteRT) for optimal performance on edge devices.
- **Package and deploy:** Package your models and dependencies into containers for deployment on various edge hardware.
- **Manage and monitor:** Manage your edge deployments, track their performance, and gather insights to improve your models over time.

Google provides the tools to deploy your AI models in these different locations, giving you more control and flexibility.

Key takeaway on Edge Computing

Edge computing allows AI to run locally on devices or servers closer to the data source, ensuring real-time responsiveness and increasing data control. Google provides tools like Lite Runtime (LiteRT) and Gemini Nano that are specifically designed for edge computing. Even when deploying on the edge, Vertex AI provides a powerful platform for building, training, and refining AI models.

3.8 Gen AI project resources: People, cost, and time

3.8.1 Roles and responsibilities

It is important to understand the resources you have available to build out your solution. Let's dig deeper into some of the roles and responsibilities at each layer of the stack.

The AI stack is designed to support different roles with specific needs. Before jumping into building out a solution, ensure you have the proper expertise and talent for what you want to build. Business users can improve their tasks, developers can build custom AI solutions, and AI practitioners can develop and deploy advanced AI models responsibly and securely. This collaborative ecosystem allows your organization to leverage the full potential of AI across various business functions.

Business leaders
Business leaders typically interact with pre-built gen AI solutions to enhance daily operations and improve customer experiences. For example, Google Workspace with Gemini can be used for content creation, data analysis, and document summarization.

Developers
Developers in your organization are responsible for building and deploying custom AI agents and integrating AI capabilities into existing applications. They can use AI Applications for custom agent creation, AI code generation, and AI-driven data processing. Developers can also leverage pre-trained APIs to rapidly integrate AI into applications, or they can use the Vertex AI platform which is designed to help developers build and deploy AI agents with tools for orchestration, grounding, and action.

AI practitioners
AI practitioners play a valuable role in customizing, deploying, and optimizing generative AI models. They leverage tools within Vertex AI to accelerate development and ensure responsible AI practices. Their expertise extends to scaling AI workloads, integrating models with BigQuery, and implementing responsible AI measures such as bias detection and adversarial testing.

3.8.2 Cost

The pricing model of gen AI can vary based on which part of the gen AI landscape you are looking at or even which specific product or company you are using. It is important to have a realistic budget and understanding of the cost for your project.

When building gen AI solutions, you pay for three primary activities:

- **Training** the model.
- **Deploying** the model to an endpoint.
- **Using** the model to make predictions.

Training and deploying models often involve paying for the compute time used as well as the storage for the training data and model outputs.

Pricing for using models

- **Usage-based** → You pay for the amount you use, often measured in tokens or characters processed. This is common for APIs like Google's PaLM & Gemini APIs.
- **Subscription-based** → You pay a recurring fee for access to the model, often with tiers based on usage limits or features.
- **Licensing fees** → One-time or recurring fees for using a model, especially for commercial purposes or embedding in products.

- **Free tiers → Some providers offer free access with limited usage for experimentation or non-commercial purposes.**

Pricing metrics for using models

Here are some metrics you may see for usage-based pricing models.

- **Tokens →** A token represents a piece of text, like a word or part of a word.
- **Characters →** Some providers, like Google, charge based on the number of characters processed
- **Requests →** Sometimes you can be charged per request, regardless of the complexity or volume of the task.
- **Compute Time →** The time taken to process your requests can also factor into the cost, especially for resource-intensive tasks.

Factors affecting cost

Here are some factors affecting cost when using models.

- **Model size and complexity →** Larger, more capable models generally cost more.
- **Context Window →** A larger context window (the amount of text the model can consider) can increase costs.
- Features → Specialized features like fine-tuning or embedding can have separate pricing.
- **Deployment →** Depending on where you deploy your model and application, you may have compute based costs.

3.8.3 Time

It probably won't surprise you to learn that the more custom your solution is, the more time and resources it takes to build. Using a prebuilt gen AI application takes seconds. Building your own custom solution with a custom AI model, can take months. Choosing to build a custom agent with AI Applications takes much less time.

Think about your project timelines and evaluate them against your needs and requirements. Determine if the timeline is realistic based on your requirements.

Key takeaway

Successful generative AI initiatives require leaders to strategically allocate resources, including the right mix of business users, developers, and AI practitioners.

Understanding the costs associated with different generative AI solutions, from pre-built applications to custom models, is crucial for effective budgeting. Aligning project timelines with the complexity of the chosen solution ensures realistic expectations and successful outcomes.

3.9 Gen AI solution needs

Before diving headfirst into your generative AI project, it's crucial to take a step back and assess your needs. Aligning your ambitions with your capabilities is key to setting realistic

expectations and ensuring a successful outcome. Let's explore the key factors to consider when evaluating the scope and feasibility of your generative AI initiatives.

3.9.1 Scale

Is this for individual use, a small team, a large company, or millions of customers?

- **Small scale** → As an individual user or small team, you can go very far with pre-built tools. Try to leverage existing gen AI powered applications.

- **Large Scale** → When building out solutions for millions of customers, you'll likely need a more robust and customized solution. Pick solutions that offer scalability and security. Factor in details to your decision like infrastructure costs, data storage, and potential latency challenges.

3.9.2 Customization

How specialized does your solution need to be? Powerful foundation models are constantly improving, making custom model development less necessary in many cases.

- Start with existing models → Explore pre-trained models available through APIs or open-source libraries. Fine-tune these models on your specific data to achieve better performance for your tasks.
- Identify your unique needs → What sets your project apart? Does it require specialized knowledge, handle complex tasks, or demand a unique user experience? This will help you determine the level of customization needed.
- **Consider data specificity** → If your project deals with specialized domains like law or medicine, consider fine-tuning with domain-specific datasets or exploring models specifically trained for those areas.
- **Consider task complexity** → Think about the complexity of the task itself. Are you aiming for simple tasks like text summarization or more intricate ones like code generation or creative writing? This influences model choice and training approaches.

3.9.3 User interaction

Think about how users will interact with your AI and design an intuitive and engaging experience.

User interface (UI)
Seamlessly integrate the AI into your existing workflows. This might involve a dedicated interface (e.g., a chatbot widget on your website) or embedding AI capabilities within your current applications.

User experience (UX)
Aim for a user-friendly experience. Is it conversational, informative, or task-oriented? Consider the level of guidance and feedback users might need.

3.9.4 Privacy

How sensitive is the data you are working with?

Data security
What measures will you implement to protect data during processing and storage?
Implement robust security measures to protect data during processing and storage. Consider encryption, access controls, and secure data centers.

Compliance
Are there specific regulations (GDPR, HIPAA, etc.) that your project needs to adhere to?

3.9.5 Other considerations

How specialized does your solution need to be? Powerful foundation models are constantly improving, making custom model development less necessary in many cases.

- **Latency** - What are the acceptable response time limits?

Tip: Consider real-time requirements. Do you need instantaneous responses, or can you tolerate some delay? If real-time interaction isn't critical, you have more flexibility in choosing models and infrastructure.

- Connectivity → Will the solution always have internet access?

Tip: Consider offline functionality. Are there scenarios where the solution needs to function without internet access?

- Accuracy → How crucial is the accuracy of the AI's output? What are the acceptable error tolerances?

Tip: Define acceptable tolerances. Determine the level of accuracy required for your AI's output. This will influence model selection, training data, and evaluation metrics.

- Explainability → Do you need to understand the reasoning behind the AI's decisions?

Tip: Transparency is key. In certain domains, understanding the AI's reasoning is crucial. Consider using models or techniques that offer explainability features, especially in healthcare or finance.

3.10 Making decisions

You understand your needs and resources. Now it's time to make decisions such as what model you will use. By carefully considering the factors covered below and conducting thorough research, you can make informed decisions about gen AI pricing and choose the most cost-effective solution for your needs.

Comparing different companies and models

- **Evaluate model capabilities**: Look for benchmarks and comparisons to assess the quality and performance of different models.

- **Compare pricing structures:** Consider your expected usage and calculate the costs based on the pricing models and metrics of each provider.
- **Factor in additional costs:** Account for potential costs like data storage, API calls, and fine-tuning.
- **Read the fine print:** Pay attention to usage limits, data privacy policies, and other terms of service.

Key resources for comparison

- **Provider websites:** Check the pricing pages of different companies.
- **Research papers and benchmarks:** Look for independent evaluations and comparisons of different models.
- **Community forums and discussions:** Engage with other users and experts to learn about their experiences and pricing insights.

3.11 Maintenance

Building a generative AI solution is a journey, not a destination. To maximize your long-term success, it's essential to consider the ongoing maintenance and evolution of your project from the beginning.

- How will your project be maintained?
- Do you have the resources in place to maintain the project over time?
- What specific maintenance needs will your project have?

Key maintenance considerations

By proactively planning for maintenance, you'll ensure the long-term success and sustainability of your generative AI solution. Google Cloud provides a fully managed environment, which allows developers to focus on agent development rather than infrastructure and maintenance. With the platform handling deployment and model improvement, developers can maximize their return on investment and avoid costly issues down the road.

- **Model Monitoring and Retraining**
  Establish a system to continuously monitor your model's performance and retrain it periodically with updated data.

- Data Updates
  Plan for regular data updates to keep your model fresh and relevant. This might involve adding new data sources, cleaning existing data, or adapting to evolving data formats.

- Software Updates and Bug Fixes
  Stay informed about updates to your chosen AI platform, libraries, or frameworks, and implement them promptly to ensure optimal performance and security.

- Hardware and Infrastructure
  Consider the maintenance needs of your hardware and infrastructure. This includes server maintenance, security updates, and capacity planning to accommodate growing data and user demands.

- Security and Compliance

Maintain a vigilant approach to security. Regularly review and update your security measures to protect against evolving threats. Ensure ongoing compliance with data privacy regulations as they change.

3.12 Question Paper – 3

Q. At which of the following stages in the gen AI development process do you need to consider infrastructure requirements if you aren't building on a platform that handles it for you?

A) **Model refinement:** Improving your AI models based on feedback and new data.
B) **Data collection:** Gathering and preparing the data used to train your AI models.
C) **Model deployment:** Making your AI models available for use in applications or services.
D) **Model monitoring:** Tracking the performance of your deployed AI models and identifying potential issues.
E) **Model training:** Training your AI models on the prepared data.

Ans: A, B, C, D, E

3.13 Question Paper – 3

1. At which of the following stages in the gen AI development process do you need to consider infrastructure requirements if you aren't building on a platform that handles it for you?
    A) **Model refinement:** Improving your AI models based on feedback and new data.
    B) **Data collection:** Gathering and preparing the data used to train your AI models.
    C) **Model monitoring:** Tracking the performance of your deployed AI models and
    D) **Model training:** Training your AI models on the prepared data.

2. What are the two key elements that distinguish AI agents from standalone AI models?
    A) Reasoning loop and tools.
    B) Natural language processing and machine learning.
    C) Data access and user interface.
    D) Automation and personalization.

3. A news organization wants to develop an AI agent that delivers personalized news to each user. The agent should be able to:

    - Learn the user's interests and reading habits.
    - Filter and prioritize news articles based on relevance.

• Summarize key information from multiple sources.
• Recommend related articles and diverse perspectives.
• Adapt to the user's feedback and evolving preferences.

How would the "agents" layer contribute to the functionality of this personalized news reader application?

A) The agents layer would define the specific tasks the AI performs, such as filtering articles, summarizing information, and making recommendations.
B) The agents layer would design the user interface of the news reader application.
C) The agents layer would provide the infrastructure for storing and processing the news data.
D) The agents layer would provide the underlying AI models for natural language processing and understanding.

4. A game developer wants to create more realistic and engaging non-player characters (NPCs) in their game. They envision NPCs that can:
   - Engage in dynamic conversations with the player.
   - React to the player's actions and choices.
   - Adapt their behaviour based on the game's environment and storyline.
   - Exhibit unique personalities and backstories.

   Which layer of the gen AI landscape would be MOST crucial in defining the behaviors and capabilities of these AI powered NPCs?

   A) Models
   B) Infrastructure
   C) Agents
   D) Platforms

5. A data science team is training a new generative AI model on a massive dataset of images. They need access to powerful hardware and software resources to handle the computationally intensive training process.

   Which layer of the gen AI landscape would provide the necessary computational power and storage for this data science team?

   A) Infrastructure
   B) Platforms
   C) Agents
   D) Applications

6. A travel agency wants to use an AI agent to help customers plan their vacations. The agent should be able to:
   - Gather customer preferences (budget, destination interests, travel dates).
   - Search for flights, accommodations, and activities.
   - Create personalized itineraries with options and recommendations.
   - Book flights and hotels based on customer choices.
   - Provide ongoing support throughout the trip.

How would the "agents" layer and the "applications" layer work together to create this AI powered travel planning experience?

A) The agents layer would define the AI's capabilities (searching, booking, recommending), while the applications layer would provide the user-facing tool (website or app) to interact with the agent.
B) The agents layer would provide the user interface for the travel planning application.
C) The applications layer would provide the computational resources for the AI agent to operate.
D) The applications layer would determine the specific tasks the AI agent can perform, such as searching for flights or booking hotels.

7. Which component can take care of the following task: You need to track experiments, manage different version of the model, and keep a record of changes.
    A) Model Monitoring
    B) Workflow Orchestration
    C) Model Registry
    D) None of the above

8. Which component can take care of the following task: Your deployed model's performance is declining and you need to identify the cause and retrain it.
    A) Model Monitoring
    B) Workflow Orchestration
    C) Model Registry
    D) None of the above

9. Which component can take care of the following task: You want to automate your entire ML pipeline from data processing to model deployment
    A) Model Monitoring
    B) Workflow Orchestration
    C) Model Registry
    D) None of the above

10. Which Tool can be used to train the Gemini Foundation Model
    A) Vertex AI
    B) NotebookLM
    C) Gemini Foundation Model
    D) Cloud Run Function

11. You need a model to translate languages for your global e-commerce platform. You want a readily available, high-performing solution and have a moderate budget. Where can you find such a model
    A) In Vertex AI
    B) In Vertex AI Model Garden
    C) In Gemini
    D) NotebookLM

12. What is the role of MLOps in Vertex AI?
    A) To learn how to best achieve a goal based on inputs and tools.
    B) To provide the physical hardware for AI systems.
    C) To allow users to interact with AI capabilities.
    D) To automate, standardize, and manage the ML project lifecycle.

13. Why is high-performance storage important for generative AI?
    A) To provide a user-friendly interface for interacting with AI models.
    B) To store and efficiently access the massive datasets used in AI training.
    C) To automate the deployment and management of AI models.
    D) To ensure the responsible use of AI systems.

14. What is the purpose of AutoML in Vertex AI?
    A) To manage and monitor the performance of deployed AI models.
    B) To automate the entire ML workflow, from data preparation to deployment.
    C) To provide pre-trained models for various use cases.
    D) To allow users to build and train AI models with minimal technical expertise.

15. What are GPUs and TPUs in the context of AI infrastructure?
    A) Storage devices for storing large AI datasets.
    B) Software applications for managing AI models.
    C) Networking protocols for connecting AI systems.
    D) Specialized processors designed for parallel processing in AI tasks.

16. Which of the following statements best describes Model Garden on Vertex AI?
    A) A platform for managing and monitoring AI infrastructure.
    B) A service designed for finding and deploying pre-trained AI models.
    C) A service for discovering, customizing, and deploying existing AI models.
    D) Platform for developing AI models from scratch and experimenting further.

17. You need a model that can generate realistic images from text descriptions for your new marketing campaign. You have some coding skills but are short on time. Which path would you choose?
    A) Use a pre-trained API to generate images quickly.
    B) Build a custom image generation model from scratch.
    C) Fine-tune a pre-trained text-to-image model from Model Garden.
    D) Use ChatGPT Foundation model

18. What is the primary advantage of edge computing for AI applications?
    A) Lower development costs
    B) Reduced need for data preprocessing
    C) Real-time responsiveness and reduced latency
    D) Increased data storage capacity

19. What is Gemini Nano designed for?
    A) Deploying AI on edge devices and smartphones
    B) Managing and monitoring AI infrastructure

C) Running complex simulations on supercomputers
D) Training large language models in data centres

20. Your marketing team needs to quickly generate engaging product descriptions for an upcoming ecommerce sale. Which approach is the most efficient way to achieve this?
    A) Hire a team of copywriters to write each description manually.
    B) Fine-tune a pre-trained language model from Model Garden on existing product data.
    C) Use a gen AI powered application like Google Workspace with Gemini to draft the descriptions.
    D) Build a custom model from scratch to generate product descriptions.

21. Who is responsible for building and deploying custom AI agents and integrating AI capabilities into applications?
    A) AI practitioners
    B) Business leaders
    C) Developers
    D) Network engineers
22. **What best defines a generative AI agent?**
    A) A large language model trained on massive datasets
    B) An application that achieves a goal by observing the world and acting using available tools
    C) A chatbot limited to answering user questions
    D) A cloud infrastructure component

23. **Which of the following is NOT a common application area for gen AI agents?**
    A) Customer service
    B) Employee productivity
    C) Creative tasks
    D) Physical hardware manufacturing

24. **How do agents relate to gen AI powered applications?**
    A) Agents replace the user interface entirely
    B) Agents act as the infrastructure layer
    C) Agents are intelligent components within a larger application
    D) Agents are standalone models without applications

25. What is a key benefit of using Google Cloud AI Applications for building agents?
    A) They eliminate the need for models
    B) They require extensive low-level coding
    C) They allow building agents using natural language
    D) They only support pre-trained agents

26. Which capability allows agents to support intuitive chatbot interfaces?
    A) Automating complex workflows
    B) Understanding and responding to natural language
    C) High-performance storage

D) Edge deployment

27. **What is meant by a multi-agent system?**
    - **A)** Multiple models running on one GPU
    - **B)** Agents operating independently without coordination
    - **C)** Multiple specialized agents working together in one application
    - **D)** A system deployed only on edge devices

28. **In a travel booking application, which task is most suitable for an AI agent?**
    - **A)** Designing the user interface
    - **B)** Managing cloud billing
    - **C)** Finding flights and hotels based on user preferences
    - **D)** Hosting the application server

29. **In gen AI powered applications, what role does the application primarily serve?**
    - **A)** Training the AI models
    - **B)** Providing structure, context, and user interface
    - **C)** Performing mathematical optimization
    - **D)** Replacing agents entirely

30. **Which characteristic best describes conversational agents?**
    - **A)** They only process structured data
    - **B)** They understand user intent, not just literal text
    - **C)** They cannot call external tools
    - **D)** They only operate on scheduled triggers

31. **What is the correct sequence of steps for a conversational agent?**
    - **A)** Generate response → Call tool → Understand → Deliver output
    - **B)** Understand → Generate response → Call tool → Deliver output
    - **C)** Provide input → Understand → Call tool → Generate response → Deliver response
    - **D)** Call tool → Understand → Provide input → Deliver response

32. **Which scenario best fits a workflow agent?**
    - **A)** Chatting with a customer about a refund
    - **B)** Generating creative writing
    - **C)** Automating a multi-step business process like file transfer and notification
    - **D)** Translating languages in real time

33. **What is typically the output of a workflow agent?**
    - **A)** A conversation reply only
    - **B)** A trained model
    - **C)** A report, file, confirmation, or system update
    - **D)** A user interface

34. **Which type of agent assists developers with writing and debugging code?**
    - **A)** Creative agent
    - **B)** Data agent
    - **C)** Code agent
    - **D)** Security agent

35. **Which agent type is responsible for analyzing large datasets and identifying trends?**
    - **A)** Customer service agent
    - **B)** Data agent
    - **C)** Workflow agent
    - **D)** Conversational agent

36. **Security agents can perform which of the following tasks?**
    - **A)** Image generation
    - **B)** Detecting trends in marketing data
    - **C)** Researching threats and creating detection rules
    - **D)** Designing application UIs

37. **What is the primary role of an AI model?**
    - A) Providing networking capabilities
    - B) Acting as the "brain" of the AI system
    - C) Managing cloud storage
    - D) Deploying applications

38. **What is Model Garden in Vertex AI?**
    - A) A dataset repository
    - B) A service for discovering, customizing, and deploying existing model
    - C) A networking service
    - D) A workflow automation tool

39. **Which of the following is a first-party Google foundation model?**
    - A) LLaMA
    - B) Gemini
    - C) BERT (open-source only)
    - D) Stable Diffusion

40. **What infrastructure component is most critical for training large generative AI models?**
    - A) User interface
    - B) High-performance computing using GPUs and TPUs
    - C) Edge devices
    - D) Chat applications

41. **What is the main purpose of high-performance storage in AI systems?**
    A) Improving UI responsiveness
    B) Storing small configuration files
    C) Efficiently storing and accessing massive datasets
    D) Reducing model accuracy

42. **What is edge computing?**
    A) Running AI exclusively in the cloud
    B) Running AI closer to the data source or point of need
    C) Training models without data
    D) Centralizing all AI workloads

43. **Why is edge computing important for self-driving cars?**
    A) Lower hardware costs
    B) Better cloud integration
    C) Real-time decision making without cloud latency
    D) Easier model training

44. **Which Google tool is designed to run AI models efficiently on devices?**
    A) Vertex Pipelines
    B) Lite Runtime (LiteRT)
    C) BigQuery
    D) Cloud Storage

45. **What is Gemini Nano optimized for?**
    A) Large-scale cloud training
    B) Data warehousing
    C) On-device and edge AI deployment
    D) Infrastructure monitoring

46. **Which of the following is NOT a primary cost activity in gen AI projects?**
    A) Training the model
    B) Deploying the model
    C) Using the model for predictions
    D) Designing marketing materials

47. **What pricing model charges based on tokens or characters processed?**
    A) Subscription-based
    B) Licensing fees
    C) Usage-based
    D) Flat-rate pricing

48. **Which factor generally increases the cost of using AI models?**
    A) Smaller context window

B) Smaller model size
C) Larger and more complex models
D) Fewer requests

49. Which approach typically requires the least time to implement?

A) Training a custom foundation model
B) Building a custom agent from scratch
C) Using a pre-built gen AI application
D) Developing new infrastructure

50. Why is ongoing maintenance critical for gen AI solutions?

A) Models never change
B) Data and requirements remain static
C) Performance, security, and compliance evolve over time
D) It eliminates infrastructure costs

Chapter – 4: Transform your Work

4.1 Overview

In this course, you get answers to:

- How can gen AI be integrated into your daily workflow?
- What are some techniques to improve prompting?
- What are some of Google's gen AI products that don't require any development expertise?
- How can you develop more automated workflows for your team?

This course introduces Google's gen AI applications, such as Google Workspace with Gemini and NotebookLM. It guides you through concepts like grounding, retrieval augmented generation, constructing effective prompts and building automated workflows.

4.2 Google Workspace with Gemini

Google Workspace with Gemini can serve as a helpful writing aid and powerful search and discovery tool.

Google Workspace is a collection of cloud-based productivity and collaboration tools that help people create, communicate, and collaborate, such as Gmail, Google Docs, Google Sheets, Google Meet, and Google Slides.

You learned about the Gemini app, but remember, Gemini is a gen AI model that is then used in many other applications. Some of those applications were built in Google Workspace. You can access Gemini directly within your Workspace apps through Gemini for Workspace.

Google Workspace with Gemini allows you to use Google's generative AI capabilities. Depending on your version of Google Workspace, you may have access to a different range of features and security protections. For more information, see the Help Center article, Google Workspace with Gemini.

Google Workspace business and enterprise licenses come with extra enterprise-grade data protection for your organization. These protections extend to the Google Workspace with Gemini features and the Gemini app (with Gemini Advanced). The licenses also come with administrative features for managing your organization's usage and access.

Let us now discuss on different Google Workspace tools: -

4.2.1 Gemini side panel

One way to access Google Workspace with Gemini is through the Gemini side panel, which is present on the side of many Workspace apps. With the side panel, Gemini can assist you with summarizing, analyzing, and generating content by utilizing insights gathered from your emails, documents, and more—all without switching applications or tabs.

4.2.2 Gemini in Google Sheets

Gemini in Google Sheets helps you generate formulas using AI.

You can provide Gemini with prompts using plain language or generic cell names to create complex formulas. For example, you can ask Gemini to "create a formula to find cell C1 in range D:G and output the value in column G." Gemini then creates the formula.

For example, new employees can use the Gemini in Drive side panel to get up to speed on projects and company initiatives. They can ask Gemini questions like "summarize the goals and current status of Project Phoenix" and easily find the information they need, enabling them to quickly become familiar with the project.

4.2.3 Gemini in Gmail

Gemini in Gmail uses generative AI to help users write and refine emails more efficiently.

An HR representative can use Gemini to write a company-wide email announcing a new employee benefit. They can provide a prompt that includes information about the new benefit, who is eligible, and the effective date, and Gemini will generate a draft email.

4.2.4 Gemini in Google Docs

Gemini in Google Docs helps you streamline the process of content creation, refining, and proofreading right within your document as you write.

A marketing team member can utilize Gemini to generate captivating taglines and draft social media posts for a new athletic shoe campaign. The user can provide a prompt like "Generate three social media posts that highlight the benefits of our athletic shoes, focusing on comfort and performance." Gemini then generates text options for the posts.

4.2.5 Gemini in Google Slides

Gemini in Google Slides can help you and your team work more effectively by using AI to generate photorealistic images that enhance your presentations.

By providing Gemini with a descriptive prompt that outlines the subject, setting, or style you desire, you can quickly create images that otherwise would require time and effort to source or design. This streamlines the presentation creation process and frees up your team to focus on content and delivery.

4.2.6 Gemini in Google Meet

Gemini in Google Meet can help your global team communicate by providing real-time transcription and translation, facilitating seamless communication in multilingual meetings. It could also automate meeting summaries and action item extraction, saving time and increasing productivity for participants.

These Google Workspace with Gemini features allow you to use **AI-enhanced workflows** in your daily work to become **more productive and effective**.

4.3 Taking enterprise Workspace farther

Gemini's capabilities also extend to other powerful applications within the Google Workspace ecosystem, like Google Vids and AppSheet.

4.3.1 Google Vids

Google Vids is an online video creation and editing app available to Google Workspace users through your organization.

With Help me Create in Vids, you can use Gemini to generate a first draft of your video. As a project manager, you can use Vids to share project updates, timelines, and key insights with your stakeholders to more effectively communicate information than an email or lengthy report can.

4.3.2 AppSheet

AppSheet is a no-code app development tool included with Google Workspace enterprise editions. With Gemini in AppSheet, you can quickly create apps using AI by describing your needs in a prompt using natural language.

Gemini can create an app structure with tables, columns, and links based on your description, such as "Manage inspections of my facility." You can then review and edit the app structure before creating the app in the AppSheet app editor.

Use case on Google Workspace

Use case: Understood.org

Adding Google Workspace with Gemini functionality into your daily workflow can greatly enhance your productivity. Let’s take a look at an example from a company, Understood.org, to bring this to life.

Understood.org is a social impact organization that supports people with learning and thinking differences such as ADHD and dyslexia, a critical issue, as it impacts more than 70 million Americans.

They leveraged Google Workspace with Gemini and the Gemini app to surface trends, build donor profiles, and tailor its communications. Using natural language, researchers at the nonprofit used Gemini's outputs to tailor their donation outreach. Using Google Workspace with Gemini has resulted in significant benefits including 10 hours saved per research project and more.

Key takeaway on Google Workspace

You've explored the diverse capabilities of Gemini within Google Workspace and better understand how to enhance productivity to drive impactful outcomes, as demonstrated by Understood.org's success story.

In the next lesson, let's learn how prompting can improve our effectiveness when using gen AI technologies.

4.4 Prompting techniques

As you continue experimenting with different gen AI applications, you'll quickly discover the crucial role of prompting. Developing your prompting skills is essential for achieving optimal results and tailoring generative AI to your specific needs.

The key to using gen AI tooling is to ensure you get experience prompting. It's easy to get started, but to get the best results there are different techniques that you can implement.

4.4.1 Using examples in your prompts

When it comes to prompting techniques, there are a few different approaches.

- **Zero-shot prompting** is like asking a foundation model to complete a task with no prior examples, relying solely on its existing knowledge.
- **One-shot prompting** involves showing the foundation model just one example, allowing it to learn and apply that knowledge to similar situations.
- **Few-shot prompting**, on the other hand, provides the foundation model with multiple examples to learn from, which helps it better understand the task and improve its performance.

Let's see this in action

Imagine you manage an online clothing store and want to use AI to enhance your customer service. Here is how different prompting techniques might play out.

Zero-shot prompting

Customer asks about your return policy. The LLM, drawing on its vast knowledge base, understands the question and provides a response, even though it hasn't received specific training on your company's policies.

One-shot prompting

You prompt with a single example of a customer inquiry, such as a question about shipping times. The output follows this example and accurately tags similar customer questions.

Few-shot prompting

You prompt with a few examples of customer inquiries and effective responses. The output follows these examples and accurately tags similar customer questions. For example, the output might include a discount code when a customer reports a delayed delivery, or provide detailed instructions on how to return a damaged item.

Example of X-shot Prompting

Imagine you're in charge of a help desk system overflowing with customer support tickets. Your goal is to use the power of Gemini to categorize these tickets efficiently. The categories are:

- Billing Issue
- Account Management
- Technical Support
- Feature Request
- Refund Request

Here's the tricky ticket: "Hi, I recently upgraded, but I'm still seeing the features from the basic plan. I tried logging out and back in, but that didn't fix it. Can you help?"

The task

Open Gemini and classify this ticket using different prompting techniques below.

Zero-shot prompting

Imagine Gemini has never seen a support ticket. How would you phrase a prompt to get Gemini to categorize this ticket correctly, with no prior examples? Give it your best shot!

One-shot prompting

You decide to give Gemini some help. Provide one example ticket and its correct category to guide Gemini's classification. For example:

- Ticket: "I can't reset my password." Category: Account Management.
- Now, with this single example, how would you prompt Gemini to categorize the new ticket?

Few-shot prompting

Time to unleash the full power of examples! Provide Gemini with multiple example tickets and their categories, like these:

- Example 1: Ticket: "I was charged twice for my subscription this month." Category: Billing Issue.
- Example 2: Ticket: "I need a copy of my invoice for last month." Category: Account Management.
- Example 3: Ticket: "The new feature isn't working as expected." Category: Technical Support.

Now, craft a prompt that uses these examples to guide Gemini towards the correct category for the new ticket.

4.4.2 Role prompting

Role prompting is a technique used to guide the behavior of LLMs by assigning them a specific role or persona. This can be anything from a business analyst or Shakespearean actor to a helpful customer service agent. By instructing the model to adopt a particular role, you influence the style, tone, and focus of its responses.

Prompt 1

You are a customer service agent for a telecommunications company with ten years of experience. A customer is inquiring about their latest bill, which is higher than expected.

LLM Output

The LLM can adopt a helpful and patient tone, explaining potential reasons for the higher bill, offering solutions, and providing information about the company's policies in a clear and concise manner.

Prompt 2

You are a marketing copywriter for a new line of athletic wear. Write a product description for a pair of running shoes emphasizing their comfort and performance-enhancing features.

LLM Output

With a prompt like this, the LLM may generate persuasive and engaging copy highlighting the key benefits of the shoes, using language that appeals to your target audience. It might even craft compelling narratives or tap into emotions that resonate with the target audience, making the product more memorable.

Review: Try it: Step into the role

Open Gemini and practice assigning a persona to an AI model, and observe how it influences the output.

Task

- Choose your Persona → Select a role or persona that interests you. This could be anything from a leader or a scientist to a poet or a stand-up comedian.
- Craft your prompt → Fill in the blanks to create your role prompting prompt: "You are a [persona]. Please write a [fill in the blank]."

Consider tasks like writing a poem, and generating content like a blog post or a research paper, or business-related content like an image for a marketing campaign or a product description.

Bonus: Try the same prompt as a different persona.

This exercise will help you understand the power of role prompting and how it can be used to guide AI models towards desired outputs. Have fun exploring different personas.

4.4.3 Prompt chaining

When working in a tool like Gemini, it's important to understand how previous prompts may impact future prompts.

Prompt chaining is a powerful technique for getting more complex and nuanced results from large language models like Gemini. It's like having a conversation with the AI where each response builds upon the previous one, leading to a more sophisticated and refined outcome.

This is where the concept of a chatbot comes in. You can have a back and forth with Gemini.

Try It: The Chain Reaction

Go to Gemini and experience the power of prompt chaining to achieve a complex outcome.

The Task

- **Set a Goal**
  Think of a multi-step task or a complex question that you'd like Gemini to help you with. This could be something like:
  - Writing a detailed business proposal.
  - Planning a business trip itinerary.
  - Creating an image with various elements for a marketing campaign.
- **Start the Chain**

Begin with an initial prompt that sets the stage for your goal. For example, if you're planning a business trip, your first prompt might be:

"*I need to plan a 5-day business trip to Tokyo in August. My primary goal is to meet with clients and attend the Google Cloud Next Tokyo conference. I'd also like to explore some key landmarks if time allows. Can you help me create a preliminary itinerary?*"

- **Build the Chain**

Based on Gemini's response, ask follow-up questions or provide additional instructions to refine the output. For example:

- "Can you suggest some suitable hotels near the conference center that offer business traveller amenities?"
- "Can you also add some day trips to nearby cities?"

- **Observe and Analyze**

Pay attention to how each prompt builds upon the previous one, leading to a more comprehensive and nuanced result.

This exercise will help you understand how prompt chaining can be used to tackle complex tasks and generate more sophisticated outputs. Have fun experimenting with different chains.

Summary

Here's a simple way to think about these prompting techniques:

- **Zero-shot prompting:** "Just tell it what you want."
- **Few-shot prompting:** "Show, don't tell."
- **Role prompting:** "Set the stage, and let the AI act."

Remember, you can also combine prompting techniques to suit your needs.

4.5 Reuse prompts with Gemini and Gems

Let us discuss how to customize and automate your AI experience.

Reusing prompts

4.5.1 Reusing prompts

You may already find yourself reusing prompts or parts of prompts in your day-to-day workflow. As a start, you can save these prompts as templates in a document somewhere like a Google Doc and a Google Sheet.

Sample prompt examples

Here are a few sample prompt templates to review and leverage:

- **Marketing Manager**

"Write a catchy social media post for [platform] promoting our new [product/service]. Highlight [key features/benefits] and include a call to action to [desired action]."

- **Content Writer**

"Write a [content type: blog post, article, script] about [topic]. The target audience is [audience description]. The tone should be [tone: informative, humorous, persuasive]. Include [keywords]."

- **Customer Support**

"Respond to this customer email: [paste email content]. Be empathetic and address their concerns about [issue]. Offer a solution that aligns with our [company policy]."

- **Software Developer**

"Write Python code to [Task]. Use [Libraries/Frameworks] and follow [Coding style guidelines]."

But there are also more efficient ways of doing this. Google's products and features make gen AI accessible and easy to use for everyone, without needing to be a professional prompt engineer.

4.6 Prompt Chaining

Remember prompt chaining? It lets you continue your conversation in the same chatbot so the chatbot knows your history before continuing on with the new prompt. Well, if you don't want to keep repeating the same history, you can use the same conversation thread and keep working there. In Gemini, you can even name your chats and pin them for easy access.

Prompt chaining scenario

Imagine you're a marketing manager using Gemini to develop a campaign for a new line of organic skincare products. Let's see prompt chaining in action with two sets of prompts and responses. Note how Gemini remembers the context from the previous prompts.

Example

I'm launching a new line of organic skincare products focused on anti-aging. My target audience is women aged 35-55 who are interested in natural beauty solutions. Give me some ideas for a marketing campaign.

→ Next

Of course! Below you can find a list with several campaign ideas, including social media strategies, influencer collaborations, and content marketing suggestions: [...]

→ Next

Okay, let's focus on the content marketing strategy. Develop a content calendar for the next three months with blog post topics, social media post ideas, and potential email newsletter themes.

→ Next

Here's a detailed content calendar breakdown aligned with your target audience and campaign goals (organic skincare, anti-aging, women 35-55): [...]

4.7 Gemini Advanced

Gemini Advanced is an upgraded version of Gemini with extra features, making it far more capable at highly complex tasks, like coding, logical reasoning, following nuanced instructions, and creative collaboration. You can adjust your data retention settings to determine how long information is stored and delete your conversation history at any time. For more detailed information, see the help center article, Upgrade to Gemini Advanced.

4.7.1 Saved info

Using the same prompts in a chain won't always work because, sometimes, you just want the gen AI tool to remember the context but not everything in your whole conversation history. For example, maybe you're a sales rep preparing for calls with potential clients. You might want to practice different sales pitches but wouldn't want the AI's feedback to be influenced by your previous practice sessions or specific details about past clients. In these cases, you can actually save information in Gemini that it will always use.

This is done through your **Settings** in your **Saved info**. You can let Gemini know that you're a sales rep so you do not need to constantly reiterate this in every prompt conversation.

Or let's say a sales team leader wants to use Gemini to generate weekly sales reports. They can save their team's sales targets, key performance indicators (KPIs), and preferred reporting format in the Saved info. This ensures that Gemini always uses the correct information when generating reports, without the need to re-enter it each time.

4.7.2 Gems

But even saved information won't always work. Because that will apply to all your Gemini chats. Sometimes you need specific context for particular tasks or conversations. This is where Gems come in.

Like agents, these AI assistants can process information and reason over complex ideas within the context of your chosen task. You can have Gems dedicated to:

- **Creative writing:** This Gem could be pre-loaded with your preferred writing style, tone, and commonly used resources. It could even help you brainstorm ideas, generate different creative text formats (poems, scripts, articles), and provide feedback on your drafts.
- **Coding:** A coding Gem could be equipped with your favorite libraries, coding conventions, and even access to relevant documentation. It could assist with debugging, suggest code optimizations, and even generate code snippets based on your instructions.
- **Marketing:** This Gem could be pre-loaded with your target audience demographics, competitor information, and campaign performance data. It could help you brainstorm campaign ideas, generate different marketing copy (social media posts, email newsletters, website content), analyze campaign performance, and suggest optimizations.

Think of Gems as your personalized AI assistants within Gemini

How do Gems Work?

- **Personalized responses**

Gems can be tailored with specific instructions and information. This allows them to provide responses that are more relevant to your needs and preferences within that particular use case.

- **Streamlined workflows**

Gems can help you streamline repetitive tasks by providing templates, prompts, or guided interactions. This can save you time and effort when working on familiar projects or activities.

- **Reset context**

You can set the context for a Gem, like giving it background information or specific instructions. Then, you can start multiple chats with that Gem, each with its own focus and flow. These chats remain separate, so information from one won't spill over into another. It's like having different conversations with the same expert, each tailored to a specific purpose.

You can choose to use Google's existing Gems that were built for many different common tasks, or you can create your own gem tailored to your use case.

What information can you provide to a custom Gem?

When you create a custom Gem, you provide it with instructions on how to act, what to do, and more. You are essentially prompting the agent or Gem on how to act. It's time to take all that prompting practice to the next level: Prompting to build your agents.

This is a very good place to use role-prompting, where you prompt the Gem to act as though it is in a specific type of role, such as an air traffic controller.

You can also provide specific documents and resources to a Gem, giving it a focused knowledge base to draw from. Imagine a "code reviewer" Gem that has access to your project's coding style guidelines and best practices documentation. This would allow the Gem to provide feedback grounded in your specific project requirements.

Regardless of your use case, when creating Gems, it's important to be as clear and specific as possible in your instructions. You can also provide examples to illustrate your desired outcomes. Either way, be prepared to refine your instructions over time as you use the Gem and identify areas for improvement.

Think about how users will interact with the Gem and what kind of information they will need to provide. By following these guidelines, you can create powerful Gems that streamline tasks, improve efficiency, and provide valuable insights.

Summary

For each use case below, select the most suitable prompt strategy from the following options:

1. Prompt chaining: Continuing a conversation with the AI, maintaining context.

2. Saved info: Storing persistent information for all your Gemini interactions.
3. Gem: Creating a personalized AI assistant with specific instructions and resources.

Key takeaway

Remember to leverage prompt chaining to build on previous interactions and guide the AI, and don't forget to reuse your well-crafted prompts to save time and boost your productivity. Clear and concise prompts are key to unlocking the full potential of generative AI.

4.8 Grounding

In the previous lesson, you learned how you can attach user-provided resources to your Gem. You can also attach documents and resources in your prompts in the Gemini app with Gemini Advanced. This is called grounding.

Grounding refers to the ability of the AI model to connect its output to verifiable and specific sources of information.

This is a critical aspect of building trust and reliability in AI systems. This is something you can already do in a Gem or Gemini prompt with no coding at all to improve accuracy and relevance to the task you need your agent to solve.

For example, you can create a Gemini prompt that helps you refine your writing by editing your text to meet your organization's specific guidelines and standards. You can provide Gemini with resources, such as your company's style guides within the prompt itself, or by uploading relevant files.

By grounding with this additional information, you provide the AI model with a richer context. This can lead to more accurate and insightful results.

4.8.1 Rag: Retrieval-augmented generation

One powerful grounding technique is retrieval-augmented generation (RAG). It involves:

1. **Retrieving relevant information:** The AI model first retrieves relevant information from a vast knowledge base (like a database, a set of documents, or even the entire web). This retrieval process is often powered by sophisticated techniques, like semantic search or vector databases.
2. **Generating output:** The model then uses this retrieved information to generate the final output. This could be anything from answering a question to writing a creative story.

Model only [without RAG]

The user inputs a prompt.

The model provides output without external knowledge. The output might be outdated, and is more likely to have hallucinations.

Model with RAG

The user inputs a prompt.

A query is sent to a backend retrieval system such as a vector database or search engine to find the most relevant information, and embed it as a part of the prompt.

The model provides the output with the latest external knowledge. The output is much less likely to have hallucinations.

Key takeaways

RAG empowers AI models to go beyond simply memorizing information and instead learn to effectively access and utilize external knowledge sources to generate more insightful and reliable outputs.

- **Improved accuracy and relevance:** By accessing and utilizing relevant information directly, RAG models can produce more accurate and informative outputs.
- **Improved explainability and transparency:** RAG can increase transparency and trust in the AI system by showing the specific sources used to generate the output, ensuring that claims can be checked for accuracy.
- **Extending LLM capabilities:** RAG can constrain the output from an LLM so that it only generates a response based on the specific context it was provided.

One way to start using a RAG system without any coding or database development is with a tool called NotebookLM.

4.10 NotebookLM

4.10.1 Overview

NotebookLM is an AI-first notebook, grounded in your own documents, designed to help you gain insights faster.

It's a unique tool that is built using Gemini models and is designed to help you learn and understand information better by acting as a virtual research assistant. You can "ground" NotebookLM in specific sources like documents, presentations, or even audio and video files. This means your AI assistant will only use information from those sources to answer your questions and generate summaries.

NotebookLM is your personalized AI research partner who has read the provided source material and can answer any question about it.

It can even create different types of summaries, like bullet points, Q&A formats, or even a creative story!

NotebookLM Plus

For users who want to take their research to the next level, there's NotebookLM Plus, a subscription plan that offers benefits like increased capacity, customization of response length and style, and usage analytics.

NotebookLM Enterprise

NotebookLM Enterprise takes NotebookLM Plus to the next level by adding compliance and administrative features necessary for enterprise environments. This includes extra privacy and security features to give you enhanced control over your data. Your notebook information is only shared with your chosen collaborators, and you can manage access using predefined identity and access management (IAM) roles. To learn more, see the documentation for What is NotebookLM Enterprise?.

4.10.2 How NotebookLM works

NotebookLM operates on the concept of "notebooks" to help you organize and interact with your information. Here's a breakdown:

- **Creating a notebook**

  Think of a notebook as a container for a specific project, topic, or area of interest. You start by giving your notebook a title and then uploading the relevant source materials (documents, PDFs, even audio or video files) that you want it to learn.

- **The chat interface**
  - Within each notebook, you have a single, ongoing chat with the NotebookLM AI.
  - You can ask questions, request summaries, or give instructions, just like you would in a regular chat.
  - NotebookLM will respond using only the information from the sources you've uploaded to that specific notebook.

- **Saving insights as notes**
  - When the AI provides a response you find valuable, you can save it as a note within the notebook.
  - These notes become an organized collection of key takeaways, summaries, or important points.

- **Sharing and collaboration**
  - You can create "shared notebooks" to collaborate with others.
  - In a shared notebook, everyone can access the same sources, ask questions, and contribute to the notes.
  - This fosters collaborative learning and knowledge sharing.

Imagine you're writing a research paper. You can upload all your relevant documents to NotebookLM and ask it to:

- Summarize key findings from each source.
- Identify connections and contradictions between different perspectives.
- Generate outlines and drafts for your paper.
- Answer specific questions about the topic.

NotebookLM essentially becomes your personal research assistant, helping you navigate complex information and gain deeper insights. It's a powerful tool for students, researchers, and anyone who wants to learn more effectively.

4.10.3 How is NotebookLM different from Gemini and Gems?

While Gems excel at general tasks and workflows, NotebookLM is designed to help you deeply understand and learn from specific sets of information. It's like having a dedicated expert who has thoroughly analyzed your chosen materials and is ready to answer any question you have.

NotebookLM versus Gems

- **Hyper-focused knowledge:** Instead of drawing from a broad knowledge base, NotebookLM focuses solely on the sources you provide. This could be anything from research papers and articles to meeting notes and presentations.
- **Interactive learning:** NotebookLM goes beyond simply summarizing information. It encourages active learning by allowing you to ask questions, generate different types of summaries, and even create quizzes to test your understanding.
- **Source-based answers:** Every answer and insight provided by NotebookLM is directly grounded in your uploaded sources. This ensures accuracy and allows you to easily trace back to the original information. If you ask NotebookLM a question that isn't covered in the materials you've provided, it will honestly tell you that it can't answer. It won't invent information or speculate. This ensures that the information you get is always grounded in your sources and reliable.

4.10.4 NotebookLM Use cases

- Creating Training materials and documentation

  Onboard new team members and facilitate knowledge transfer by creating a central repository for essential resources. Create a dedicated notebook to share with new hires with relevant onboarding materials (employee handbook, product documentation, training presentations). The new employees can then use

NotebookLM to ask questions, clarify information, and gain a deeper understanding of their role and the company.

- **Researching a new Topic**

  Gather articles and videos on a subject you're interested in and have NotebookLM create a comprehensive summary or answer specific questions you have.

- **Learn by Listening to Audio**

  With audio overviews in NotebookLM you can learn about the material covered in your sources through a podcast-like experience. NotebookLM will summarize your material, connecting topics, and produce a downloadable audio with back-and-forth conversation between two AI hosts.

- **Prepare for Presentation**

  Feed NotebookLM your presentation slides and ask it to generate speaker notes or practice questions to help you rehearse.

- **Summarize Documentation**

  Investment firms and legal teams need to thoroughly analyze documents and data. Upload financial statements, legal documents, and market research reports into NotebookLM. The team can then use the AI to quickly identify key information, summarize findings, and flag potential risks or inconsistencies.

- Project Proposals and Plans

  Keep everyone aligned and informed by creating a shared NotebookLM with all project-related information.

Question and Answer

- Grounding allows an AI model to connect its responses to specific sources of information.
  - Yes
- NotebookLM can access and use information from the entire internet.
  - No
- You can use NotebookLM to create quizzes based on your uploaded documents.
  - Yes
- NotebookLM is best suited for tasks like writing creative stories or poems.
  - No
- Grounding is a feature exclusive to NotebookLM.
  - No

4.10.5 Key takeaway - NotebookLM

NotebookLM is an AI-powered notebook designed to help you quickly gain insights from your documents. It acts as a virtual research assistant, able to summarize information, answer your questions, and even create different types of summaries based on your uploaded documents, presentations, or audio and video files.

Unlike general AI tools that draw from a broad knowledge base, NotebookLM focuses solely on the sources you provide. This "grounding" ensures that the information it provides is accurate, relevant, and traceable back to your original materials.

4.11 Gemini for Google Cloud

Google is using gen AI, specifically Gemini for Google Cloud, to help customers enhance their workflows and productivity.

Just like with Google Workspace, Gemini is revolutionizing how we interact with Google Cloud. From simplifying data analysis to boosting developer productivity, Gemini for Google Cloud offers a suite of AI-powered tools designed to enhance your cloud experience.

Let's explore the power of Gemini within Google Cloud!

Always on enterprise security with Gemini for Google Cloud

Gemini for Google Cloud does not use your prompts or Gemini's responses to train models. It comes with the standard Google Cloud enterprise protections.

- **Gemini Cloud Assist**

  Gemini Cloud Assist is like having an AI expert on your team who helps you design, manage, and optimize applications on Google Cloud. It provides personalized guidance and is integrated with your Google Cloud environment to provide application lifecycle management assistance. It analyzes your cloud environment, your resources deployed, as well as metrics and logs to deliver actionable insights tailored to your needs.

- **Gemini in BigQuery**

  Gemini in BigQuery makes data analysis easier and more accessible. It can help you write code, understand your data, and even generate insights automatically, regardless of your SQL experience. This means faster and more efficient data exploration.

- **Gemini Code Assist**

  Gemini Code Assist acts as an AI pair programmer, helping developers write better code, just like when two programmers work together to solve problems. Gemini Code Assist can provide code suggestions, generate code blocks, and even offer explanations. It supports over 20 popular programming languages, code editors, and developer platforms, all of which help developers increase productivity.

- **Gemini in Colab Enterprise**

  A Colab Enterprise notebook is an interactive environment that lets you write and execute code. Gemini in Colab Enterprise can use AI to help you write Python code in your notebook by suggesting code segments as you type and generating code based on your descriptions. This streamlines your data analysis and machine learning workflows.

- **Gemini in Databases**

  Gemini in Databases helps developers and database administrators manage their databases more effectively. It uses AI to simplify many aspects of using a database, from

building applications with natural language to managing an entire fleet of databases from a single interface.

- **Gemini in Looker**

With Gemini in Looker, you can analyze data and gain insights faster. As an intelligent assistant, it helps you understand your data, create visualizations, and even generate reports, making data exploration more intuitive.

- **Gemini in Security**

Gemini in Security helps security teams detect, contain, and stop threats from spreading. It provides near-instant analysis of security findings and potential attack paths. Gemini in Security also summarizes prevalent tactics, techniques, and procedures used by threat actors, giving customers around the world detailed and timely threat intelligence.

These applications demonstrate how gen AI, specifically the Gemini model, can be integrated into various tools. For more information on Google Cloud's enterprise protections, review the article "Creating trust through transparency."

Summary

It makes data analysis easier and more accessible, regardless of your SQL experience.

- What is a key benefit of Gemini in BigQuery
  - ✓ It makes data analysis easier and more accessible, regardless of your SQL experience.
- How does Gemini Code Assist improve developer productivity?
  - ✓ It acts as an AI pair programmer, offering code suggestions, generating code blocks, and providing explanations.
- What is unique about how Gemini for Google Cloud handles user data?
  - ✓ Google doesn't use your prompts or Gemini's responses to train models, ensuring your data remains private.

You've explored the diverse capabilities of Gemini within Google Cloud, from simplifying data analysis with BigQuery to accelerating code development with Gemini Code Assist. You can see how Gemini is integrated within Google Cloud, enhancing various services to boost productivity and innovation.

4.12 Question Paper - 5

1. You're designing a language learning app and want to incorporate an AI tutor that can provide personalized feedback to students practicing their conversational skills. Which role-prompt would be most effective in guiding the AI's interactions with students?
   A) "You are a language model built to help students learn a new language."

B) "You are a helpful and encouraging tutor who provides constructive feedback on language learning exercises."
C) "You are a robot programmed to assess language proficiency."
D) "Provide personalized feedback to students practicing conversational skills."

2. You're leading a virtual team meeting in Google Meet with colleagues located in different time zones. Some attendees are struggling to understand the discussion due to language barriers. How can Gemini assist in making this Google Meet video conference more accessible for all participants?
   A) Gemini can generate live translated captions with speaker identification, making it easier for everyone to follow along, regardless of language or accent differences.
   B) Gemini can automatically summarize the discussion for those who missed the meeting.
   C) Gemini can automatically adjust the volume of each speaker to ensure everyone is heard equally.
   D) Gemini can provide translated summaries of the spoken conversation in the meeting chat.

3. What is the main advantage of using role prompting?
   A) It allows the AI model to learn from a few examples.
   B) It improves the AI model's ability to understand complex questions.
   C) It enables complex tasks like planning a business trip itinerary
   D) It influences the style, tone, and focus of the AI's responses by assigning it a specific persona.

4. Which prompting technique relies entirely on the foundation model's pre-existing knowledge, without any provided examples?
   A) Few-shot prompting
   B) One-shot prompting
   C) Multi-shot prompting
   D) Zero-shot prompting

5. You're drafting a proposal for a new client and want to ensure your writing is clear, concise, and persuasive. How can Gemini in Google Docs assist you in this task?
   A) Gemini in Docs can predict the client's response to your proposal based on their past interactions.
   B) Gemini in Docs can provide suggestions for improving your writing style, grammar, and tone, and even generate different phrasing options to enhance your message.
   C) Gemini in Docs can automatically assign tasks to team members based on their roles on the client project.
   D) Gemini in Docs can analyze your proposal's content and automatically generate relevant visuals like charts and graphs to enhance its impact.

6. Answer which features to be used: A social media manager who frequently needs to generate posts with a consistent brand voice and style across different platforms

A) Gem
B) Saved Info
C) Prompt Chaining
D) Zero-shot prompting

7. Answer which features to be used: A Sales representative practicing different sales pitches and wants feedback tailored to each pitch without interference from previous practice sessions
    A) Gem
    B) Saved Info
    C) Prompt Chaining
    D) Zero-shot prompting

8. Answer which features to be used: A Developer who frequently uses Gemini for Coding projects and only want to generate code snippets in a specific in a specific language
    A) Gem
    B) Saved Info
    C) Prompt Chaining
    D) Multi-shot prompting

9. Answer which features to be used: A writer who wants to brainstorm an idea for a creative writing project and receive feedback
    A) Gem
    B) Saved Info
    C) Prompt Chaining
    D) Multi-shot prompting

10. Answer which features to be used: A dog groomer who wants all self-care dog related questions to be answered about dog
    A) Gem
    B) Saved Info
    C) Prompt Chaining
    D) Multi-shot prompting

11. What is the primary function of Retrieval-Augmented Generation (RAG) in AI models?
    A) To restrict the AI model's output to a predefined set of responses.
    B) To enable the AI model to access and utilize external knowledge sources for generating outputs.
    C) To replace human intervention in the AI model's output generation process.
    D) To enhance the AI model's ability to memorize vast amounts of information.

12. How does grounding improve the reliability of a Gem in Gemini?
    A) It allows the Gem to access and process information from any website on the internet.
    B) It prevents the Gem from generating any responses that are inaccurate.

C) It connects the Gem's output to specific and verifiable sources of information.
D) It enables the Gem to learn from user interactions and improve its responses over time.

13. You're a freelance writer who specializes in creating different types of content, including blog posts, social media captions, and website copy. You find yourself frequently switching between different writing styles and tones depending on the project. You want to streamline your workflow and ensure consistency in your writing across various formats. Which feature of Gemini best suits your needs?
    A) Gemini for Google Cloud
    B) Gemini Saved info
    C) Gemini Deep Research
    D) Gems

14. **What is the primary advantage of using Saved info in Gemini compared to reusing prompts by copying and pasting them from a document?**
    A) Saved info is where you save your Gemini passwords.
    B) Saved info is automatically shared with all your colleagues, promoting collaboration.
    C) Saved info allows you to store prompts with richer formatting, such as bolding and bullet points.
    D) Saved info ensures that specific information, like your role or company details, is consistently applied across all your Gemini interactions.

15. **What is a key advantage of using NotebookLM Enterprise for team projects?**
    A) You can manage notebook access using predefined identity and access management (IAM) roles in NotebookLM Enterprise.
    B) Google has control over your data in NotebookLM Enterprise.
    C) Google has access to all your data in NotebookLM Enterprise.
    D) NotebookLM Enterprise is the same as the NotebookLM personal product.

16. **You're a product manager preparing for a meeting to discuss a new product launch. You have market research reports, competitor analysis, and customer feedback surveys stored in your Google Drive. How can NotebookLM help you prepare for this meeting?**
    A) NotebookLM can predict the questions your team will ask based on their personalities and prepare answers in advance.
    B) NotebookLM can automatically send calendar invites to all meeting participants with a suggested agenda.
    C) NotebookLM can summarize the key findings from your documents, identify potential challenges, and provide talking points for discussion, helping you lead a productive meeting.
    D) NotebookLM can automatically generate a presentation with slides and talking points for your meeting.

17. **Which of the following statements about the Gemini for Google Cloud enterprise security model is true?**
    A) Google controls access and permissions for customer data.
    B) Customer data is included in publicly accessible datasets.
    C) Customer data is shared with third-party developers.
    D) Customer data is encrypted and access controls are in place.

18. **A team of developers wants to improve their coding efficiency and collaboration. Which Gemini for Google Cloud tool would be most beneficial for them?**
    A) Gemini in Security Command Center
    B) Gemini Cloud Assist
    C) Gemini Code Assist
    D) Gemini in Databases

19. **Google Workspace with Gemini primarily helps users by acting as a:**
    A) Code deployment platform
    B) Writing aid and search/discovery tool
    C) Database management system
    D) Cybersecurity monitoring tool

20. **Which of the following applications is NOT part of Google Workspace?**
    A) Gmail
    B) Google Docs
    C) Google Slides
    D) Vertex AI

21. **What is the main benefit of using the Gemini side panel in Workspace apps?**
    A) Writing production-grade code
    B) Switching between multiple applications
    C) Summarizing and generating content without leaving the app
    D) Training AI models

22. **Gemini in Google Sheets allows users to:**
    A) Design dashboards automatically
    B) Generate formulas using natural language prompts
    C) Translate spreadsheets into multiple languages
    D) Secure spreadsheets with encryption

23. **Which Workspace tool uses Gemini to generate photorealistic images for presentations?**
    A) Google Docs
    B) Google Sheets
    C) Google Slides
    D) Google Meet

24. What feature of Gemini in Google Meet enhances global collaboration?

A) Code suggestions
B) Automated testing
C) Real-time transcription and translation
D) Data visualization

25. Google Vids primarily helps users by:

A) Writing legal documents
B) Creating and editing videos using AI
C) Managing databases
D) Training ML models

26. AppSheet is best described as:

A) A low-code AI training platform
B) A no-code app development tool
C) A cloud security solution
D) A data analytics engine

27. Zero-shot prompting refers to:

A) Giving multiple examples to the model
B) Giving one example to the model
C) Giving no examples and relying on prior knowledge
D) Resetting the conversation context

28. Few-shot prompting improves performance by:

A) Increasing training data
B) Providing multiple task examples
C) Changing the AI model
D) Using saved information

29. Role prompting influences AI output by:

A) Limiting access to data
B) Assigning a specific persona to the model
C) Increasing response length
D) Changing the deployment region

30. Prompt chaining is best described as:

A) Using saved prompts across projects
B) Resetting AI context after each prompt
C) Building complex outputs through sequential prompts
D) Training AI with examples

31. What is the key advantage of using Saved Info in Gemini?

A) It applies context to a single chat only
B) It stores information for all future interactions

C) It replaces prompt chaining
D) It restricts AI responses

32. Gems in Gemini are best described as:

A) Pre-trained AI foundation models
B) Personalized AI assistants for specific tasks
C) Security features
D) Data storage units

33. Grounding improves AI responses by:

A) Connecting outputs to verifiable sources
B) Increasing creativity
C) Reducing response time
D) Removing context

34. What problem does Retrieval-Augmented Generation (RAG) primarily solve?

A) High computing costs
B) Hallucinations and outdated information
C) Slow model training
D) Lack of user interface

35. NotebookLM differs from Gemini and Gems because it:

A) Focuses only on user-provided sources
B) Is designed only for developers
C) Cannot generate summaries
D) Uses internet-wide knowledge

36. Which of the following is a key NotebookLM use case?

A) Social media marketing
B) AI model training
C) Cloud infrastructure management
D) Research and document analysis

37. A key benefit of Gemini for Google Cloud is that:

A) User prompts are used for training models
B) It lacks enterprise security
C) It integrates AI into cloud workflows securely
D) It replaces Google Cloud services

38. What is the Default prompting in Gemini

A) Zero-Shot
B) One-Shot
C) Few-shot
D) No-Shot

39. **How can you reuse a prompt in Gemini. Answer all that apply.**
    A) Manually save prompts in a **Google Doc or Google Sheet**
    B) Reuse Prompts with Prompt Chaining
    C) Use Saved Info (Persistent Reuse Across Chats)
    D) Create a Gem (Best for Structured Reuse)

40. **Which statement is not correct related to NotebookLM**
    A) NotebookLM has different versions like NotebookLM, NotebookLM Plus and NotebookLM Enterprise
    B) NotebookLM enrich its output by accessing Web
    C) NotebookLM can process text, audio, Video and images
    D) NotebookLM is one of the tool of Gemini enterprise

Chapter – 5: Gen AI Agents: Transform Your Organization

5.1 Overview

This course explores how organizations can use custom gen AI agents to help tackle specific business challenges. You gain hands-on practice building a basic gen AI agent, while exploring the components of these agents, such as models, reasoning loops, and tools.

5.2 A brief history of Agents

Today's generative AI agents can achieve a goal by observing the world and acting upon it using the tools it has at its disposal. This wasn't always the case. Early virtual agents were limited to predefined paths and lacked true understanding. But thanks to advances like generative AI and retrieval augmented generation (RAG), today's agents can understand your intent, access relevant information, and provide much more helpful and natural interactions. All due to the connections of the three components of an agent.

5.2.1 Components of an Agent

- **Foundation Model**
  This is the underlying language model (LLM) that powers the agent. It could be a small or large language model, a Google model like Gemini, or a model from another provider. The key is to select a model with training data relevant to the agent's intended use case.
- **Tools**
  Tools enable the agent to interact with the outside world. These can include extensions that connect to APIs, functions that act as mock API calls, and data stores like vector databases. Tools allow the agent to not only observe the world but also act upon it.
- **Reasoning loop**
  This is the core of the agent, responsible for making decisions and taking actions. It's an iterative process where the agent considers its goal, the available tools, and the information it has gathered. Frameworks like ReAct (Reason and Act) are commonly used to guide the reasoning process.

Key takeaway

Now that you have reviewed the individual components of an agent, this course will dive deeper into each component and some of the tools Google Cloud has to support you in building out your agents.

5.2.2 Using models

Keep in mind that generative AI agents are powered by generative AI models. These models are the brains of the operation.

Previous courses in this learning path have covered aspects of building and tuning AI models. In this lesson, we will focus on how to effectively use your gen AI model in your agent.

Once your AI models are trained and ready to go, there are still ways to improve your desired results. One such way is by adjusting certain parameters and settings.

5.2.3 Sampling parameters and settings

Sampling parameters act as settings that influence the AI model's behavior, giving you more customized results. Think of these as knobs and dials you can adjust with your prompt input to impact the model's output. By tweaking these settings, you can ensure the model's output aligns with your specific needs, whether it's generating more creative text, providing more concise summaries, or staying within a certain tone.

Let's explore some of the most common parameters you can adjust.

- Token Count

Think of tokens as the basic building blocks of text for the model. Rather than reading whole sentences at once, the model breaks text down into these chunks. A token can be a whole word, part of a word, or even a single space or punctuation mark.

Models have limits on how many tokens they can handle. A higher token count allows for longer conversations but requires more processing power. As a rule of thumb, one token is roughly four characters in English. So, 100 tokens are about 60 to 80 words.

- **Temperature**

  This parameter controls the "creativity" of the model, because it adjusts the randomness of word choices during text generation, influencing the diversity and unpredictability of the output. A higher temperature makes the output more random and unpredictable, while a lower temperature makes it more focused, deterministic and repeatable.

- **Top-p [Nucleus Sampling]**

  Top-p" stands for the cumulative probability of the most likely tokens considered during text generation. This is another way to control the randomness of the model's output. It concentrates the probability on the most likely tokens, making the output more coherent

and relevant. A lower top-p value leads to more focused responses (i.e. only the most probable tokens), while a higher value allows for more diversity (i.e. extending to lower probability tokens as well).

- Safety Settings

These settings allow you to filter out potentially harmful or inappropriate content from the model's output. You can adjust the level of filtering based on your specific needs and preferences.

- **Output Length**

This determines the maximum length of the generated text. You can set it to a specific number of words or characters or allow the model to generate text until it reaches a natural stopping point.

By experimenting with these parameters, you can significantly influence the AI model's behavior. For example, if you need a concise and factual answer, you might use a lower temperature and a smaller output length. If you're looking for a more creative and open-ended response, you could increase the temperature and top-p values.

How do you access these settings and sampling parameters?

he way you access and change these settings and parameters varies based on where the models are coming from and how you are accessing the models themselves.

Many models can be accessed via APIs, which stands for application programming interfaces. Essentially, an API is a way for different software systems to communicate and exchange information. It acts like a messenger, allowing one application to request services or data from another. For example, when you use an app to check the weather, it's likely using an API to get that information from a weather service. In the case of generative AI, your application would use an API to send a prompt to the model and receive a response. These APIs often allow you to customize the model's behaviour by taking settings and parameters as input along with your prompt.

Key takeaway

Generative AI models are the brains behind AI agents, capable of learning patterns and generating creative text formats. By adjusting sampling parameters like temperature, token count, and top-p, you can fine-tune model behavior to achieve desired outputs.

5.3 Google’s generative AI APIs

In this section, you'll explore Google's specific tools—Google AI Studio and Vertex AI Studio—to put those concepts into practice and start experimenting with generative AI APIs.

Google’s generative AI APIs offer pre-trained large language models that can be fine-tuned for specific tasks. These APIs include capabilities for text completion, multi-turn chat, code generation, and image generation. For example, the Imagen API can be used to generate and customize images.

Google provides two tools where you can easily start to access and experiment with generative AI model APIs.

- Google AI studio
- Vertex AI Studio

5.3.1 Google AI studio

Google AI Studio is a web-based tool that allows developers, students, and researchers to try Gemini models and begin building with the Gemini Developer API. It is designed for ease of use and accessibility, targeting a broad audience, including non-technical users who want to leverage AI capabilities without deep expertise in machine learning.

5.3.2 Vertex AI studio

Vertex AI Studio is a Google Cloud console tool for rapidly prototyping and testing generative AI models. It provides developers with a space to test models using prompt samples, design and save prompts, and tune foundation models.

5.3.3 Google AI studio Vs Vertex AI Studio

Both Google AI Studio and Vertex AI Studio allow you to experiment with and utilize Google's Gemini API, but they cater to different needs and levels of expertise. Here's a breakdown of their key differences:

Key takeaway on Gemini Development tools

Google provides tools like Vertex AI Studio and Google AI Studio for experimenting with generative AI APIs and customizing models for specific tasks. These tools empower both developers and non-technical users to harness the power of generative AI and integrate it into their applications.

5.4 Parameter playground

Having learned about the different sampling parameters and the tools to access them, this lesson will now give you a hands-on opportunity to experiment in Google AI Studio and examine how these settings impact a model's output firsthand.

he best way to understand these parameters is to try them yourself. Google AI Studio provides a user-friendly interface for experimenting with different settings and observing their effects on the model's output. This hands-on experience will help you develop an intuition for how to adjust your inputs to the AI model and achieve the desired results.

Explore the impact of different parameters on Gemini's output.

Practice Session

Open Google AI Studio

To begin experimenting with AI in Google AI Studio, complete the following steps:

1. If you haven't already, log into your Google account and open Google AI Studio. (Note: You'll need to accept the terms and conditions as prompted in order to proceed.). Link: https://aistudio.google.com/apps
2. Select **Try Gemini**.

Craft a Creative Prompt

Imagine you're preparing for an important presentation. Your goal is to deliver a compelling pitch that convinces your audience to invest in your team or company's new product or service.

1. Locate the prompt input area and **Run** button on the page. This is used to send your prompt to the Gemini model.
2. Enter a prompt into the prompt input for help creating a pitch about a fictional product or service you'd like to create. You can include details like the product's key features, the target audience, or the desired outcome.
    - *For example: Write a pitch for a new app offering fast and affordable on-campus food delivery for university students.*
3. Press **Run** to generate a response to your prompt.
4. Review the response from the Gemini API.

Experiment with Temperature

Now, you can generate multiple versions of your pitch, each time adjusting a new parameter.

1. In the **Run settings** panel, change the **Temperature** by moving the slider or entering a new value into the input box. *Try lower values for a more formal and factual tone and higher values for more passion and enthusiasm.*
2. Input the prompt you previously provided and select **Run** again.
3. Review the response from the Gemini API.

Experiment with more sampling Parameters

1. In the **Run** settings panel, select **Advanced settings** to expand the list of options.
2. Change the **Output length** to a different word limit and retry your prompt.
3. Then, try changing **Top P** and retry your prompt. Experiment with lower values for a more focused and concise pitch and higher values for a more elaborate pitch with greater variety.

Observe and Analyze

Pay close attention to how each parameter change affects the tone, style, and persuasiveness of the pitch:

- Does the pitch sound confident and authoritative?
- Is it concise and to the point or more detailed and elaborate?
- Does it evoke the desired emotions in the audience?

Key takeaway

Hands-on experimentation with sampling parameters in a tool like Google AI Studio is the best way to understand how to control a generative AI model's output. By adjusting settings like Temperature, Top p, and Output length, you can directly influence the tone, style, and creativity of the model's response to achieve your specific goals.

5.5 Adding prompt engineering to your reasoning loops

5.5.1 The reasoning loop

The reasoning loop is a key component of a generative AI agent that governs how the agent takes in information, performs internal reasoning, and uses that reasoning to inform its next action or decision. It is an iterative, introspective process that continues until the agent achieves its goal or reaches a stopping point. The complexity of the reasoning loop can vary greatly depending on the agent and the task it is performing.

Here's a breakdown of the key aspects of the reasoning loop:

- **Iterative Process**

  The reasoning loop is not a one-time operation, but rather a cyclical process where the agent continuously evaluates its progress and determines the next best action to take. This loop involves steps for action, tool selection, and observation.

- **Internal Reasoning**

  The agent uses its underlying language model to think through the steps it needs to take to complete a task. The language model provides the agent with reasoning and logic capabilities.

- **Decision Making**

  Based on its internal reasoning, the agent decides on the next course of action. This involves choosing the appropriate tools to use and determining the necessary inputs for those tools.

- **Reasoning Framework**

  The reasoning loop utilizes various prompt engineering frameworks and techniques to guide its reasoning and planning.

Key takeaway

The reasoning loop is a central, iterative process that enables a generative AI agent to take in information, perform internal reasoning, and make decisions. By understanding the reasoning loop, you can begin to add logic to your agents to make them more powerful and effective.

5.5.2 Prompt engineering techniques

There are many different prompting techniques that one can use, but we will highlight two of the most common for this course: ReAct prompting and chain-of-thought (CoT) prompting.

- **ReAct prompting**

  ReAct is a prompting framework that allows the language model to reason and take action on a user query, with or without in-context examples.

- **Chain-of-Thought [CoT]**

  Chain-of-thought prompting is a technique where you guide a language model through a problem-solving process by providing examples with intermediate reasoning steps, helping it learn to approach new problems in a more structured and logical way.

ReAct prompting

ReAct, which stands for "reasoning and acting," is like giving an LLM a brain and a pair of hands. It allows the LLM to not only think about a problem but also take actions to solve it. This is a big step up from traditional prompting, where the LLM could only passively generate text.

Imagine you're asking an LLM to find you a good Italian restaurant nearby. With ReAct, the LLM can: **Think → Act → Observe → Respond**

- Think → Okay, I need to find Italian restaurants that are close by and have good reviews.
- Act → I'll search online for Italian restaurants in this area.
- Observer → Ah, here are some options with high ratings.
- Think → Which one is the best fit based on the user's preferences?
- Act → I'll check the menus and see which one has the most appealing dishes.
- Observer → This one has delicious-looking pasta!
- Respond → I recommend [restaurant name]. It has great reviews, is located nearby, and has a fantastic pasta selection.

This "thought-action-observation" loop allows the LLM to interact with the real world and gather information, making it much more powerful than a traditional LLM.

Why is ReAct important?

- **Dynamic problem solving:** ReAct allows LLMs to tackle complex tasks that require interacting with external resources and adapting to new information.
- **Reduced hallucination:** By grounding the LLM's reasoning in real-world data, ReAct can help reduce the risk of generating incorrect or nonsensical information.
- **Increased trustworthiness:** The ability to see the LLM's reasoning process and how it interacts with external sources makes its responses more transparent and trustworthy.

Key components of ReAct

- **Think:** The LLM generates a thought about the problem, similar to CoT.
- **Act:** The LLM decides what action to take, such as searching the web, accessing a database, or using a specific tool; the LLM specifies the input for the action, like a search query or database command.
- **Observe:** The LLM receives feedback from the action, such as search results or database entries.
- **Respond:** The LLM generates a response, which could involve providing an answer to the user, taking further actions, or formulating a new thought for the next iteration.

ReAct in action

ReAct has been successfully used in various applications, including:

- **Question answering:** LLMs can use ReAct to access external knowledge sources and answer questions more accurately.
- **Fact verification:** LLMs can verify claims by searching for evidence online.
- **Decision making:** LLMs can use ReAct to gather information and make informed decisions in interactive environments.

The agent provides a prompt to the model, asking it to generate the next step and its corresponding output, which may include a thought, an action (including a tool choice), an action input, and an observation.

Chain-of-thought (CoT)

Remember prompt chaining? Where you keep prompting in the same thread so the LLM keeps your chat history and learns more as you go? Well, you can do something very similar behind the scenes for your user using chain-of-thought (CoT) prompting. CoT enables reasoning capabilities through intermediate steps.

Think of CoT as a way to make LLMs even smarter by teaching them to think step-by-step, just like a human would. Instead of just giving the LLM a prompt and expecting an answer, you guide it through the reasoning process. You provide examples of how to solve similar problems, showing the steps involved. This helps the LLM learn to approach problems in a more logical and structured way. It is similar to teaching a student to think out loud. You're teaching the LLM to think out loud. By showing it the intermediate steps, you're helping it develop a "chain of thought" that leads to the correct answer.

Why is CoT important?

CoT is a powerful technique that helps LLMs think a bit more like humans. By guiding them through the reasoning process, we can unlock their full potential and achieve even more impressive results.

- **Improved reasoning:** CoT helps LLMs solve complex problems that require logical thinking.
- **Better accuracy:** By breaking down problems into smaller steps, CoT can lead to more accurate results.
- **Enhanced explainability:** CoT makes it easier to understand how the LLM arrived at an answer, which is important for building trust and transparency.

Key components of CoT

Just like there are different ways to solve a problem, there are different ways to implement CoT. Some popular techniques include:

- **Self-consistency:** Encouraging the LLM to generate multiple solutions and choose the most consistent one.
- **Active-prompting:** Allowing the LLM to ask clarifying questions or request additional information.
- **Multimodal CoT:** Combining text with other forms of data, like images or videos, to enhance reasoning.

5.5.3 Choosing a prompting framework

There are many other frameworks and techniques that can be used for prompt engineering and within your reasoning loop. All these different techniques can be combined and leveraged in different ways depending on the use case.

The best framework for you will depend on your use case. For example, while both ReAct and CoT enhance LLM reasoning, they have different strengths:

- CoT focuses on internal reasoning, guiding the LLM through a chain of thought.

- ReAct focuses on external interaction, allowing the LLM to gather information and take actions in the real world.

In fact, ReAct and CoT can be combined for even more powerful results. By integrating both techniques, we can create LLMs that are capable of both deep reasoning and dynamic interaction with the world around them.

Key takeaway

Prompt engineering techniques like ReAct and chain-of-thought (CoT) prompting can enhance the reasoning capabilities of agents, leading to more accurate, reliable, and human-like interactions. By understanding prompt engineering techniques, you can effectively guide the development of AI agents that can transform your organization.

Example

- Finding a highly-rated Italian restaurant near a current location
  - Answer: ReAct
- Summarizing a long article about the history of artificial intelligence
  - Answer: CoT
- Brainstorming ideas of a new product
  - Answer: CoT
- Searching for Research paper for a particular scientific topic
  - Answer: ReAct

5.6 Adding a Tooling

This section will demonstrate how generative AI agents interact with the real world through the power of tools.

5.6.1 Types of agent tools

Agent tooling equips agents with the resources they need to be effective. Think of it as providing the agent with the right skills, connections, and knowledge to achieve its goals. These tools allow agents to access information, perform actions, and interact with various systems.

We can categorize agent tools into four key types.

- **Extensions (APIs)**

  Extensions bridge the gap between an agent and external APIs. APIs (application programming interfaces) are sets of rules that govern how software interacts. Extensions provide a standardized way for agents to use APIs, regardless of the API's specific design. This simplifies API interaction, making it easier for agents to access external services and data.

  Example: An agent designed to book travel might use an extension to interact with a travel company's API. The extension handles the complexities of communicating with travel company's systems, allowing the agent to focus on the task of finding and booking flights.

- **Functions**

Functions are like specialized tools within the agent's toolbox. They represent specific actions the agent can perform. An agent's reasoning system selects the appropriate function based on the task at hand. Functions can encapsulate complex logic or interactions, making them reusable and manageable.

Example: A "calculate_price" function might take flight details and passenger information as input and return the total cost. The agent can call this function whenever it needs to calculate a price.

- **Data stores**

Data stores provide agents with access to information. This can include real-time data, historical data, or knowledge bases. Data stores ensure that the agent's responses are accurate, relevant, and up-to-date.

Example: An agent might use a data store to access current weather conditions, stock prices, or a database of customer information.

- **Plugins**

Plugins extend the agent's capabilities by adding new skills or integrations. They can connect the agent to specific services, provide access to specialized tools, or enable interaction with particular platforms.

Example: A plugin could enable an agent to interact with a calendar application, allowing it to schedule appointments. Another plugin might integrate with a payment gateway, enabling the agent to process transactions.

5.6.2 How the reasoning loop works with tools

The ReAct (reasoning and acting) cycle prompting technique describes how agents use tools:

1. **Reasoning [Tool Selection]**
   The agent analyzes the task and determines which tools are needed. It considers the available extensions, functions, and data stores to choose the most appropriate resources.
2. **Acting [Tool execution]**
   The agent executes the selected tool. This might involve calling an API via an extension, invoking a function, or querying a data store. The agent provides the necessary inputs to the tool.
3. **Observation**
   The agent receives the output from the tool. This output becomes the "observation" in the ReAct cycle.
4. **Iteration [Dynamic iteration]**
   Based on the observation, the agent reasons about the next steps. It might need to select different tools, refine its approach, or gather more information. This cycle repeats until the task is complete.

5.6.3 Example of Tools usage: Scheduling a garden consultation

- **Reasoning**

The agent needs to find an available time slot for a garden consultation. It selects the scheduling plugin, which integrates with the gardener's calendar.

- **Acting**

The agent uses the scheduling plugin to check the gardener's availability for the next week, specifying the desired consultation duration (e.g., 30 minutes) and the client's preferred days (e.g., weekdays). It also accesses the customer database (via a data store) to retrieve the client's name and contact information to include in the appointment details.

- **Observation**

The scheduling plugin returns a list of available time slots.

- **Iteration**

The agent presents the available slots to the client. If the client requests a specific time that's unavailable, the agent might use a function to suggest alternative times or offer to put the client on a waiting list. Once the client confirms a time, the agent uses the scheduling plugin again to book the appointment and sends a confirmation email to both the client and the gardener. This cycle continues until the consultation is successfully scheduled.

5.6.4 Key takeaway

Agent tooling is all about enabling agents to take action, whether that's accessing databases, making purchases, interacting with social media, controlling smart home devices, scheduling appointments, booking travel, or performing other necessary tasks. By understanding the range of tools available, you can effectively equip your AI agents with the resources they need to perform complex tasks and interact intelligently with the world.

5.7 Google's agent tooling

Whether building your agents inside or outside of Google Cloud, the options for tooling are extensive. You can custom-build tools, leverage third-party solutions, or, as we'll focus on here, utilize the powerful suite of tools available within Google Cloud.

While this lesson won't cover every single Google Cloud product and feature, it will highlight some key options to spark ideas for the kinds of integrations you can create. It's important to note that many of these tools can be used even if you choose to build your core agent logic outside of Google Cloud.

5.7.1 Key Google Cloud tool for agents

Here are some of the most relevant Google Cloud services for agent tooling:

- **Cloud Storage**

  A highly scalable and durable object storage service. Use Cloud Storage to store and retrieve data that your agent needs.

- **Databases [Cloud SQL, Firestore, Spanner]**

  Google Cloud offers a variety of database solutions to suit your needs. Your agent can use these databases to store and retrieve information, manage user data, or track its own progress.

- **Cloud Run Functions**

  Create serverless functions that act as specialized tools for your agent. Cloud Run functions can be used to connect to databases, call external APIs, perform complex calculations, or handle other specific tasks. They are easily triggered by your agent and scale automatically.

- **Cloud Run**

  For more complex agent tools that require containerized environments, Cloud Run provides a serverless platform for deploying and running stateless containers. This is ideal for custom tools that have specific dependencies or require more control.

- **Vertex AI**

  Agents can use other agents as tooling. You can use Vertex AI to create models or agents that are called as tooling by other agents.

5.7.2 Pre-built AI APIs

Along with these services, Google Cloud also offers a wide range of pre-built APIs that provide powerful functionalities for your agent. These APIs can be used for tasks like understanding natural language, analyzing images, translating text, and more.

- **Speech-to-Text API**

  The **Speech-to-Text API** accurately converts speech into text. This is useful for transcribing meetings, customer service calls, or even video content.

- **Text-to-Speech API**

  The **Text-to-Speech API** converts text into natural-sounding speech, which is helpful for creating voice user interfaces in devices and applications, and for personalized communication.

- **Translation API**

  The **Translation API** can translate text, documents, websites, and even audio and video files into more than 135 languages.

- **Document Translation API**

The **Document Translation API** can translate formatted documents while keeping the original layout.

- **Document AI API**

The **Document AI API** allows businesses to extract data from various document formats. It can be used to automate data capture and document processing. This is useful for many kinds of documents, from invoices to contracts to medical records. Document AI can also summarize large documents using generative AI.

- **Cloud Vision API**

The **Cloud Vision API** allows developers to understand the content of an image by applying powerful machine learning models. It can detect objects, faces, and landmarks, and even read text within images (OCR). This API is useful for a wide range of applications, including image tagging, content moderation, and visual search.

- **Cloud Video Intelligence API**

The **Cloud Video Intelligence API** allows developers to analyze video content and extract meaningful information. It can identify objects, actions, and events within a video, and even transcribe spoken words. This API can be used for applications like content recommendation, video search, and media analysis.

- **Natural Language API**

The **Natural Language API** helps derive insights from unstructured text using Google's machine learning. This API can be used to understand the sentiment of text, classify content, and extract important entities such as people, places, and events. This API can also analyze text in multiple languages.

5.7.3 Other APIs

Google Cloud also offers a vast API Library with APIs linked to many other Google products such as Google Maps, Google Workspace, YouTube, Google Photos, and more. For additional information, check out the API Library [https://console.cloud.google.com/apis/library].

5.7.4 Agent tooling example: Meeting location planner

Scenario: A user uploads a document (e.g., a meeting invitation or a list of potential conference venues) and asks the agent to suggest the most convenient meeting location for attendees.

Using Document AI to understand document content and the Google Maps API for location analysis, the agent can provide users with helpful meeting location recommendations based on unstructured data.

- **Document processing (Document AI API)**
  The agent uses the Document AI API to process the uploaded document and identify addresses mentioned within the document.

- **Geocoding (Google Maps Geocoding API)**
  The agent takes the extracted addresses and uses the Google Maps Geocoding API to convert them into latitude and longitude coordinates to calculate distances and travel times.

- **Location analysis (custom logic)**
  The agent uses custom logic implemented within a Cloud Function to analyze the geocoded addresses and determine a suggested location.

5.7.5 Key takeaway

Google Cloud provides an extensive suite of tools and APIs for building and powering your agents. By leveraging services like Cloud Functions, Databases, and pre-built APIs such as Document AI and Google Maps, you can create agents that perform complex, multi-step tasks. This allows you to orchestrate multiple tools to solve real-world problems and build powerful, intelligent agents.

5.8 Building applications from your agents

5.8.1 Introduction

Whether you're using the Gemini API directly with its default settings or with your own customized parameters through AI Studio, let's hear from the course creators discussing how to integrate this into your own applications.

5.8.2 Using the API

To use the API within your application, follow the directions for each tool:

Google AI Studio

- **Gemini API quickstart**

  This quickstart shows you how to install our libraries and make your first Gemini API request.

  Before you begin

  You need a Gemini API key. If you don't already have one, you can get it for free in Google AI Studio.
  [https://aistudio.google.com/app/apikey?_gl=1*w1hnql*_ga*MTI2MzA2ODI4MC4xNzY2MjUyODEz*_ga_P1DBVKWT6V*czE3NjgzNzU1NDYkbzMkZzEkdDE3NjgzNzY5MjEkajMxJGwwJGgyMTIyMjQzNzc2]

- **Install the Google GenAI SDK**

  Python

  Using Python 3.9+, install the google-genai package using the following pip command:

  pip install -q -U google-genai

  Apps Script

1. To create a new Apps Script project, go to script.new.
2. Click **Untitled project**.
3. Rename the Apps Script project **AI Studio** and click **Rename**.
4. Set your API key

    a. At the left, click **Project Settings** .

    b. Under **Script Properties** click **Add script property**.

    c. For **Property**, enter the key name: GEMINI_API_KEY.

    d. For **Value**, enter the value for the API key.

    e. Click **Save script properties**.
5. Replace the Code.gs file contents with the following code:

Make your first request

Here is an example that uses the generateContent method to send a request to the Gemini API using the Gemini 2.5 Flash model.

If you set your API key as the environment variable GEMINI_API_KEY, it will be picked up automatically by the client when using the Gemini API libraries. Otherwise you will need to pass your API key as an argument when initializing the client.

Note that all code samples in the Gemini API docs assume that you have set the environment variable GEMINI_API_KEY.

```
// See https://developers.google.com/apps-script/guides/properties
// for instructions on how to set the API key.
const apiKey = PropertiesService.getScriptProperties().getProperty('GEMINI_API_KEY');
function main() {
  const payload = {
    contents: [
      {
        parts: [
          { text: 'Explain how AI works in a few words' },
        ],
      },
```

```
    ],
  };


  const url = 'https://generativelanguage.googleapis.com/v1beta/models/gemini-3-flash-preview:generateContent';
  const options = {
    method: 'POST',
    contentType: 'application/json',
    headers: {
      'x-goog-api-key': apiKey,
    },
    payload: JSON.stringify(payload)
  };
  const response = UrlFetchApp.fetch(url, options);
  const data = JSON.parse(response);
  const content = data['candidates'][0]['content']['parts'][0]['text'];
  console.log(content);
}
```

What's next

Now that you made your first API request, you might want to explore the following guides that show Gemini in action:

- Text generation
- Image generation
- Image understanding
- Thinking
- Function calling
- Long context
- Embeddings

Generate content with the Gemini API in Vertex AI

Use generateContent or streamGenerateContent to generate content with Gemini.

The Gemini model family includes models that work with multimodal prompt requests. The term multimodal indicates that you can use more than one modality, or type of input, in a prompt. Models that aren't multimodal accept prompts only with text. Modalities can include text, audio, video, and more.

Create a Google Cloud account to get started

To start using the Gemini API in Vertex AI, create a Google Cloud account.

After creating your account, use this document to review the Gemini model request body, model parameters, response body, and some sample requests.

When you're ready, see the Gemini API in Vertex AI quickstart to learn how to send a request to the Gemini API in Vertex AI using a programming language SDK or the REST API.

Supported models

All Gemini models support content generation.

Parameter list

See examples for implementation details.

Request body

```
{
 "cachedContent": string,
 "contents": [
  {
   "role": string,
   "parts": [
    {
     // Union field data can be only one of the following:
     "text": string,
     "inlineData": {
      "mimeType": string,
      "data": string
     },
     "fileData": {
      "mimeType": string,
      "fileUri": string
```

```
      },
      // End of list of possible types for union field data.

      "thought": boolean,
      "thoughtSignature": string,
      "videoMetadata": {
        "startOffset": {
          "seconds": integer,
          "nanos": integer
        },
        "endOffset": {
          "seconds": integer,
          "nanos": integer
        },
        "fps": double
      },
      "mediaResolution": MediaResolution
    }
  ]
 }
],
"systemInstruction": {
 "role": string,
 "parts": [
  {
   "text": string
  }
 ]
},
"tools": [
 {
```

```
    "functionDeclarations": [
      {
        "name": string,
        "description": string,
        "parameters": {
          object (OpenAPI Object Schema)
        }
      }
    ]
  }
],
"safetySettings": [
  {
    "category": enum (HarmCategory),
    "threshold": enum (HarmBlockThreshold)
  }
],
"generationConfig": {
  "temperature": number,
  "topP": number,
  "topK": number,
  "candidateCount": integer,
  "maxOutputTokens": integer,
  "presencePenalty": float,
  "frequencyPenalty": float,
  "stopSequences": [
    string
  ],
  "responseMimeType": string,
  "responseSchema": schema,
  "seed": integer,
```

```
  "responseLogprobs": boolean,
  "logprobs": integer,
  "audioTimestamp": boolean,
  "thinkingConfig": {
    "thinkingBudget": integer,
    "thinkingLevel": enum
  },
  "mediaResolution": MediaResolution
 },
 "labels": {
  string: string
 }
}
```

The request body contains data with the following parameters:

The request body contains data with the following parameters:

Parameters	
cachedContent	Optional: string The name of the cached content used as context to serve the prediction. Format: projects/{project}/locations/{location}/cachedContents/{cachedContent}
contents	Required: Content The content of the current conversation with the model. For single-turn queries, this is a single instance. For multi-turn queries, this is a repeated field that contains conversation history and the latest request.
systemInstruction	Optional: Content Available for gemini-2.0-flash and gemini-2.0-flash-lite. Instructions for the model to steer it toward better performance. For example, "Answer as concisely as possible" or "Don't use technical terms in your response".

The text strings count toward the token limit.

The role field of systemInstruction is ignored and doesn't affect the performance of the model.

Note: Only **text** should be used in **parts** and content in each **part** should be in a separate paragraph.

tools — Optional. A piece of code that enables the system to interact with external systems to perform an action, or set of actions, outside of knowledge and scope of the model. See Function calling.

toolConfig — Optional. See Function calling.

safetySettings — Optional: SafetySetting

Per request settings for blocking unsafe content.

Enforced on GenerateContentResponse.candidates.

generationConfig — Optional: GenerationConfig

Generation configuration settings.

labels — Optional: string

Metadata that you can add to the API call in the format of key-value pairs.

contents

The base structured data type containing multi-part content of a message.

This class consists of two main properties: role and parts. The role property denotes the individual producing the content, while the parts property contains multiple elements, each representing a segment of data within a message.

Parameters

role string

The identity of the entity that creates the message. The following values are supported:

- user: This indicates that the message is sent by a real person, typically a user-generated message.
- model: This indicates that the message is generated by the model.

The model value is used to insert messages from the model into the conversation during multi-turn conversations.

parts Part

A list of ordered parts that make up a single message. Different parts may have different IANA MIME types.

For limits on the inputs, such as the maximum number of tokens or the number of images, see the model specifications on the Google models page.

To compute the number of tokens in your request, see Get token count.

parts

A data type containing media that is part of a multi-part Content message.

Parameters

text	Optional: string A text prompt or code snippet.
inlineData	Optional: Blob Inline data in raw bytes. For gemini-2.0-flash-lite and gemini-2.0-flash, you can specify up to 3000 images by using inlineData.
fileData	Optional: fileData Data stored in a file.
functionCall	Optional: FunctionCall. It contains a string representing the FunctionDeclaration.name field and a structured JSON object containing any parameters for the function call predicted by the model. See Function calling.
functionResponse	Optional: FunctionResponse.

The result output of a FunctionCall that contains a string representing the FunctionDeclaration.name field and a structured JSON object containing any output from the function call. It is used as context to the model.

See Function calling.

thought — Optional: boolean

Indicates whether the part represents the model's thought process or reasoning.

thoughtSignature — Optional: string (bytes format)

An opaque signature for the thought so it can be reused in subsequent requests. A base64-encoded string.

videoMetadata — Optional: VideoMetadata

For video input, the start and end offset of the video in Duration format, and the frame rate of the video . For example, to specify a 10 second clip starting at 1:00 with a frame rate of 10 frames per second, set the following:

- "startOffset": { "seconds": 60 }
- "endOffset": { "seconds": 70 }
- "fps": 10.0

The metadata should only be specified while the video data is presented in inlineData or fileData.

mediaResolution — Optional: MediaResolution

Per-part media resolution for the input media. Controls how input media is processed. If specified, this overrides the mediaResolution setting in generationConfig. LOW reduces tokens per image/video, possibly losing detail but allowing longer videos in context. Supported values: HIGH, MEDIUM, LOW.

blob

Content blob. If possible send as text rather than raw bytes.

Parameters

mimeType — string

The media type of the file specified in the data or fileUri fields. Acceptable values include the following:

Click to expand MIME types

For gemini-2.0-flash-lite and gemini-2.0-flash, the maximum length of an audio file is 8.4 hours and the maximum length of a video file (without audio) is one hour. For more information, see Gemini audio and video requirements.

Text files must be UTF-8 encoded. The contents of the text file count toward the token limit.

There is no limit on image resolution.

data bytes

The base64 encoding of the image, PDF, or video to include inline in the prompt. When including media inline, you must also specify the media type (mimeType) of the data.

Size limit: 7 MB for images

FileData

URI or web-URL data.

Parameters

mimeType string

IANA MIME type of the data.

fileUri string

The URI or URL of the file to include in the prompt. Acceptable values include the following:

- Cloud Storage bucket URI: The object must either be publicly readable or reside in the same Google Cloud project that's sending the request. For gemini-2.0-flash and gemini-2.0-flash-lite, the size limit is 2 GB.
- HTTP URL: The file URL must be publicly readable. You can specify one video file, one audio file, and up to 10 image files per request. Audio files, video files, and documents can't exceed 15 MB.
- YouTube video URL:The YouTube video must be either owned by the account that you used to sign in to the Google Cloud console or is public. Only one YouTube video URL is supported per request.

When specifying a fileURI, you must also specify the media type (mimeType) of the file. If VPC Service Controls is enabled, specifying a media file URL for fileURI is not supported.

functionCall

A predicted functionCall returned from the model that contains a string representing the functionDeclaration.name and a structured JSON object containing the parameters and their values.

Parameters

name	string The name of the function to call.
args	Struct The function parameters and values in JSON object format. See Function calling for parameter details.

functionResponse

The resulting output from a FunctionCall that contains a string representing the FunctionDeclaration.name. Also contains a structured JSON object with the output from the function (and uses it as context for the model). This should contain the result of a FunctionCall made based on model prediction.

Parameters

name	string The name of the function to call.
response	Struct The function response in JSON object format.

videoMetadata

Metadata describing the input video content.

Parameters

startOffset	Optional: google.protobuf.Duration The start offset of the video.
endOffset	Optional: google.protobuf.Duration The end offset of the video.
fps	Optional: double

The frame rate of the video sent to the model. Defaults to 1.0 if not specified. The minimum accepted value is as low as, but not including, 0.0. The maximum value is 24.0.

safetySetting

Safety settings.

Parameters

category Optional: HarmCategory

The safety category to configure a threshold for. Acceptable values include the following:

Click to expand safety categories

- HARM_CATEGORY_SEXUALLY_EXPLICIT
- HARM_CATEGORY_HATE_SPEECH
- HARM_CATEGORY_HARASSMENT
- HARM_CATEGORY_DANGEROUS_CONTENT

threshold Optional: HarmBlockThreshold

The threshold for blocking responses that could belong to the specified safety category based on probability.

- OFF
- BLOCK_NONE
- BLOCK_LOW_AND_ABOVE
- BLOCK_MEDIUM_AND_ABOVE
- BLOCK_ONLY_HIGH

method Optional: HarmBlockMethod

Specify if the threshold is used for probability or severity score. If not specified, the threshold is used for probability score.

harmCategory

Harm categories that block content.

Parameters

HARM_CATEGORY_UNSPECIFIED	The harm category is unspecified.

HARM_CATEGORY_HATE_SPEECH	The harm category is hate speech.
HARM_CATEGORY_DANGEROUS_CONTENT	The harm category is dangerous content.
HARM_CATEGORY_HARASSMENT	The harm category is harassment.
HARM_CATEGORY_SEXUALLY_EXPLICIT	The harm category is sexually explicit content.

harmBlockThreshold

Probability thresholds levels used to block a response.

Parameters

HARM_BLOCK_THRESHOLD_UNSPECIFIED	Unspecified harm block threshold.
BLOCK_LOW_AND_ABOVE	Block low threshold and higher (i.e. block more).
BLOCK_MEDIUM_AND_ABOVE	Block medium threshold and higher.
BLOCK_ONLY_HIGH	Block only high threshold (i.e. block less).
BLOCK_NONE	Block none.
OFF	Switches off safety if all categories are turned OFF

harmBlockMethod

A probability threshold that blocks a response based on a combination of probability and severity.

Parameters

HARM_BLOCK_METHOD_UNSPECIFIED	The harm block method is unspecified.
SEVERITY	The harm block method uses both probability and severity scores.
PROBABILITY	The harm block method uses the probability score.

generationConfig

Configuration settings used when generating the prompt.

Parameters

temperature Optional: float

The temperature is used for sampling during response generation, which occurs when topP and topK are applied. Temperature controls the degree of randomness in token selection. Lower temperatures are good for prompts that require a less open-ended or creative response, while higher temperatures can lead to more diverse or creative results. A temperature of 0 means that the highest probability tokens are always selected. In this case, responses for a given prompt are mostly deterministic, but a small amount of variation is still possible.

If the model returns a response that's too generic, too short, or the model gives a fallback response, try increasing the temperature. If the model enters infinite generation, increasing the temperature to at least 0.1 may lead to improved results.

1.0 is the recommended starting value for temperature.

- Range for gemini-2.0-flash-lite: 0.0 - 2.0 (default: 1.0)
- Range for gemini-2.0-flash: 0.0 - 2.0 (default: 1.0)

For more information, see Content generation parameters.

topP Optional: float

If specified, nucleus sampling is used.

Top-P changes how the model selects tokens for output. Tokens are selected from the most (see top-K) to least probable until the sum of their probabilities equals the top-P value. For example, if tokens A, B, and C have a probability of 0.3, 0.2, and 0.1 and the top-P value is 0.5, then the model will select either A or B as the next token by using temperature and excludes C as a candidate.

Specify a lower value for less random responses and a higher value for more random responses.

- Range: 0.0 - 1.0
- Default for gemini-2.0-flash-lite: 0.95
- Default for gemini-2.0-flash: 0.95

candidateCount Optional: int

The number of response variations to return. For each request, you're charged for the output tokens of all candidates, but are only charged once for the input tokens.

Specifying multiple candidates is a Preview feature that works with generateContent (streamGenerateContent is not supported). The following models are supported:

- gemini-2.0-flash-lite: 1-8, default: 1
- gemini-2.0-flash: 1-8, default: 1

maxOutputTokens Optional: int

Maximum number of tokens that can be generated in the response. A token is approximately four characters. 100 tokens correspond to roughly 60-80 words.

Specify a lower value for shorter responses and a higher value for potentially longer responses.

For more information, see Content generation parameters.

stopSequences Optional: List[string]

Specifies a list of strings that tells the model to stop generating text if one of the strings is encountered in the response. If a string appears multiple times in the response, then the response truncates where it's first encountered. The strings are case-sensitive.

For example, if the following is the returned response when stopSequences isn't specified:

public static string reverse(string myString)

Then the returned response with stopSequences set to ["Str", "reverse"] is:

public static string

Maximum 5 items in the list.

For more information, see Content generation parameters.

presencePenalty Optional: float

Positive penalties.

Positive values penalize tokens that already appear in the generated text, increasing the probability of generating more diverse content.

The maximum value for presencePenalty is up to, but not including, 2.0. Its minimum value is -2.0.

frequencyPenalty Optional: float

Positive values penalize tokens that repeatedly appear in the generated text, decreasing the probability of repeating content.

This maximum value for frequencyPenalty is up to, but not including, 2.0. Its minimum value is -2.0.

responseMimeType Optional: string (enum)

The output response MIME type of the generated candidate text.

The following MIME types are supported:

- application/json: JSON response in the candidates.
- text/plain (default): Plain text output.
- text/x.enum: For classification tasks, output an enum value as defined in the response schema.

Specify the appropriate response type to avoid unintended behaviors. For example, if you require a JSON-formatted response, specify application/json and not text/plain.

text/plain isn't supported for use with responseSchema.

responseSchema Optional: schema

The schema that generated candidate text must follow. For more information, see Control generated output.

To use this parameter, you must specify a supported mime type other than text/plain for the responseMimeType parameter.

seed Optional: int

When seed is fixed to a specific value, the model makes a best effort to provide the same response for repeated requests. Deterministic output isn't guaranteed. Also, changing the model or parameter settings, such as the temperature, can cause variations in the response even when you use the same seed value. By default, a random seed value is used.

responseLogprobs Optional: boolean

If true, returns the log probabilities of the tokens that were chosen by the model at each step. By default, this parameter is set to false.

logprobs Optional: int

Returns the log probabilities of the top candidate tokens at each generation step. The model's chosen token might not be the same as the top candidate token at

each step. Specify the number of candidates to return by using an integer value in the range of 1-20.

You must enable responseLogprobs to use this parameter.

audioTimestamp Optional: boolean

Available for the following models:

- Gemini 2.0 Flash-Lite
- Gemini 2.0 Flash

Enables timestamp understanding for audio-only files.

This is a preview feature.

thinkingConfig Optional: object

Configuration for the model's thinking process for Gemini 2.5 and higher models.

The thinkingConfig object contains the following fields:

- thinkingBudget: integer. By default, the model automatically controls how much it thinks up to a maximum of 8,192 tokens.
- thinkingLevel: enum. Controls the amount of internal reasoning the model performs before generating a response. Higher levels may improve quality on complex tasks but increase latency and cost. Supported values are LOW and HIGH.

mediaResolution Optional: MediaResolution

Controls how input media is processed. LOW reduces tokens per image/video, possibly losing detail but allowing longer videos in context. Supported values: HIGH, MEDIUM, LOW.

Response body

```
{
 "candidates": [
  {
   "content": {
    "parts": [
     {
      "text": string
     }
```

```
  ]
},
"finishReason": enum (FinishReason),
"safetyRatings": [
  {
    "category": enum (HarmCategory),
    "probability": enum (HarmProbability),
    "blocked": boolean
  }
],
"citationMetadata": {
  "citations": [
    {
      "startIndex": integer,
      "endIndex": integer,
      "uri": string,
      "title": string,
      "license": string,
      "publicationDate": {
        "year": integer,
        "month": integer,
        "day": integer
      }
    }
  ]
},
"avgLogprobs": double,
"logprobsResult": {
  "topCandidates": [
    {
      "candidates": [
```

```
          {
            "token": string,
            "logProbability": float
          }
        ]
      }
    ],
    "chosenCandidates": [
      {
        "token": string,
        "logProbability": float
      }
    ]
  }
 ],
 "usageMetadata": {
  "promptTokenCount": integer,
  "candidatesTokenCount": integer,
  "totalTokenCount": integer
 },
 "modelVersion": string
}
```

Response element	Description
modelVersion	The model and version used for generation. For example: gemini-2.0-flash-lite-001.
text	The generated text.
finishReason	The reason why the model stopped generating tokens. If empty, the model has not stopped generating the tokens. Because the response uses the prompt for

	context, it's not possible to change the behavior of how the model stops generating tokens. • FINISH_REASON_STOP: Natural stop point of the model or provided stop sequence. • FINISH_REASON_MAX_TOKENS: The maximum number of tokens as specified in the request was reached. • FINISH_REASON_SAFETY: Token generation was stopped because the response was flagged for safety reasons. Note that Candidate.content is empty if content filters block the output. • FINISH_REASON_RECITATION: The token generation was stopped because the response was flagged for unauthorized citations. • FINISH_REASON_BLOCKLIST: Token generation was stopped because the response includes blocked terms. • FINISH_REASON_PROHIBITED_CONTENT: Token generation was stopped because the response was flagged for prohibited content, such as child sexual abuse material (CSAM). • FINISH_REASON_IMAGE_PROHIBITED_CONTENT: Token generation was stopped because the image provided in the prompt was flagged for prohibited content. • FINISH_REASON_NO_IMAGE: Token generation was stopped because an image was expected in the prompt, but none was provided. • FINISH_REASON_SPII: Token generation was stopped because the response was flagged for sensitive personally identifiable information (SPII). • FINISH_REASON_MALFORMED_FUNCTION_CALL: Candidates were blocked because of malformed and unparsable function call. • FINISH_REASON_OTHER: All other reasons that stopped the token • FINISH_REASON_UNSPECIFIED: The finish reason is unspecified.
category	The safety category to configure a threshold for. Acceptable values include the following: Click to expand safety categories
probability	The harm probability levels in the content. • HARM_PROBABILITY_UNSPECIFIED

- NEGLIGIBLE
- LOW
- MEDIUM
- HIGH

blocked	A boolean flag associated with a safety attribute that indicates if the model's input or output was blocked.
startIndex	An integer that specifies where a citation starts in the content. The startIndex is in bytes and calculated from the response encoded in UTF-8.
endIndex	An integer that specifies where a citation ends in the content. The endIndex is in bytes and calculated from the response encoded in UTF-8.
url	The URL of a citation source. Examples of a URL source might be a news website or a GitHub repository.
title	The title of a citation source. Examples of source titles might be that of a news article or a book.
license	The license associated with a citation.
publicationDate	The date a citation was published. Its valid formats are YYYY, YYYY-MM, and YYYY-MM-DD.
avgLogprobs	Average log probability of the candidate.
logprobsResult	Returns the top candidate tokens (topCandidates) and the actual chosen tokens (chosenCandidates) at each step.
token	Generative AI models break down text data into tokens for processing, which can be characters, words, or phrases.
logProbability	A log probability value that indicates the model's confidence for a particular token.
promptTokenCount	Number of tokens in the request.
candidatesTokenCount	Number of tokens in the response(s).

totalTokenCount	Number of tokens in the request and response(s).

Note: For billing purposes, tokens consumed by document inputs to Gemini 3 Pro and later models are counted as image tokens.

Examples

Text Generation

Generate a text response from a text input.

Gen AI SDK for PythonPython (OpenAI)Go

```
from google import genai
from google.genai.types import HttpOptions

client = genai.Client(http_options=HttpOptions(api_version="v1"))
response = client.models.generate_content(
    model="gemini-2.5-flash",
    contents="How does AI work?",
)
print(response.text)
# Example response:
# Okay, let's break down how AI works. It's a broad field, so I'll focus on the ...
#
# Here's a simplified overview:
# ...
```

Using multimodal prompt

Generate a text response from a multimodal input, such as text and an image.

Gen AI SDK for PythonPython (OpenAI)Go

```
from google import genai
from google.genai.types import HttpOptions, Part

client = genai.Client(http_options=HttpOptions(api_version="v1"))
response = client.models.generate_content(
    model="gemini-2.5-flash",
```

```
    contents=[
        "What is shown in this image?",
        Part.from_uri(
            file_uri="gs://cloud-samples-data/generative-ai/image/scones.jpg",
            mime_type="image/jpeg",
        ),
    ],
)
print(response.text)
# Example response:
# The image shows a flat lay of blueberry scones arranged on parchment paper. There are ...
```

Streaming text response

Generate a streaming model response from a text input.

Gen AI SDK for PythonPython (OpenAI)Go

```
from google import genai
from google.genai.types import HttpOptions

client = genai.Client(http_options=HttpOptions(api_version="v1"))

for chunk in client.models.generate_content_stream(
    model="gemini-2.5-flash",
    contents="Why is the sky blue?",
):
    print(chunk.text, end="")
# Example response:
# The
#  sky appears blue due to a phenomenon called **Rayleigh scattering**. Here's
#  a breakdown of why:
# ...
```

Model versions

To use the auto-updated version, specify the model name without the trailing version number, for example gemini-2.0-flash instead of gemini-2.0-flash-001.

For more information, see Gemini model versions and lifecycle.

What's next

- Learn more about the Gemini API in Vertex AI.
- Learn more about Function calling.
- Learn more about Grounding responses for Gemini models.

Was this helpful?

Send feedback

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5.8.3 Cloud Run Function

Small pieces of code that run in the cloud. They're triggered by specific events, like a new image being uploaded or a message being posted. You can use Cloud Run functions to automatically process these events using the Gemini API. For example, you could have a Cloud Run function that uses Gemini to automatically generate captions for uploaded images. The key benefit is that you don't have to manage any servers. Google takes care of all the infrastructure.

5.8.4 Cloud Run

You can integrate the Gemini API into virtually any application that can make HTTP requests. This can be a wide range of environments, including web applications, mobile apps, desktop software, and even embedded systems.

You can also leverage your API or code through some of Google's no code and low code tooling, such as:

- **Apps Script**

  A cloud-based platform that lets you automate Google Workspace tasks by leveraging a combination of JavaScript code and Google's built-in services. You can use the Gemini API with Apps Script to build add-ons that enhance features in Google Workspace using generative AI.

- **Appsheet**

  AppSheet is a no-code platform for building custom business apps. You can use Apps Script to extend the capabilities of your AppSheet applications using custom logic or by integrating APIs. With the Gemini API, for example, you can automate tasks, generate content, or provide intelligent assistance directly within your AppSheet app.

5.8.5 Multi-agent applications

When leveraging agents into your application, consider situations when you may need multiple agents.

These systems utilize multiple agents, each potentially specialized for a specific task, to create more efficient and capable applications. For instance, a travel booking app might use one agent to search for flights, another to find hotels, and a third to suggest local attractions.

These agents can work independently or interact with each other to provide a seamless booking experience. This modular approach not only improves efficiency but also allows for greater flexibility and scalability. Even more interestingly, an agent itself can be a tool within another agent. A sentiment analysis agent, for example, could be used by a customer service agent to gauge user satisfaction during an interaction.

Key takeaway

Integrating generative AI models into your applications can be achieved through various methods, including leveraging APIs, Cloud Run functions, Cloud Run, and no-code tools like Apps Script and AppSheet. These methods offer flexibility for incorporating generative AI capabilities into diverse environments and applications. Multi-agent systems can further enhance efficiency and capability by utilizing multiple specialized agents.

Understanding these integration options can empower you to build and deploy generative AI solutions that meet your specific business needs.

5.9 Retrieval-augmented generation (RAG) and tooling

5.9.1 Pre-RAG history

Before retrieval-augmented generation (RAG), models could not directly learn from tool data. They could use tools to fetch information, but lacked the ability to process and integrate that data into their knowledge. Their understanding was limited to their training data, and tool-retrieved data was used only for the immediate query, not for learning. RAG solves this by enabling models to retrieve and learn from data provided by tools.

5.9.2 How the model works with tools

Retrieval-augmented generation (RAG) enhances the capabilities of large language models (LLMs) by grounding their responses in external knowledge sources. This process allows the model to access and process information beyond its training data, leading to more accurate, relevant, and up-to-date responses.

After the user submits a query or request to the LLM, here's how RAG works with tools.

- **Retrieval**

  The LLM, equipped with retrieval tools, identifies relevant information from external sources. These tools can include:

  Data stores: These can be internal databases or other sources of structured and unstructured data.

  - **Vector databases:** These databases store embeddings (numerical representations) of text, allowing the LLM to find semantically similar information to the user's query. The LLM uses the user query to create an embedding and searches the vector database for matching documents or passages.
  - **Search engines:** The LLM can use search engines (via extensions or APIs) to find relevant web pages, articles, or other online content.
  - **Knowledge graphs:** These structured databases store information about entities and their relationships. The LLM can query knowledge graphs to retrieve facts and relationships relevant to the user's query.

  The LLM uses these tools to retrieve a set of potentially relevant documents or passages.

- **Augmentation**

The retrieved information is then incorporated (or "augmented") into the prompt that is fed to the LLM. This augmented prompt now contains both the user's original query and the relevant context retrieved from external sources.

- **Generation**

The LLM processes the augmented prompt and generates a response. Because the prompt includes relevant external information, the LLM can generate a more informed, accurate, and contextually appropriate response. It can also cite the sources of its information, increasing transparency and trustworthiness.

5.9.3 Iteration

In some RAG systems, the LLM might iterate on the retrieval process. For example, if the initial retrieval doesn't yield satisfactory results, the LLM might refine its query or use different retrieval tools to find more relevant information. It could also ask clarifying questions of the user. This iterative process allows the LLM to continuously improve the quality and relevance of its responses.

5.9.4 Example on RAG: Answering a question about a recent event

- **User Query**
  What were the main developments at the recent climate conference in Dubai?"
- **Retrieval**

  The LLM uses a combination of tools:

  - It queries a vector database of news articles related to climate change.
  - It uses a search engine to find recent news reports and official conference documents.
- **Augmentation**

  The LLM combines the user's query with the retrieved information (summaries of key discussions, agreements, and announcements from the conference).

- **Generation**

  The LLM generates a response summarizing the main developments at the climate conference, citing the news articles and conference documents it used as sources. Because it has access to up-to-date information, the LLM's response is more accurate and comprehensive than if it had relied solely on its training data.

- **Iteration [Optional]**
  If the initial search results are insufficient, the LLM might try different search terms or query other data sources to get more complete information.

By combining the power of LLMs with the ability to access and process external knowledge, RAG systems create more powerful and reliable AI agents.

5.9.5 Data Stores

RAG allows LLMs to access and process information beyond their training data, making them more accurate and relevant. A key component of this process are data stores in AI Applications. They act as structured and unstructured knowledge bases that your agent can draw upon.

You can access a variety of data store options to connect your agent to your data through AI Applications, including:

- **Websites:** Access and process information directly from web pages, enabling your agent to stay up-to-date with current events or access publicly available information.
- **Structured data:** Connect to information organized in tables or JSON format, such as product catalogues, customer databases, and internal knowledge bases. Vertex AI can automatically understand the structure of your data or allow you to define it.
- **Unstructured data:** Utilize files in various formats such as HTML, PDF, and DOCX, making it possible for your agent to access information from a wide range of sources.

➢ **Key takeaway**

Data stores in AI Applications are essential for building powerful and reliable AI agents. They provide a way to manage and organize your agent's knowledge, enabling it to provide accurate, relevant, and up-to-date responses. By leveraging the power of RAG and data stores, you can create agents that are truly helpful and informative.

5.9.6 RAG in action

Previously, you learned about the core concepts of retrieval-augmented generation (RAG) and how it works with tools. In this lesson, you'll get hands-on with Google AI Studio to see how grounding a model with real-time web data can transform its output.

Test out using grounding with Google Search to see how RAG can impact outputs by providing more accurate and relevant information, drawing from real-time web data.

➢ **Open Google AI Studio**

To begin experimenting with AI in Google AI Studio, complete the following steps:

1. If you haven't already, log into your Google account and open Google AI Studio. (Note: If prompted, you'll need to accept the terms and conditions order to proceed.)
2. Select Try Gemini.

➢ **Update the Model**

- In **Run settings**, select the current model.
- Choose **Gemini 2.5 Flash**.

➢ **Run a Prompt**

- Locate the prompt input area and **Run** button on the page. This is used to send your prompt to the Gemini model.
- Enter a prompt into the prompt input such as *"What are the latest trends in generative AI?"*.
- Press **Run** to generate a response to your prompt.

➢ **Get Grounded Result**

- To activate grounding, go to **Run settings > Tools** and toggle on the **Grounding using Google Search** option.
- Re-enter the same prompt you just provided and press **Run** to generate a new result.

What did you notice about the difference between the first and second responses?

Key takeaway

Hands-on experience with RAG demonstrates its power to make a model's responses more accurate and up-to-date. By grounding your prompts in real-time data from a tool like Google Search, you can ensure the model provides relevant information beyond its original training data.

5.10 The power of search agents

5.10.1 Vertex AI Search

Vertex AI Search offers both search and recommendation solutions.

Search solutions

Search allows you to create a powerful search experience for your public website. It can index and search across a variety of data types, including structured data in BigQuery and unstructured documents stored in Google Cloud Storage. This ensures your users can easily find the information they need on your website, regardless of how it's stored.

Search also extends to include some specific forms of search such as:

➢ **Document Search**

If your primary need is to search across a large repository of unstructured documents stored in Google Cloud Storage, this is the ideal solution. It's optimized for understanding and retrieving information from text-heavy data, making it perfect for internal knowledge bases, document archives, and more.

- **Media Search**

For organizations dealing with rich media libraries, media search provides specialized capabilities. It's designed to understand and search within images, videos, and audio files. This allows users to find specific media assets based on their content, metadata, or even spoken words within audio or video.

- **Healthcare Search**

Addressing the unique needs of the healthcare industry, this option enables searching across healthcare data while supporting regulatory compliance. It allows professionals to access patient records, research articles, and medical guidelines efficiently and securely.

- **Search for Commerce**

Specifically designed for e-commerce, this option focuses on building a search app for your retail catalog. It optimizes for product discovery, understanding product attributes, and handling complex product queries.

5.10.2 Recommendation solutions

The general purpose recommendation engine can be used to recommend similar content within websites, documents, and other structured content. It analyzes user behavior and content attributes to provide personalized recommendations, increasing user engagement and content discovery.

This also extends to include some specific forms of recommendations such as:

- **Media Recommendations**

Tailored for consumer-focused media applications like audio/video streaming and digital publishing, this option provides specialized recommendations. It understands user preferences and consumption patterns within media content to suggest relevant audio, video, articles, and more.

- **Retail Recommendations**

Optimized for e-commerce, this solution drives sales by providing personalized product recommendations. It leverages user browsing history, purchase data, and product attributes to suggest items that are likely to be of interest to each individual shopper.

5.10.3 How Vertex AI Search works

Regardless of the specific search or recommendation option you choose, Vertex AI Search operates on a foundation of intelligent data connection, grounding, and generative AI. It seamlessly connects to your existing data stores, whether they are structured databases, unstructured document repositories, or a combination of both. This connection is crucial as it allows Vertex AI Search to act as an agent, observing the user's query or context (the environment) and acting by retrieving relevant information or suggesting relevant items (using the data stores as tools) to achieve the goal of providing the right information or recommendation at the right time.

A key strength of Vertex AI Search lies in its ability to ground gen AI LLM responses with your first-party data, curated third-party data, and even Google's knowledge graph (Grounding with Google Search), minimizing "hallucinations" and ensuring trustworthy information.

This grounding is where the connection to RAG comes in. By using your own data sources as the foundation for LLM responses, Vertex AI Search implements a RAG approach, ensuring that the information provided is relevant, accurate, and grounded in the context of your specific data.

Vertex AI Search also gives you the options of adding **extra generative AI features** to your search functionality.

- **Search summaries:** Vertex AI Search can generate concise and informative summaries of search results, saving users time and effort. These summaries can be tailored to different needs, providing a quick overview of a document, a comparison of multiple products, or a synthesis of key findings from a set of search results.
- **Answers and follow ups:** Vertex AI Search can add AI generated answers to your search results. Users can ask questions in natural language to your website's search and get AI generated answers based on the search results. Users can then follow up with further questions.

Vertex AI Search is **built for enterprise** and offers granular access controls to help ensure data security. It also provides advanced analytics to understand search trends and user behavior and scalable infrastructure to handle large volumes of data and search requests. It's designed to meet the demanding requirements of organizations of all sizes. Vertex AI Search is built to easily integrate with existing enterprise systems through APIs and SDKs. Whether you're building a customer-facing search experience or an internal knowledge base for employees, Vertex AI Search can be seamlessly integrated into your workflow.

Key takeaway

By building Vertex AI Search capabilities into your application, you can empower your customers to find information faster, understand it better, and make more informed decisions. This capability significantly enhances the overall search experience and drives efficiency.

Vertex AI Search for commerce enhances online shopping by understanding customer intent. It goes beyond keywords with features like image search and personalized recommendations, which boosts customer satisfaction and drives sales.

5.11 Gen AI for customer engagement

5.11.1 Google's Customer Engagement Suite

Websites, even with the best search, are not necessarily enough for your company. Customers frequently don't want to search. They want to connect with your company directly to get the answers and support they are looking for.

These points of communication can be a pain point between customers and companies, but they are also an extremely important point of connection. Positive customer engagement could be the difference between the success and failure of your company.

Google's Customer Engagement Suite has tools designed to support your company in engaging with customers effectively. The tools are built using conversational AI, some of which is also generative AI.

With the **Customer Engagement Suite**, you can build:

- Conversational Agent
  Conversational Agents to act as effective chatbots communicating with your customers.
- Agent Assist
  Agent Assist to support your live human contact center agents.
- Conversational Insights
  Conversational Insights to gain insights into all your communications with customers (through chatbot agents or human agents).

  All of this can be built on top of Google's **Contact Center as a Service (CCaaS)**, an enterprise-grade contact center solution that is native to the cloud. Let's dive deeper into the details of how these all work and where generative AI fits in.

5.11.2 Conversational Agents

You may be a bit skeptical of customer service chatbots. But the types of conversational agents you can build today have come a long way.

Remember, there are two primary methodologies you can use when building conversational agents: deterministic and generative.

While deterministic AI alone can be rigid, and generative AI alone may lack structure, their combination creates a powerful solution. A hybrid approach allows you to build conversational agents that can handle a wide range of customer needs effectively.

With Conversational Agents in Google Cloud, you have flexible options for building and integrating your agents with your existing tools. Create simple agents with generative AI by using natural language to instruct your agent or complex hybrid agents with custom rules and logic.

- **Deterministic**

Deterministic is more associated with some historical agents. It is a rule-based, very defined system for your chatbot agents to follow. It will use very defined logic such as if the user presses this number, go to this route. Everything you want a deterministic agent to do needs to be explicitly defined. Deterministic agents usually require low to medium code to build.

- **Generative**

Generative is based on new generative AI technology. It uses large language models to give a real conversational feel to your chatbot. These agents will determine what to do on their own based on your prompt unless you specifically tell it otherwise. Generative agents usually require prompting and either no code or low code to build.

- **Hybrid**

Google Cloud Conversational Agents empowers you to build hybrid agents that combine the strengths of deterministic and generative AI. This approach offers strict control while leveraging generative AI's flexibility to better address customer needs.

5.11.3 Agent Assist

There will always be use cases where conversational agents are not enough or tricky situations where the human touch is needed. This is where live agents come in. But using human agents comes with its own challenges. Not all human agents are the same and they have different levels of experience. There can be a lot of training needed, especially when a new agent starts.

This is where Agent Assist comes in. It is a tool that supports live human agents with in-the-moment assistance, generated responses, and real-time coaching to help them resolve customer issues faster and with greater accuracy. Using AI and generative AI, Agent Assist can recommend agent responses to customers, suggest the appropriate knowledge base content to solve a customer's issue, transcribe or translate calls in real time, summarize conversations and more.

5.11.4 Conversational Insights

Whether using live or virtual agents, these connections with your customers provide lots of value and insights for your company. Conversational Insights analyzes conversational data from across the customer journey to provide contact center leaders and managers with the

data-driven insights to boost efficiency, improve agent performance, and create better customer experiences. This tool uses machine learning analytics to provide you with information such as agent and caller sentiment, entity identification, and call topics. It can automatically identify interesting customer interactions in need of further review.

With Generative FAQ in Insights, you can also see the common questions that customers are asking your contact center and how these questions are being answered. This information can help you identify gaps in your FAQs, track trending questions, and improve customer service responses.

5.11.5 Contact center as a service (CCaaS)

Today's contact centers are complex, needing to handle 24/7 communication across multiple channels (phone, text, email, etc.) while maintaining security and customer privacy.

CCaaS, part of Google's Customer Engagement suite, simplifies this by providing a complete contact center solution. It manages the infrastructure, integrates with CRMs, and offers omnichannel support (consistent experience across all channels, like websites, apps, phone, and text). CCaaS handles simultaneous multichannel communication, channel switching, multimodal interactions (text, voice, images), and agent routing, freeing businesses to focus on customer experience. CCaaS seamlessly integrates with other Customer Engagement Suite tools, including Conversational Agents for automated support, Agent Assist for real-time agent guidance, and Conversational Insights for valuable data analysis.

5.11.6 Key takeaway

Customer Engagement Suite is a cohesive way to build extremely powerful agents for your company that leverage generative AI technology along with other powerful technology to create a seamless experience for your customers.

5.12 Build your own agent

Creating playbooks

When building a generative AI agent with Conversational Agents, you create what is called a playbook for how you want your agent to act. In the playbook, you define your agent's goal, such as providing customer support, answering user questions, or even generating creative content. You then provide detailed instructions on how the agent should act and any rules you would like it to follow. In the playbook, you also have the option to link it to external tools such as data stores. Once your playbook is defined, you are ready to go and can start testing your agent and interacting with it.

Try it: Build your own agent

Cymbal Manufacturing is a leading producer of high-quality machinery. The company wants to implement an AI-powered chatbot on their website to assist customers with common inquiries, provide product information, and offer basic troubleshooting.

In this activity, you use **Google AI Studio** to build a context-aware chatbot. You explore how to define the chatbot's persona, knowledge, and behavior using system instructions, focusing on a use case for Cymbal Manufacturing.

What are system instructions?

System instructions in AI Studio are very similar to playbooks in Conversational Agents. With system instructions, you can provide context, persona, and constraints to an agent *before* any user input is provided to guide the model's behavior and ensure its responses align with your desired outcome.

- **Consistency:** Ensure the chatbot maintains a consistent tone and persona.
- **Accuracy:** Ground the model in specific knowledge, reducing hallucinations.
- **Relevance:** Keep responses focused on the intended domain (e.g., Cymbal Manufacturing).
- **Safety:** Prevent the model from generating inappropriate or unhelpful content.

Important note

Building true enterprise-grade conversational agents requires more advanced tools and techniques. For production-ready agents with robust capabilities like adversarial defense, we recommend exploring Google Cloud Conversational Agents.

➢ **STEP 1**

Set up Google AI Studio

Go to Google AI Studio via **aistudio.google.com**.

- On the Home screen, select **Chat with models**.
- In the **Run settings** menu on the right, click the current model name.
- Select **Gemini 2.5 Flash**.

➢ **STEP 2**

Generate system instructions

You need system instructions for your Cymbal Manufacturing chatbot, but what should you even write? Where should you get started? Prompt engineers often use generative AI to create system instruction prompts. In this step, you use this method to generate the system instructions for the Cymbal Manufacturing chatbot.

- In the left panel, select **Enable saving** (under **Playground**).
- If prompted, select **Allow access** and follow the authentication steps to provide Google Drive access.
- In the **Run settings** menu on the right, select **System instructions**.
- In the **Title** field, type *Agent instruction writer*.
- In the **System Instructions** input box, paste the following prompt to create the expert system instruction writer:

You are an expert at building virtual agent assistants. For the provided company, role, and other information, provide a system prompt that a developer can use. You always fulfill the user's request to the best of your ability. Create a distinct goal and set of instructions for this system prompt.

- The system instructions will automatically save. Close the **System instructions** panel.

➢ **STEP 3**

Generate system instructions for your chatbot

You now prompt the model to generate a system prompt for the Cymbal Manufacturing chatbot.

- In the prompt input box, enter the following text:

Your task: Build a system prompt for a virtual agent optimized for an LLM that would be appropriate for to assist an end user with the following job: *End user's company: Cymbal Manufacturing*

Use case to be developed: Cymbal Manufacturing builds storage containers primarily. However, Cymbal is also capable of doing custom machining based on request. If the user needs custom machining done, they should call the company directly. You may answer any question which seems reasonable within the scope of container manufacturing, but do not take too many guesses. Our containers are typically blue or green, but they can occasionally be red. Make sure to always keep a respectful tone and admit when you don't know the answer to a user's question. Cymbal's phone number is (333) 333-3333.

- Select **Run**.
- Select **Copy to clipboard** to copy the output for use in the next step.

You now have a goal and instructions for your Cymbal Manufacturing chatbot.

➢ **STEP 4**

Create the Cymbal Manufacturing chatbot

Now that you have system instructions for your Cymbal Manufacturing chatbot, it is time to create it.

At the top, select **New chat** (+).

- In the **Run settings** menu on the right, select **System instructions**.
- Click the dropdown and select **Create new instruction**.
- Type *Cymbal Manufacturing chatbot*.
- If you didn't already, copy the system instructions output from Step 3 above, which includes the goal and instructions for your Cymbal Manufacturing chatbot.
- Paste the system instructions output from the previous step into the **System instructions** for the Cymbal Manufacturing chatbot.
- The system instructions will automatically save. Close the **System instructions** panel.

➢ **STEP 5**

Test your conversational agent

In the prompt box at the bottom of your **Cymbal Manufacturing chatbot**, write different prompts to converse with the Cymbal Manufacturing conversational agent, selecting "Run" as needed.

List of things you can try:

- Say *hello*.
- Ask about the storage container product such as the color or size.
- Ask about custom manufacturing.
- Ask about something completely random like the weather.

Note: Remember this is a very basic sample of a conversational agent and how you might start to create one. It does not represent the depth of a fully developed agent using a tool such as Conversational Agents.

Tip

You may not realize it, but you actually were using a prompting technique called metaprompting in this lesson. In steps two and three, you generated the prompt that you then used as the system instructions for your final conversational agent. Metaprompting is about creating prompts that guide the AI to generate, modify, or interpret other prompts.

Metaprompting is a powerful tool for interacting with AI, enabling more dynamic, flexible, and adaptable prompt creation and interpretation. It's a key technique for unlocking the full potential of large language models.

Key takeaway

In Conversational Agents, you can create a playbook for how you want your agent to act. When building a generative AI agent with Conversational Agents, you can link to external tools like data stores. Google Cloud Conversational Agents would generally be the recommended way of building a customer chatbot agent. However, in this lesson, we used Google AI Studio as it's more accessible to a wider audience.

Metaprompting is a powerful tool for interacting with AI, enabling more dynamic, flexible, and adaptable prompt creation and interpretation. It's a key technique for unlocking the full potential of large language models.

5.13 Gemini Enterprise

5.13.1 Overview

One of the most powerful ways to transform your organization isn't necessarily through products for your external customers, but actually through supporting your internal knowledge workers and employees.

Often, employees search across multiple sources for information within their organization. Examples of such sources include software used for analytics, productivity, content, customer relationship management, communications, portfolio management, supply chain, and enterprise planning.

One way to support knowledge workers is by making information across your enterprise more discoverable.

Gemini Enterprise is designed to help you and your team use your company's information more effectively. It uses AI to create customized agents that can access and understand data from various sources, regardless of where that data is stored. These agents can then be integrated into your organization's internal websites or dashboards. Think of it as a way to give your employees their own personal research assistants, but for work.

Gemini Enterprise is a launch point for all your enterprise-ready expert agents.

- **Connect to NotebookLM Enterprise**

  Connect to NotebookLM Enterprise so employees can upload information to analyze, get insights, and listen to audio summaries of data.

- **Connect to Multimodal Search Agents**

  Connect to multimodal search agents grounded in your data (structured and unstructured) across multiple systems (Google Drive, Confluence, Jira, ServiceNow, and more) so your employees can find relevant information within your company.

- **Add Generative AI Assistants**

  Add generative AI assistants that are grounded in your enterprise data and can be prompted to take actions through connectors (tools). For example, creating Google Calendar events or editing issues in Jira Cloud.

- **Add custom agents**

  Add custom agents through Conversational Agents.

5.13.2 Gemni Enterprise Use Case

Let's explore how Gemini Enterprise can help a personal wealth management advisor generate new trade ideas and effortlessly conduct deep research on a trending topic.

- **Client Trade Idea Generation**

  Problem: A financial advisor needs to quickly translate new market research into a validated, personalized trade idea for a client with specific personal needs.

Solution:

1. The advisor uses Gemini Enterprise to retrieve and compare the latest investment reports to identify market indicators.
2. They attach NotebookLM Client Notes to ensure advice is tailored to the client.
3. The advisor uses an agent to evaluate the research against client notes. It finds that the client's portfolio doesn't have enough regional or sector-level diversification.
4. The advisor uploads a spreadsheet, runs the analysis through the company's risk calculator, then asks Gemini to draft the final client email.

- **Initiating Deep Market Research**

Problem: Advisor needs to draft a comprehensive, well-sourced report on a trending, complex political topic impacting financial markets.

Solution:

1. The advisor first limits the data sources to trusted, curated repositories, including a special bank of Government Reports.
2. Instead of immediately writing, the agent generates a comprehensive plan for its deep research, which the advisor verifies.
3. The agent executes the plan, searches thousands of sources, asks new questions, and works until satisfied.
4. Gemini Enterprise produces a detailed report, complete with all source links and even an accompanying audio summary for quick consumption.

5.13.3 How is Gemini Enterprise different from NotebookLM?

While Gemini Enterprise can connect to NotebookLM Enterprise, they serve different core purposes. Think of NotebookLM Enterprise as your specialized AI tool for diving deep into specific documents and web sources – asking questions, summarizing, and creating new content based *only* on those sources. Gemini Enterprise, on the other hand, is your comprehensive enterprise AI assistant. It uses AI agents and unified search to automate tasks and find information across *all* your connected business systems, not just specific documents you upload.

5.13.4 Key takeaway

Gemini Enterprise s a central platform for AI agents that are designed for your specific business needs. It brings together the power of Google's AI with your company's data to make it easier for employees to do their jobs. It is designed to be a secure way for businesses to deploy AI agents using Google Cloud's infrastructure. You can think of it like your personal AI assistant for work. It helps employees find information, understand data, and automate

tasks. It is designed to increase productivity for teams in a variety of fields like marketing, sales, HR, software, and research and development.

5.14 Strategy: Planning ahead

5.14.1 Plan for generative AI integration

To effectively integrate gen AI into your organization, consider the following key steps:

- **Establish a Clear Vision**

  Define how gen AI integrates with your overall corporate strategy. Secure buy-in from leadership and appoint a champion to drive adoption.

- **Prioritize high-impact Use cases**

  Focus on opportunities that align with business goals, considering factors like technical feasibility, data readiness, and potential for transformation.

- **Invest on Capabilities**

Develop the necessary skills within your organization through a combination of upskilling, recruitment, and training. Foster a culture of learning and experimentation to empower teams to leverage gen AI effectively.

- **Drive Organizational Change**

Proactively manage the shifts in workflows, roles, and mindsets that accompany gen AI implementation. Encourage agile collaboration and knowledge sharing to scale best practices.

- **Measure and demonstrate Values**

Track, measure, and assess the impact of gen AI initiatives. Use data-driven insights to continuously improve and demonstrate the value of your efforts.

- **Champion Responsible AI**

Develop and implement a robust framework for responsible AI, ensuring fairness, transparency, privacy, and security in all generative AI initiatives.

5.14.2 Plan for impact

We have discussed the importance of measuring and demonstrating value from gen AI projects, but until now, we haven't really discussed how.

- **Define key metrics**

Start by identifying the metrics most relevant to your goals. Are you aiming for a return on investment (ROI), increased revenue, reduced costs, improved efficiency, enhanced customer experience, or improved security?

The key is to select metrics that align with your overall business objectives. If it's a return on investment (ROI) that's important to your organization, calculate the financial benefits of your gen AI initiatives against their costs. If it's increased revenue, measure the direct impact of gen AI on sales and profits.

- **Collect and analyze data**

Once you've defined your key metrics, it's time to gather data. Set up tracking mechanisms, conduct surveys, or analyze existing data to understand the impact of your gen AI initiatives.

For example, if you're tracking customer experience, you might compare satisfaction

scores before and after implementing a gen AI powered chatbot. Look for patterns and trends to identify areas where gen AI is making a difference.

- **Iterate and improve**

The beauty of gen AI is its adaptability. Use the insights from your data analysis to refine your gen AI strategy and optimize performance. If you're not seeing the desired ROI, consider why you aren't seeing the desired ROI, what are the practical issues or pain points? Then adjust your approach. This may include fine-tuning your models, updating training data, improving prompting techniques, or integrating gen AI more effectively into your workflows.

5.14.3 Plan for change

Even when your gen AI solutions are working and successful, you still need to be prepared to adapt. Technology is constantly changing with new models, tools, and techniques emerging rapidly. Your customers and employees expect you to keep up.

Staying up-to-date requires a proactive approach to learning and development. It also entails fostering a culture of continuous learning and improvement within your organization. Here are some tips on the best ways to stay ahead of the curve with gen AI.

- **Regularly review and refine your strategy** based on the latest advancements and your organization's specific needs.
- **Stay informed** about new models, tools, and techniques by actively following industry news, research publications, and expert opinions.
- **Engage with the generative AI community** through conferences, workshops, and online forums to learn from others and share your knowledge.
- **Invest in training and development** programs to upskill your workforce and equip them with the knowledge and skills needed to thrive in the generative AI era.
- **Attract and retain top talent** by creating a stimulating and rewarding work environment that fosters creativity and professional growth.

5.12 Question Paper - 5

1. Identify the type of Tool: Connect to external Services [via API]
   A) Data Stores
   B) Extensions
   C) Plugins
   D) Functions

2. Identify the type of Tool: Define Specific Actions or Tasks
   A) Data Stores

B) Extensions
C) Plugins
D) Functions

3. Identify the type of Tool: Provide Access to Information
    A) Data Stores
    B) Extensions
    C) Plugins
    D) Functions

4. Identify the type of Tool: Add new skills and intergration
    A) Data Stores
    B) Extensions
    C) Plugins
    D) Functions
5. Which of the following scenarios best illustrates the concept of a multi-agent system in the context of generative AI?
    A) An AI agent that uses multiple tools, like a database and a web search API, to complete a task.
    B) A single AI agent capable of performing multiple tasks, such as writing emails and summarizing documents.
    C) An AI agent that can be customized with different parameters to adapt to different tasks.
    D) A system where multiple specialized AI agents collaborate to achieve a complex goal, such as booking a trip.

6. Imagine an AI agent tasked with scheduling a team meeting. The agent needs to access the team members' calendars, propose suitable time slots, and send out invitations. Which combination of agent tools would be most essential for this task?
    A) Plugins and data stores.
    B) Extensions and plugins.
    C) Data stores and functions.
    D) Functions and databases.

7. What is the primary distinction between Google AI Studio and Vertex AI Studio in terms of their target users?
    A) Google AI Studio requires coding experience, while Vertex AI Studio offers a no-code interface for interacting with models.
    B) Google AI Studio is for accessing image generation models, while Vertex AI Studio is for text-based models.
    C) Google AI Studio is for building custom models, while Vertex AI Studio is for using pre-trained models.
    D) Google AI Studio is designed for beginners and experimentation, while Vertex AI Studio caters to professionals and larger-scale projects.

8. Which parameter in Google AI Studio would you adjust to control the randomness and creativity of Gemini's output?

A) Output length
B) Safety settings
C) Temperature
D) Token count

9. Which of the following statements best describes the function of a "reasoning loop" within a generative AI agent?
    A) It establishes a one-time set of instructions for the agent to follow without deviation.
    B) It restricts the agent's actions to a predefined set of responses, ensuring predictable behaviour.
    C) It guides the agent through an iterative process of observation, internal reasoning, and decision-making to solve tasks.
    D) It enables the agent to interact with users in a conversational manner, responding to prompts and questions.

10. Identify the RAG Step for the following use case: Fetching Relevant documents from a knowledge base based on user query
    A) Generation
    B) Retrieval
    C) Augmentation
    D) Iteration
11. Identify the RAG Step for the following use case: Creating a response to the User's query in natural Language
    A) Generation
    B) Retrieval
    C) Augmentation
    D) Iteration

12. Identify the RAG Step for the following use case: Adding context to the query by including additional information from the knowledge base.
    A) Generation
    B) Retrieval
    C) Augmentation
    D) Iteration

13. Which feature of Vertex AI Search for commerce specifically helped customers find products they couldn't describe with words in this use case?
    A) Its ability to understand synonyms and misspellings in a text query.
    B) Its AI-powered image search, which finds visually similar products from a photo.
    C) Its personalized recommendation engine that suggests new products based on a customer's style profile.
    D) Its search summaries, which provide concise overviews of a product category.

14. Identify the features to the correct Google Customer Engagement Suite tool: Uses Deterministic and Generative AI
    A) Conversational Agent
    B) Agent Assist
    C) Conversational Insights
    D) CCaaaS [Contact Center as a Service]

15. Identify the features to the correct Google Customer Engagement Suite tool: Support Live Human Agents
    A) Conversational Agent
    B) Agent Assist
    C) Conversational Insights
    D) CCaaaS [Contact Center as a Service]

16. Identify the features to the correct Google Customer Engagement Suite tool: Analyze Customer Sentiments
    A) Conversational Agent
    B) Agent Assist
    C) Conversational Insights
    D) CCaaaS [Contact Center as a Service]

17. Identify the features to the correct Google Customer Engagement Suite tool: Provides Omnichannel support
    A) Conversational Agent
    B) Agent Assist
    C) Conversational Insights
    D) CCaaaS [Contact Center as a Service]

18. According to the provided information, how does a large language model (LLM) interact with data stores in a retrieval-augmented generation (RAG) workflow?
    A) The LLM exclusively relies on data stores for generating responses, disregarding its internal knowledge.
    B) The LLM queries data stores to retrieve information relevant to a user's request, enriching its response.
    C) The LLM passively receives information from data stores, waiting for updates to be pushed to it.
    D) The LLM uses data stores to permanently expand its base knowledge and parameters.

19. What are the key steps in planning for impact when implementing generative AI solutions?
    A) Focus on technology adoption and assume positive impact will follow.
    B) Define key metrics, collect and analyze data, and iterate and improve.
    C) Collect and analyze data with a focus on short-term goals in the beginning.
    D) Prioritize AI capabilities without considering ethical implications.

20. Which core capability of Vertex AI Search helps mitigate the issue of hallucinations in generative AI, ensuring that search results are grounded in reliable information?
    A) Advanced analytics
    B) Granular access controls
    C) Grounding
    D) Multimodal search

21. What is a key advantage of using a hybrid approach (combining deterministic and generative AI) when building conversational agents with Google Cloud's Customer Engagement Suite?
    A) It enables strict control over agent behaviour while leveraging the flexibility of generative AI.
    B) It eliminates the need for human agents in customer interactions.
    C) It restricts agent responses to predefined scripts, ensuring consistency.
    D) It allows agents to handle only simple, predictable customer queries.

22. What is the primary purpose of Gemini Enterprise within an organization?
    A) To provide a customer-facing interface for interacting with AI-powered chatbots.
    B) To enhance the organization's cybersecurity defences by detecting and preventing data breaches.
    C) To create a centralized repository for storing and managing all company data.
    D) To provide a centralized platform for organizing and accessing AI agents that can utilize company data to assist employees.

23. Which of the following options best identifies the key components of an AI agent?
    A) API connections, cloud functions, and a sophisticated database.
    B) A reasoning loop, tools, and a foundational model.
    C) A detailed user manual, customer service chatbots, and advanced marketing techniques.
    D) A complex algorithm, a user interface, and a data storage system.

24. What is the primary goal of a modern generative AI agent?
    A) To follow predefined decision trees
    B) To achieve a goal by observing the world and acting using tools
    C) To replace all human decision-making
    D) To operate without external data

25. Which advancement enabled agents to access relevant information beyond training data?
    A) Deterministic workflows
    B) Prompt chaining
    C) Retrieval-Augmented Generation (RAG)
    D) Rule-based scripting

26. Which component serves as the core intelligence of an AI agent?

A) Tools
B) Data stores
C) Foundation Model
D) Plugins

27. What is the primary purpose of tools in an AI agent?

A) Improve language fluency
B) Replace reasoning loops
C) Enable interaction with external systems
D) Store long-term memory

28. Which framework is commonly used to guide an agent's reasoning and actions?

A) Waterfall
B) ReAct (Reason and Act)
C) Agile
D) Scrum

29. What is the key difference between Google AI Studio and Vertex AI Studio?

A) Only Vertex AI Studio supports Gemini
B) AI Studio is for experimentation; Vertex AI Studio supports production workflows
C) Vertex AI Studio does not support APIs
D) AI Studio requires coding expertise

30. Which parameter most directly controls creativity in Gemini's output?

A) Output length
B) Top P
C) Temperature
D) Grounding

31. What happens when a lower Temperature value is used?

A) Output becomes more random
B) Output becomes more emotional
C) Output becomes more factual and deterministic
D) Output length increases

32. What is the purpose of the reasoning loop in an AI agent?

A) One-time decision making
B) Continuous evaluation and action until a goal is achieved
C) Storing user conversations
D) Training the model

33. Which of the following is NOT a key aspect of the reasoning loop?

A) Iterative process
B) Internal listening
C) Decision making

D) Model retraining

34. Which agent tool type standardizes access to external APIs?

A) Functions
B) Data stores
C) Extensions
D) Plugins

35. What is the role of functions in agent tooling?

A) Store embeddings
B) Perform reusable, specific actions
C) Translate languages
D) Control UI elements

36. Which tool provides structured and unstructured knowledge to an agent?

A) Extensions
B) Plugins
C) Functions
D) Data stores

37. In the ReAct cycle, what does "Observation" represent?

A) Internal reasoning
B) Tool execution
C) Output returned by a tool
D) User feedback

38. Which Google Cloud service is best suited for serverless agent tools?

A) Cloud SQL
B) Cloud Storage
C) Cloud Run Functions
D) BigQuery

39. Which API is best for extracting structured data from invoices or contracts?

A) Cloud Vision API
B) Document AI API
C) Natural Language API
D) Translation API

40. What is a key benefit of Retrieval-Augmented Generation (RAG)?

A) Faster training
B) Reduced need for prompts
C) Grounding responses in external, up-to-date data
D) Eliminating hallucinations completely

41. . Which data store type enables semantic similarity search?

A) Relational databases
B) Knowledge graphs
C) Vector databases
D) JSON stores

42. What is Vertex AI Search primarily used for?

A) Training LLMs
B) Building search and recommendation experiences
C) Deploying containers
D) Generating embeddings

43. Which feature of Vertex AI Search helps reduce hallucinations?

A) Deterministic routing
B) Prompt tuning
C) Grounding with enterprise and Google data
D) Model fine-tuning

44. What distinguishes generative conversational agents from deterministic ones?

A) They require more rules
B) They rely on predefined flows
C) They use LLMs to decide responses dynamically
D) They cannot integrate APIs

45. What is the purpose of Agent Assist?

A) Replace human agents
B) Train new LLMs
C) Support live agents with real-time AI assistance
D) Monitor customer sentiment only

46. What does Conversational Insights primarily provide?

A) Chatbot development tools
B) Real-time translations
C) Voice recognition
D) Analytics on customer-agent interactions

47. What is a playbook in Conversational Agents?

A) A training dataset
B) A system instruction defining agent behavior and goals
C) A deployment script
D) A billing configuration

48. Why is measuring gen AI initiatives important?

A) To increase model size
B) To justify licensing costs
C) To demonstrate value and improve continuously

D) To reduce user interactions

49. What is Gemini Enterprise designed to support?

A) Public chatbots only
B) Internal knowledge workers using enterprise data
C) Model fine-tuning
D) Gaming applications

50. How does Gemini Enterprise differ from NotebookLM Enterprise?

A) Gemini Enterprise works only with uploaded documents
B) NotebookLM automates workflows across systems
C) Gemini Enterprise connects agents across enterprise systems
D) NotebookLM supports APIs

Answer Sheet: Question Paper-1

1. A
2. A, C
3. D
4. D
5. B
6. A
7. C
8. C
9. A
10. C
11. A
12. A
13. A
14. B
15. B
16. B
17. C
18. B
19. C
20. B
21. D
22. C
23. B
24. C
25. C
26. C
27. D
28. B
29. C
30. B
31. C
32. B
33. C
34. A
35. C
36. B, C, D
37. A
38. B
39. A
40. B

Answer Sheet: Question Paper -2

1. D

2. C
3. B
4. A
5. C
6. D
7. B
8. B
9. B
10. B
11. A
12. A
13. D
14. A
15. B
16. C
17. C
18. D, E
19. B
20. B
21. C
22. B
23. B
24. C
25. C
26. C
27. B
28. C
29. D
30. D
31. C
32. C
33. C
34. C
35. B
36. C
37. C
38. C
39. B
40. B
41. C
42. B
43. B
44. C
45. B
46. B
47. C
48. C
49. C

50. C

Answer Sheet: Question Paper -3

1. A, B, C, D
2. A
3. A
4. C
5. A
6. A
7. C
8. A
9. B
10. A
11. B
12. D
13. B
14. D
15. D
16. C
17. C
18. C
19. A
20. C
21. C
22. B
23. D
24. C
25. C
26. B
27. C
28. C
29. B
30. B
31. C
32. C
33. C
34. C
35. B
36. C
37. B
38. B
39. B
40. B
41. C
42. B
43. C

44. **B**
45. **C**
46. **D**
47. **C**
48. **C**
49. **C**
50. **C**

Answer Sheet: Question Paper-4

1. **B**
2. **A**
3. **D**
4. **D**
5. **B**
6. **A**
7. **A**
8. **B**
9. **C**
10. **B**
11. **B**
12. **C**
13. **D**
14. **D**
15. **A**
16. **C**
17. **D**
18. **C**
19. **B**
20. **D**
21. **C**
22. **B**
23. **C**
24. **C**
25. **B**
26. **B**
27. **C**
28. **B**
29. **B**
30. **C**
31. **B**
32. **B**
33. **A**
34. **B**
35. **A**
36. **D**
37. **C**

38. **A**
39. **A, B, C, D**
40. **B**

Answer Sheet: Question Paper -5

1. **B**
2. **D**
3. **A**
4. **C**
5. **D**
6. **B**
7. **D**
8. **C**
9. **C**
10. **B**
11. **A**
12. **C**
13. **B**
14. **A**
15. **B**
16. **C**
17. **D**
18. **B**
19. **B**
20. **C**
21. **A**
22. **D**
23. **B**
24. **B**
25. **C**
26. **C**
27. **C**
28. **B**
29. **B**
30. **C**
31. **C**
32. **B**
33. **D**
34. **C**
35. **B**
36. **D**
37. **C**
38. **C**
39. **B**
40. **C**
41. **C**
42. **B**
43. **C**

44. C
45. C
46. D
47. B
48. C
49. B
50. C

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